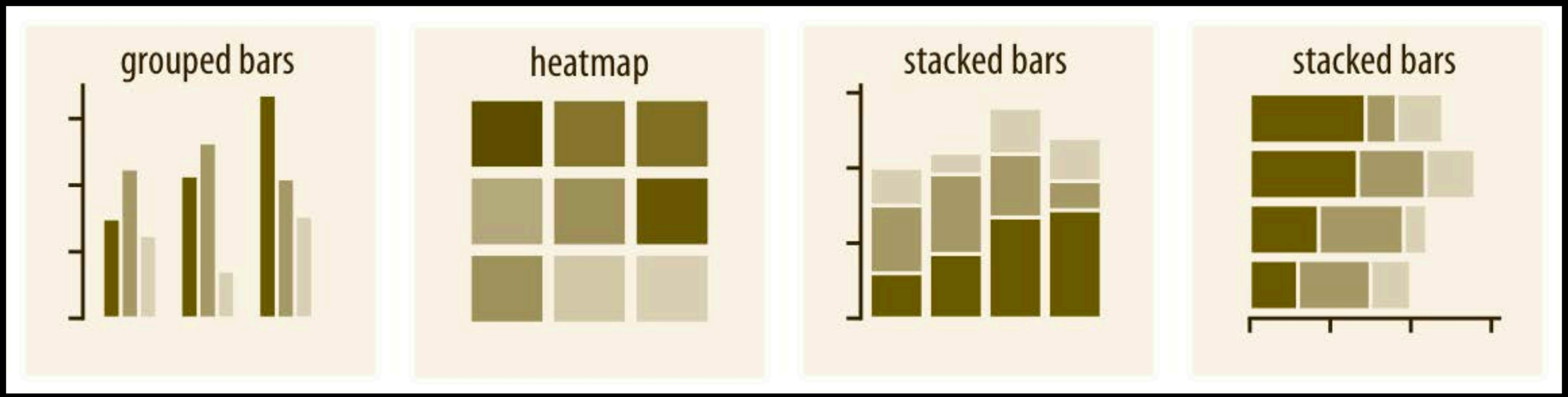
ANT 6973: DATA VISUALIZATION AND EXPLORATION

# VISUALIZING DISTRIBUTIONS AND UNCERTAINTY

# PREVIOUSLY...

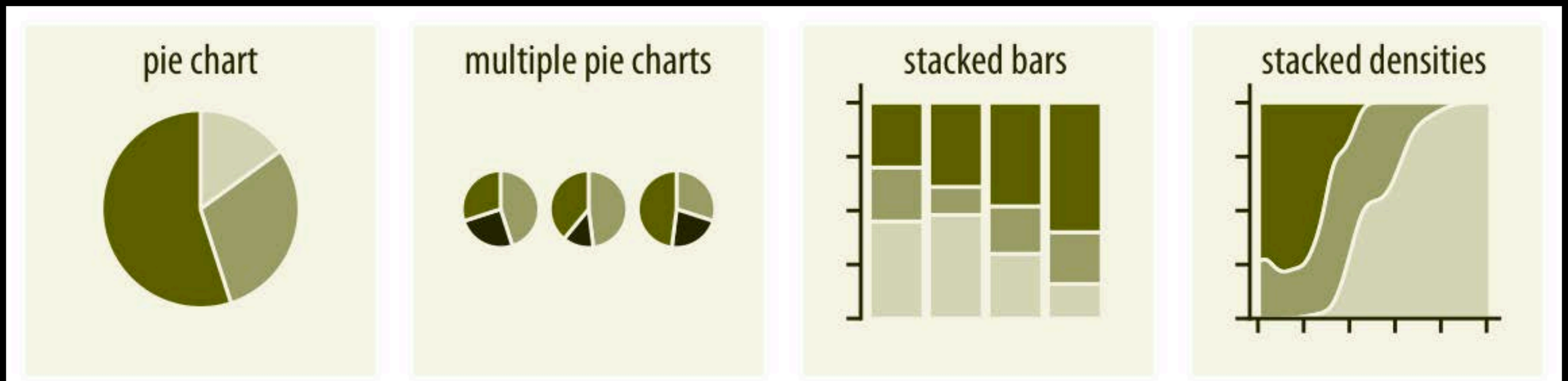
# PREVIOUSLY...

- Visualizing amounts with bar charts (and variants)

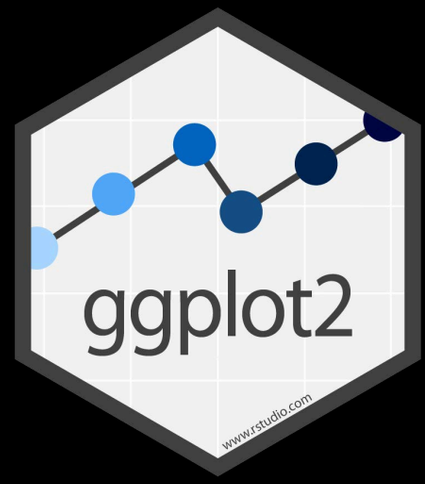


# PREVIOUSLY...

- Visualizing proportions with pies, stacked bars, and stacked areas



# PREVIOUSLY...




- Stats
- Position adjustments
- Coordinates
- Facets
- Scales
- Themes

```
ggplot(data = <DATA>) +
  <GEOM_FUNCTION>(
    mapping = aes(<MAPPINGS>),
    stat = <STAT>,
    position = <POSITION>
  ) +
  <COORDINATE_FUNCTION> +
  <FACET_FUNCTION> +
  <SCALE_FUNCTION> +
  <THEME_FUNCTION>
```

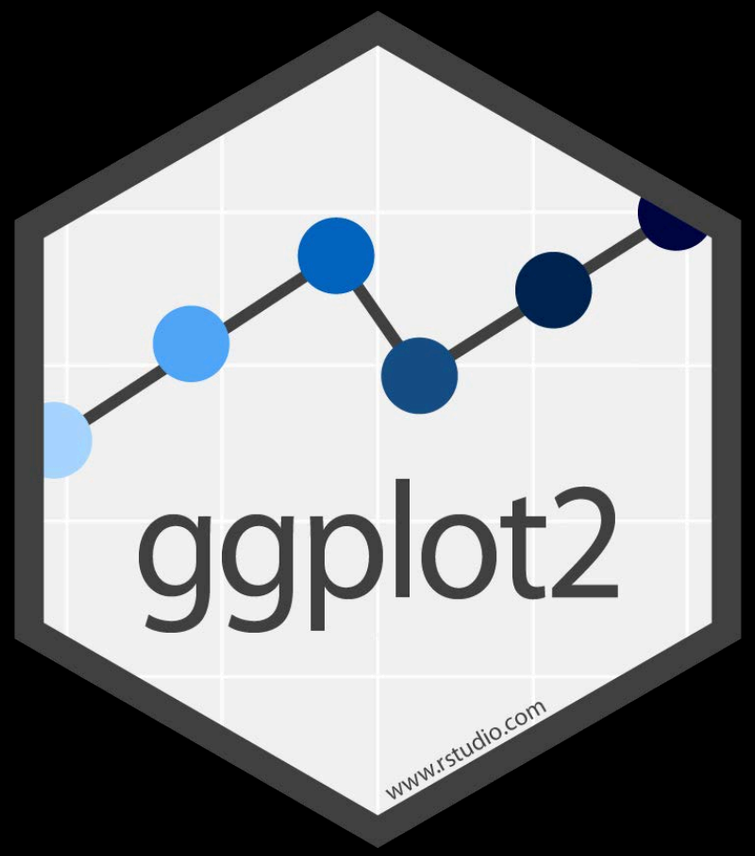
**Required**

Not required, sensible defaults supplied

# TODAY'S TOPICS

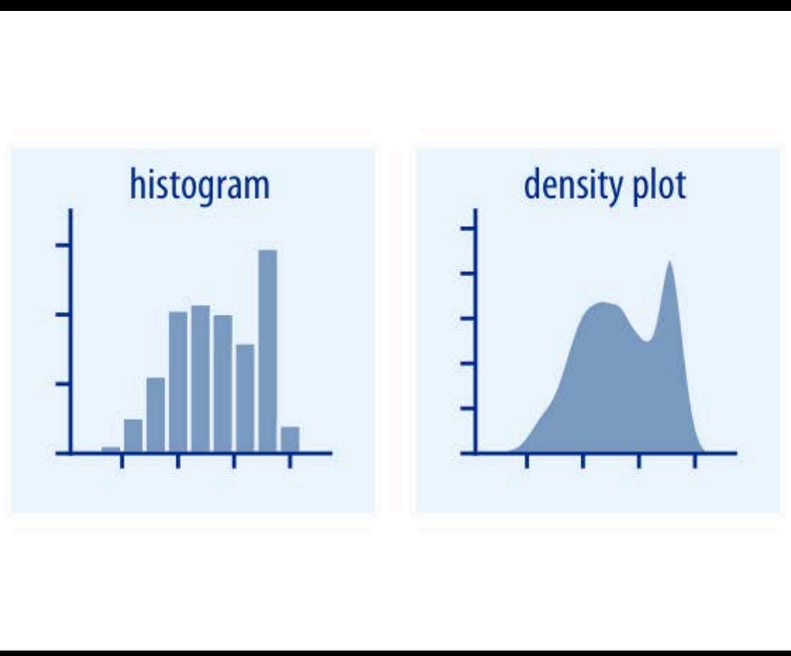
- Visualizing single distributions
- Visualizing multiple distributions
- Visualizing uncertainty
- Activities:
    - Body mass distributions of late Quaternary mammals

# VISUALIZING DISTRIBUTIONS

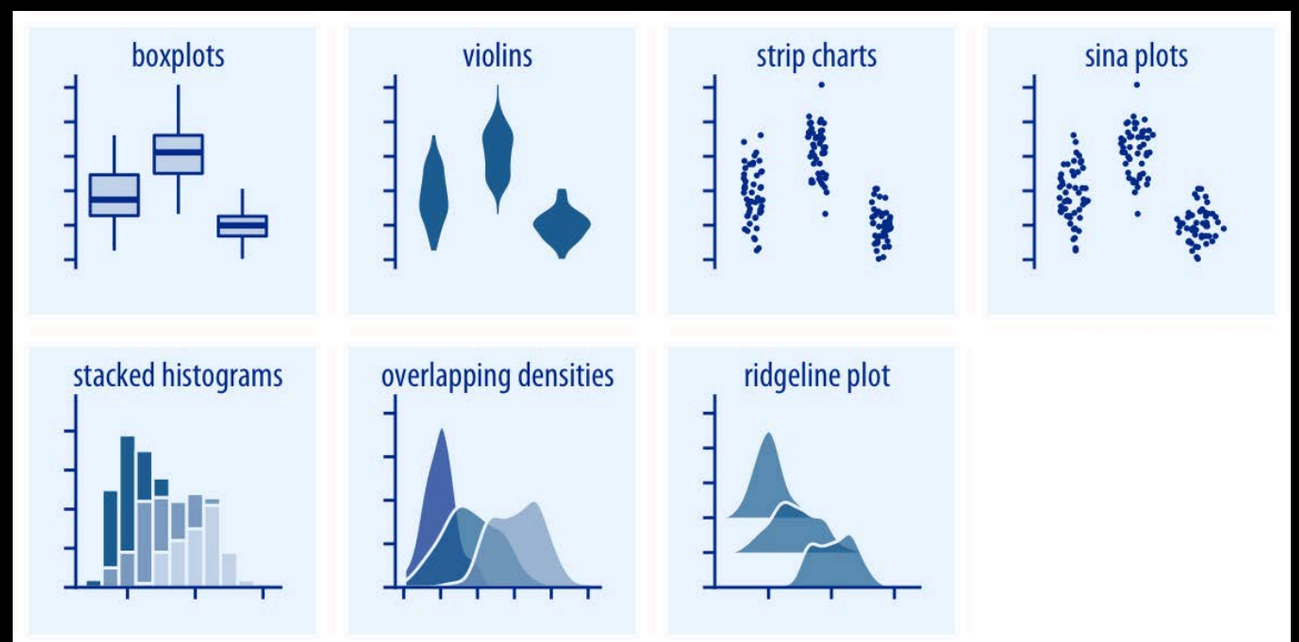


# VISUALIZING DISTRIBUTIONS

## Single distributions




## Multiple distributions



# HISTOGRAMS

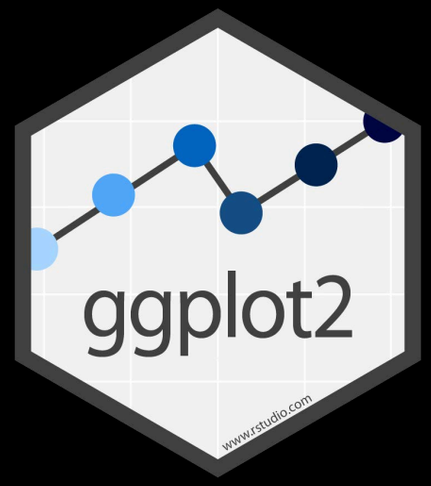
# HISTOGRAMS

- Shows the distribution of a single variable
- Breaks the data apart into specific bins and shows *count* of cases in each bin

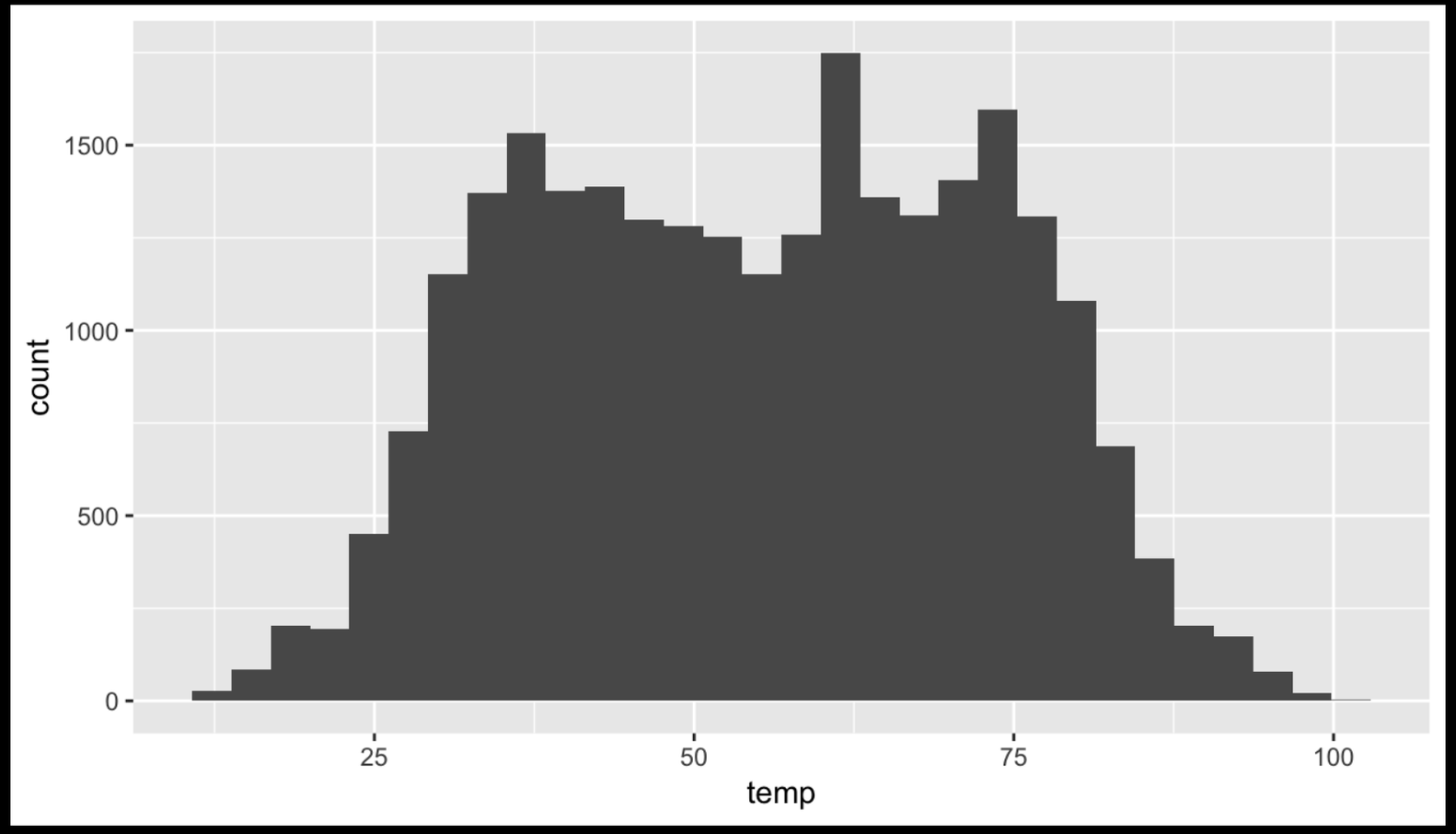


# HISTOGRAMS

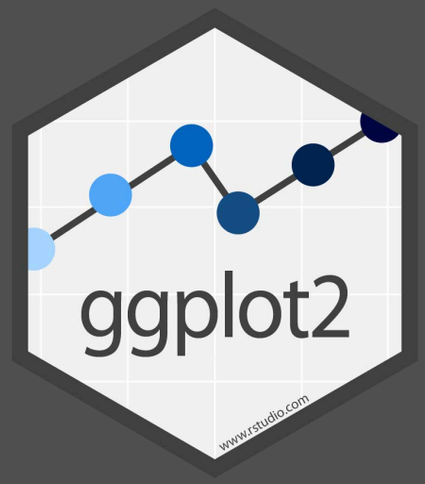
Major difference with bar charts




- Geometric object is `geom_histogram()`
- One numeric variable to mapped to x in `aes()`
- y-aesthetic computed automatically (don't map anything to y)



# YOUR TURN




- Download the late Quaternary mammals data set from the course website ("late_quaternary_mammals.RDS"). This is a file format for single R objects.

- Read the file into R using:

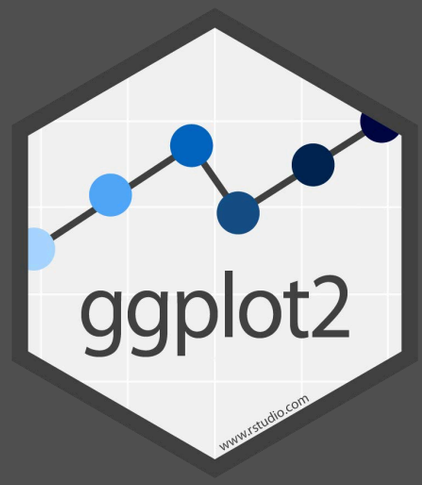
```
lqm ← readRDS("your-path/late_quaternary_mammals.RDS")
```



# YOUR TURN

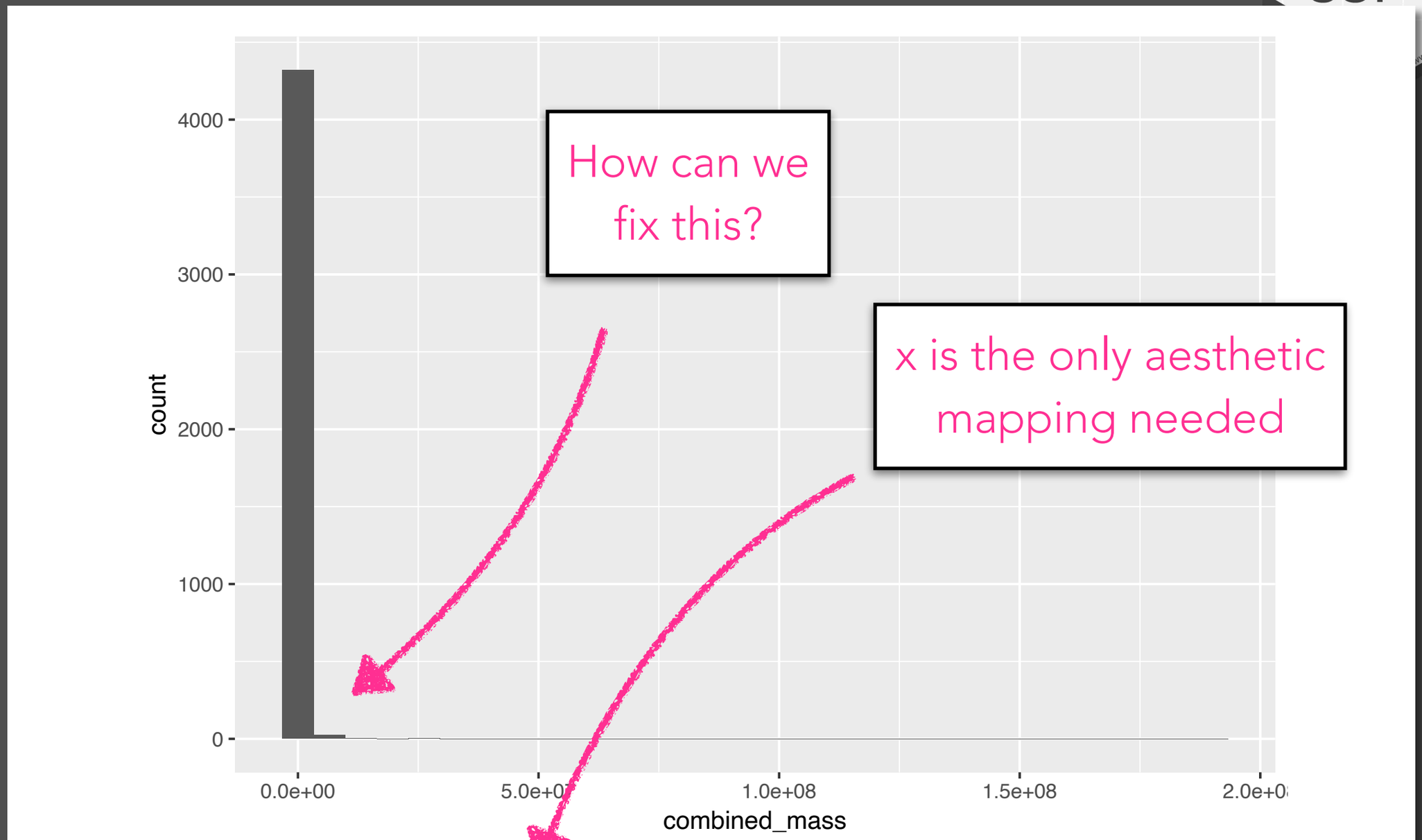
$log_{10}$(combined_mass)

Average of M/F body masses (g)



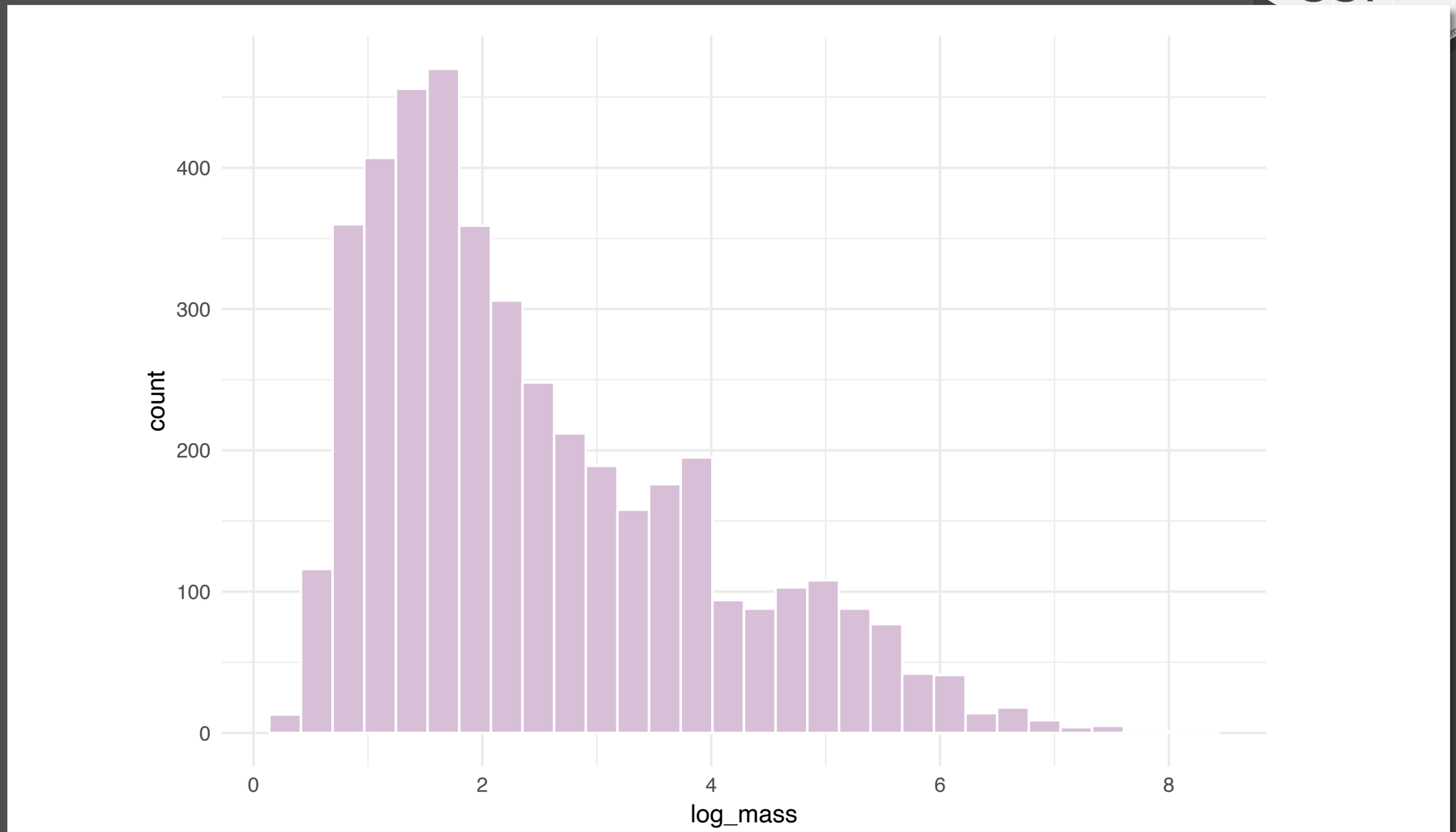

| continent | status | order | family | genus | species | log_mass | combined_mass | reference |
|---|---|---|---|---|---|---|---|---|
| Africa | extant | Artiodactyla | Bovidae | Addax | nasomaculatus | 4.85 | 70000.3 | 60 |
| Africa | extant | Artiodactyla | Bovidae | Aepyceros | melampus | 4.72 | 52500.1 | 63, 70 |
| Africa | extant | Artiodactyla | Bovidae | Alcelaphus | buselaphus | 5.23 | 171001.5 | 63, 70 |
| Africa | extant | Artiodactyla | Bovidae | Ammodorcas | clarkei | 4.45 | 28049.8 | 60 |
| Africa | extant | Artiodactyla | Bovidae | Ammotragus | lervia | 4.68 | 48000.0 | 75 |
| Africa | extant | Artiodactyla | Bovidae | Antidorcas | marsupialis | 4.59 | 39049.9 | 60 |
| Africa | extinct | Artiodactyla | Bovidae | Antidorcas | bondi | 4.53 | 34000.0 | 1 |
| Africa | extinct | Artiodactyla | Bovidae | Antidorcas | australis | 4.60 | 40000.0 | 2 |
| Africa | extant | Artiodactyla | Bovidae | Bos | taurus | 5.95 | 900000.0 | -999 |
| Africa | extant | Artiodactyla | Bovidae | Capra | walie | 5.00 | 100000.0 | -999 |

# YOUR TURN



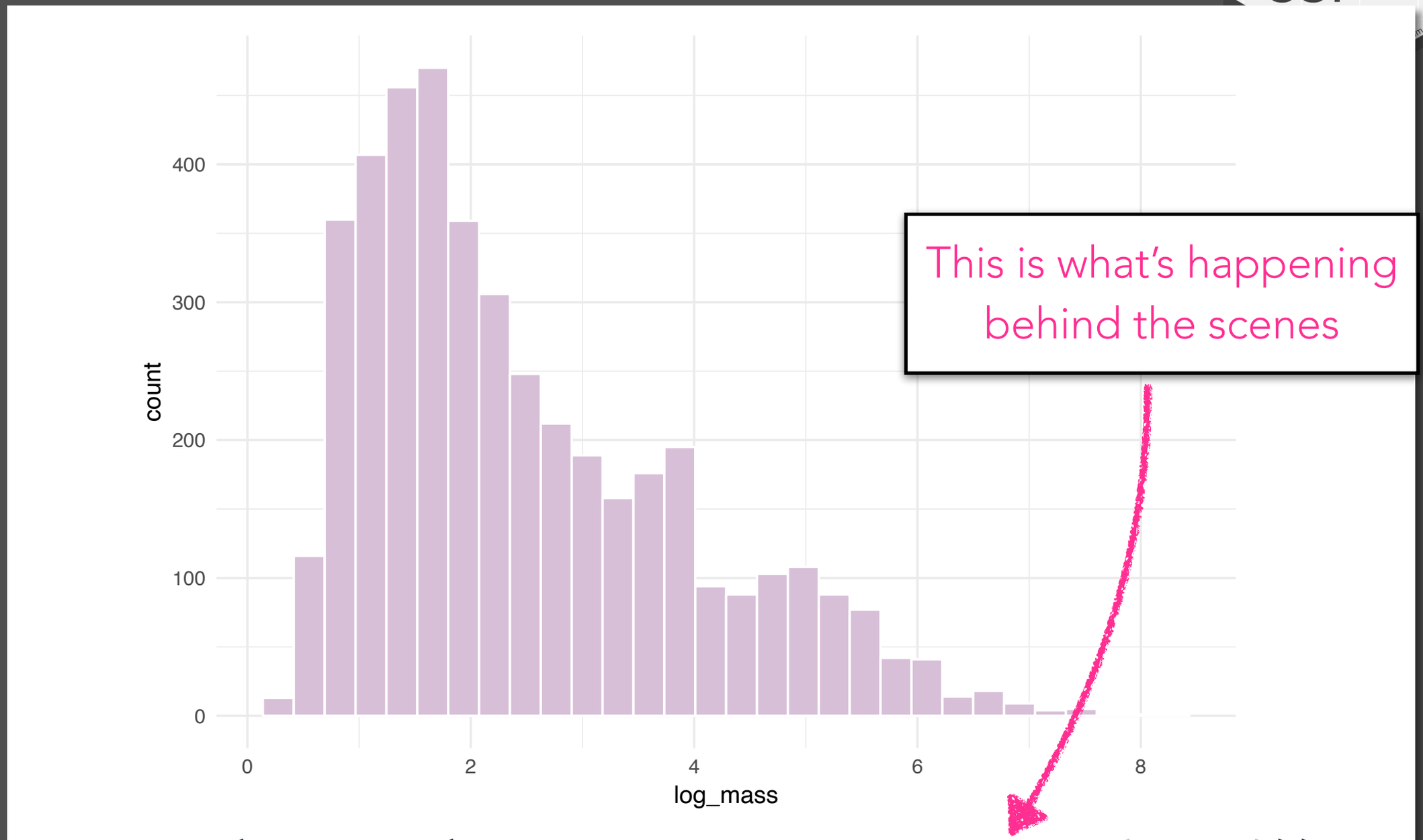

```
ggplot(lqm, aes(x = combined_mass)) +
  geom_histogram()
```

# YOUR TURN



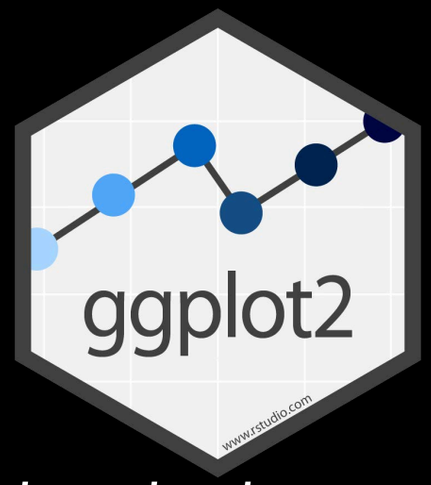

```
ggplot(lqm, aes(x = log_mass)) +
  geom_histogram(fill = "thistle", color = "white")
```

# YOUR TURN




```
ggplot(lqm, aes(x = log_mass, y = after_stat(count))) +
  geom_histogram(fill = "thistle", color = "white")
```

# HISTOGRAMS




- *When making a histogram, always explore multiple bin widths.*

- Binning behavior can be modified using:

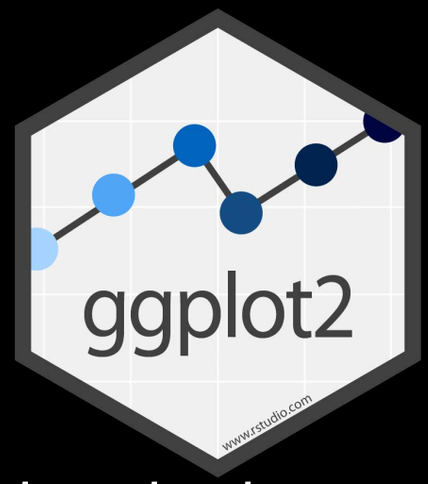
  - bins: `geom_histogram(bins = 30)`
  - binwidth: `geom_histogram(binwidth = 1)`
  - breaks: `geom_histogram(breaks = 0:10)`

# of bins

width of one bin in units of x

sequence of bin boundaries

# HISTOGRAMS



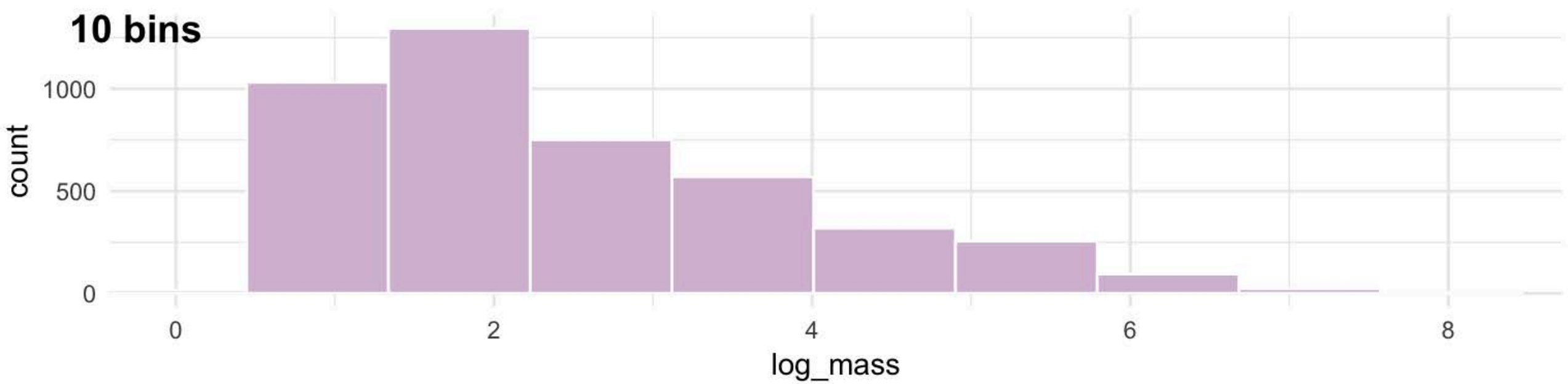

- *When making a histogram, always explore multiple bin widths.*

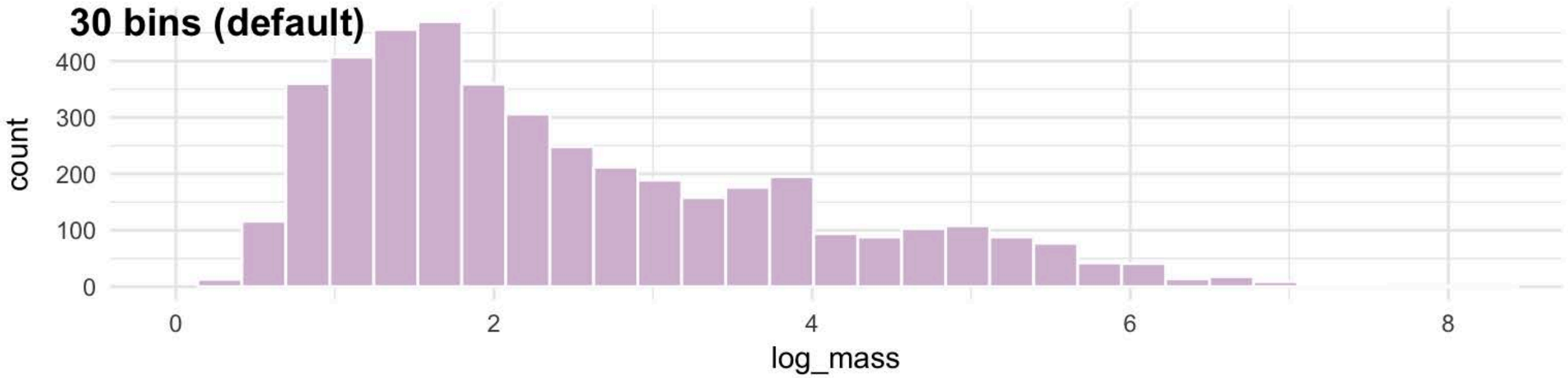
- Binning behavior can be modified using:

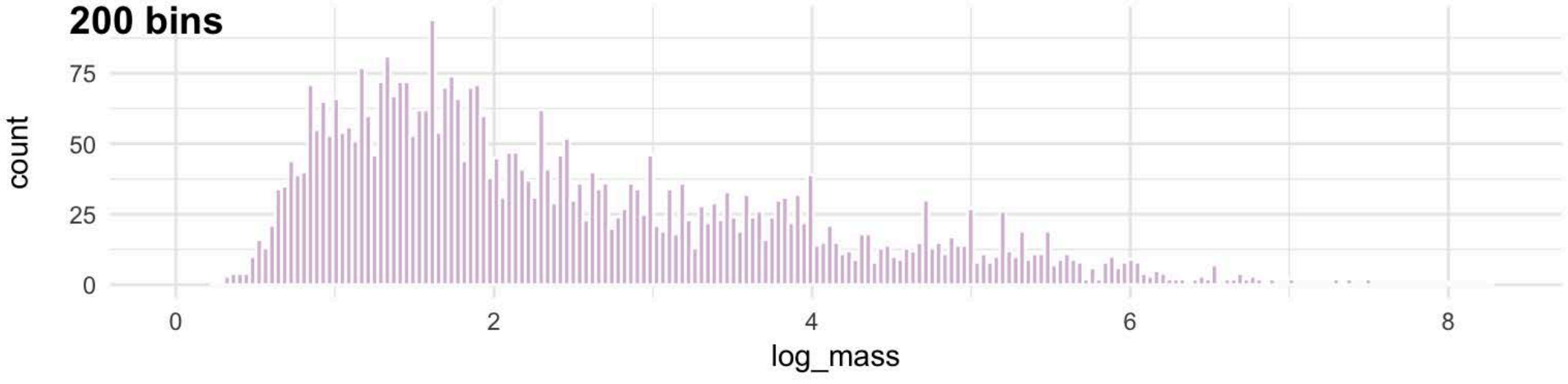
  - bins: `geom_histogram(bins = 30)`

  - binwidth: `geom_histogram(binwidth = 1)`

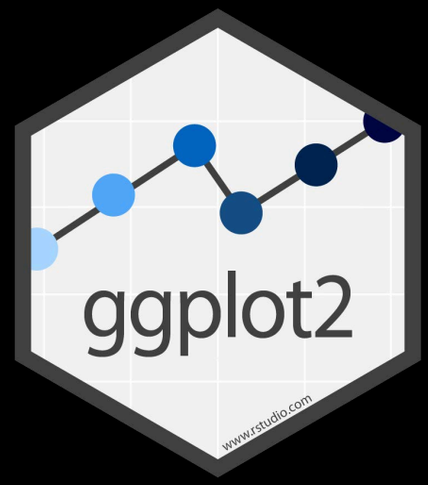
  - breaks: `geom_histogram(breaks = 0:10)`


10 bins
count
1000
500
0
0
2
4
6
8
log_mass


30 bins (default)
count
400
300
200
100
0
0
2
4
6
8
log_mass


200 bins
count
75
50
25
0
0
2
4
6
8
log_mass

# HISTOGRAMS

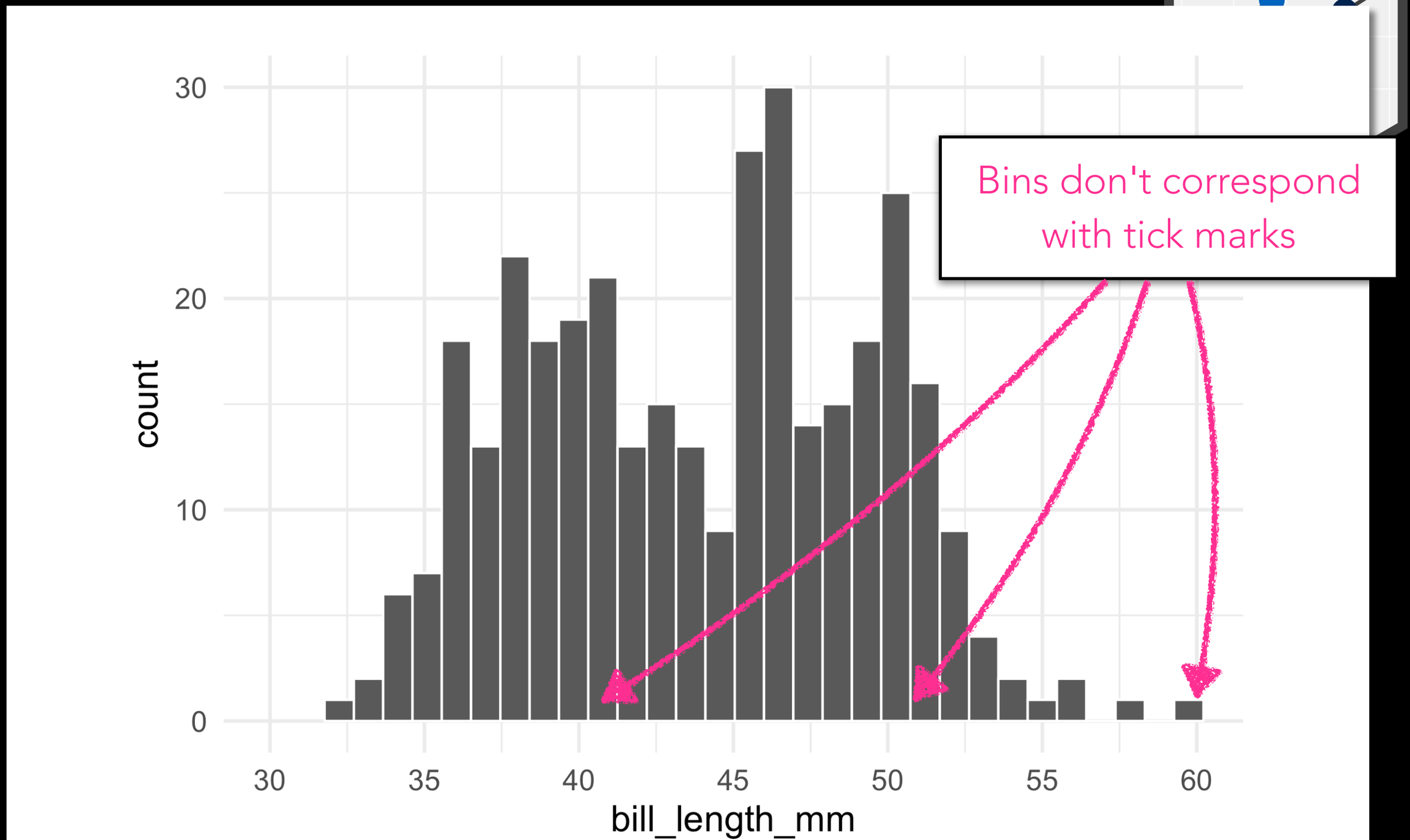



- *Also pay attention to the bin positions!*
  The defaults don't always make sense.

- Binning positions can be modified with ONE (not both) of the following:

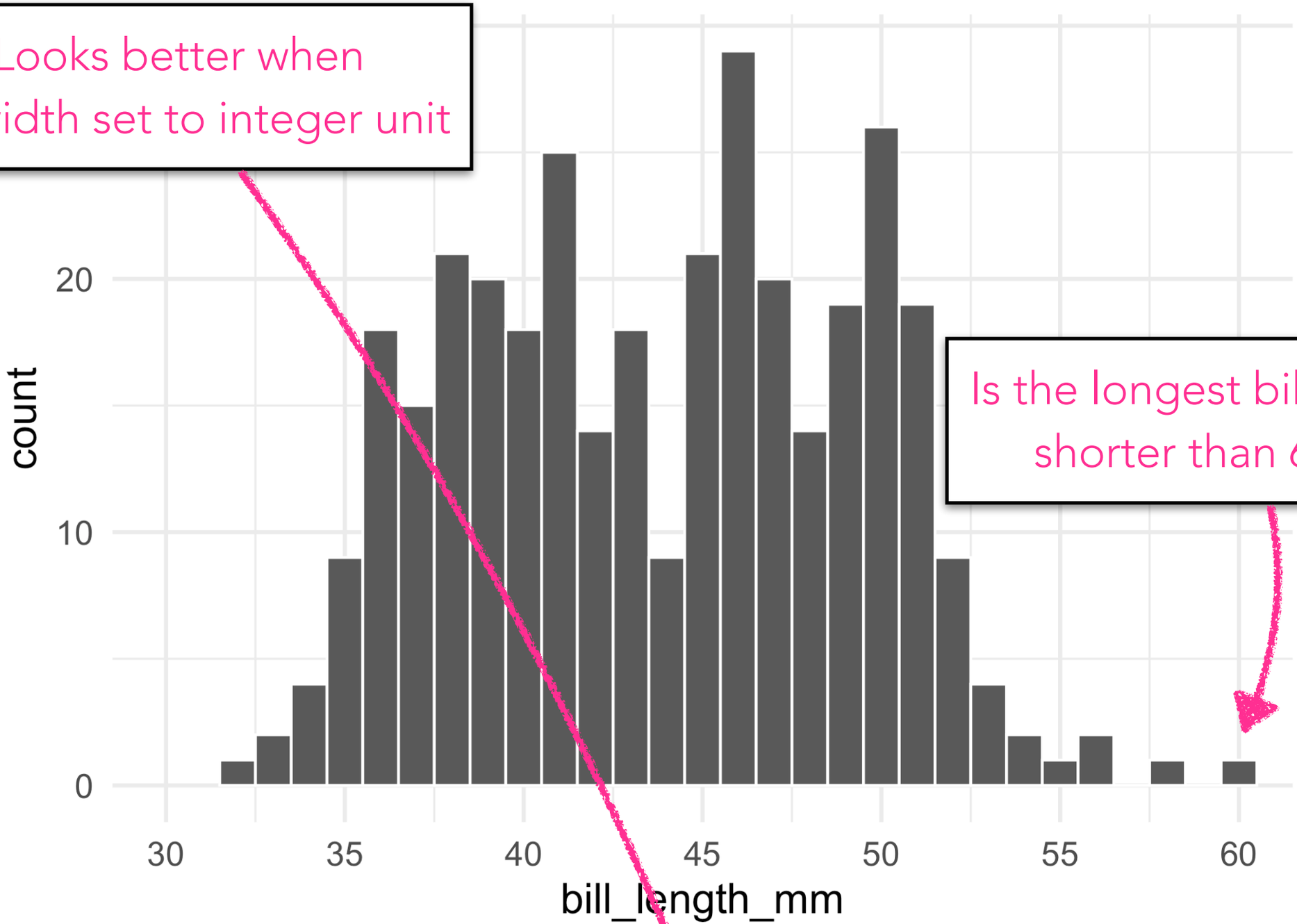
  - center: an x value for the center of one of the bins (the rest will be calculated)

  - boundary: an x value for the boundary between any two bins (the rest will be calculated)




```
ggplot(penguins, aes(x = bill_length_mm)) +
  geom_histogram(color = "white", bins = 30)
```

Looks better when binwidth set to integer unit

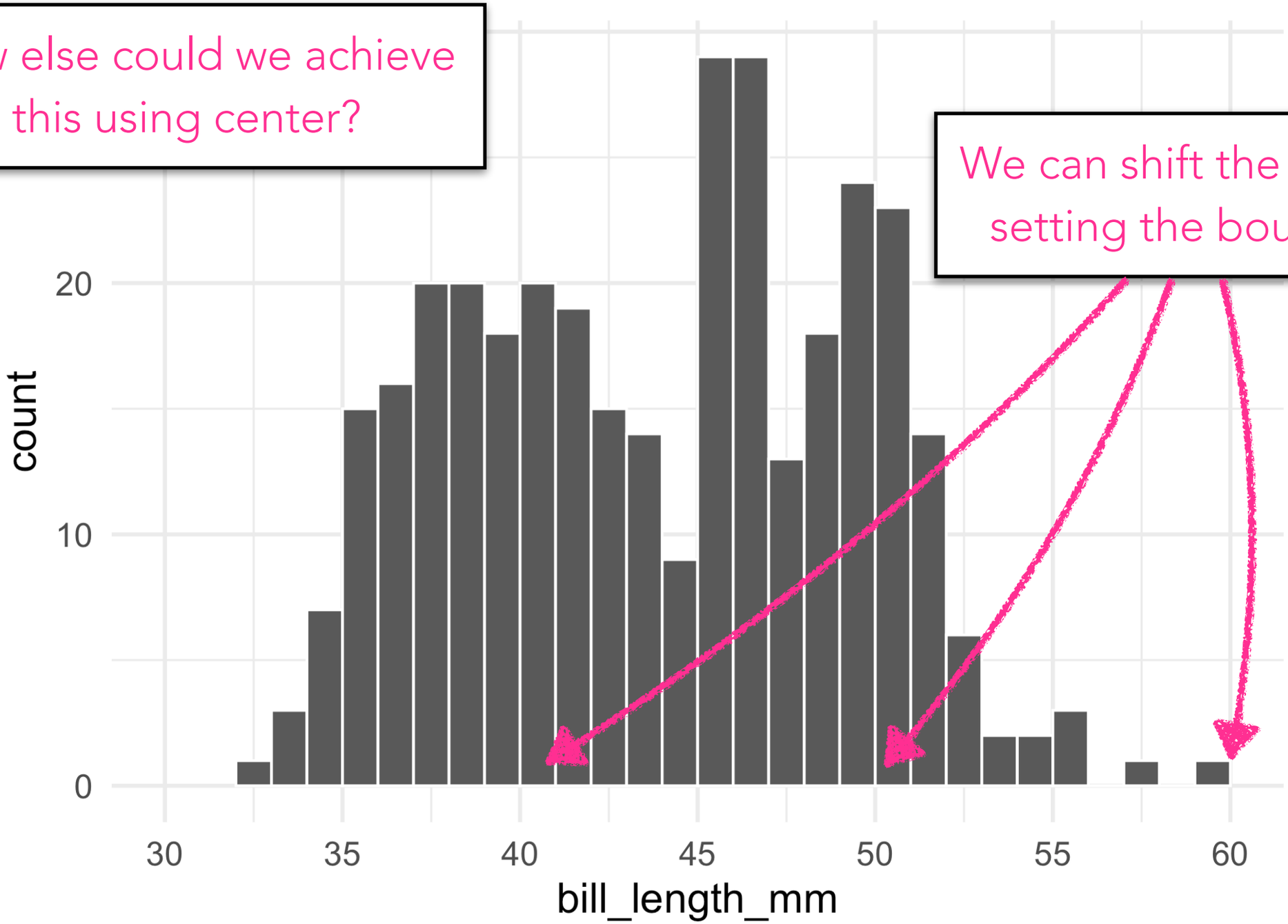



Is the longest bill longer or shorter than 60 mm?

```
ggplot(penguins, aes(x = bill_length_mm)) +
  geom_histogram(color = "white", binwidth = 1)
```

How else could we achieve this using center?

We can shift the bars by setting the boundary



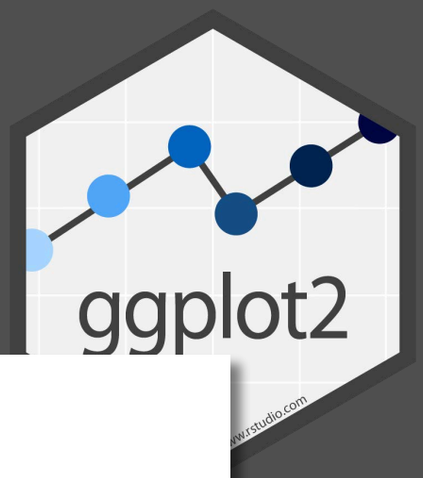

```
ggplot(penguins, aes(x = bill_length_mm)) +
  geom_histogram(color = "white", binwidth = 1, boundary = 40)
```

# HISTOGRAMS

- Histograms are generally bad at showing multiple distributions together

- For example…

# YOUR TURN




## Try to predict what this will do…

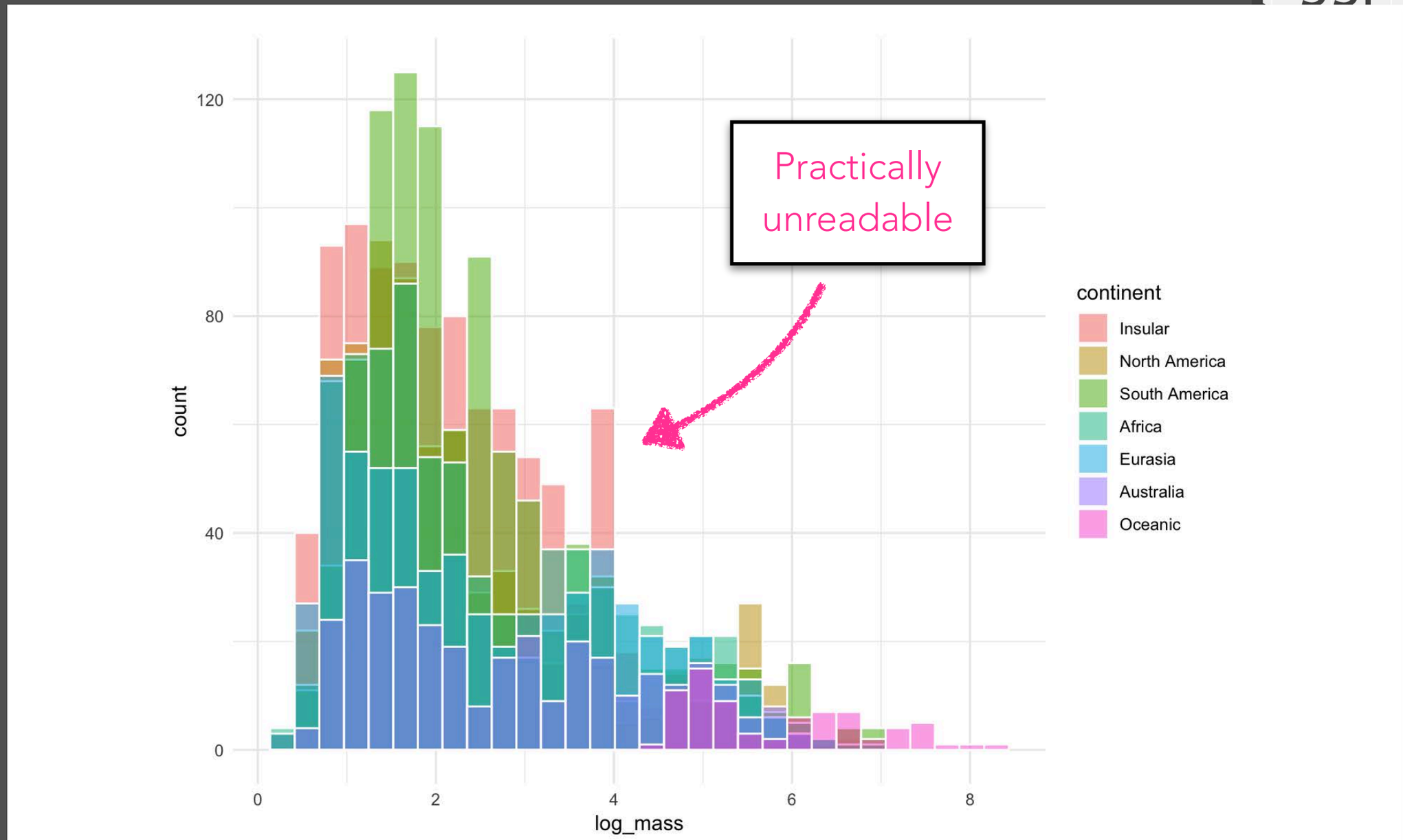
Different fill color for each continent

Makes histograms overlap

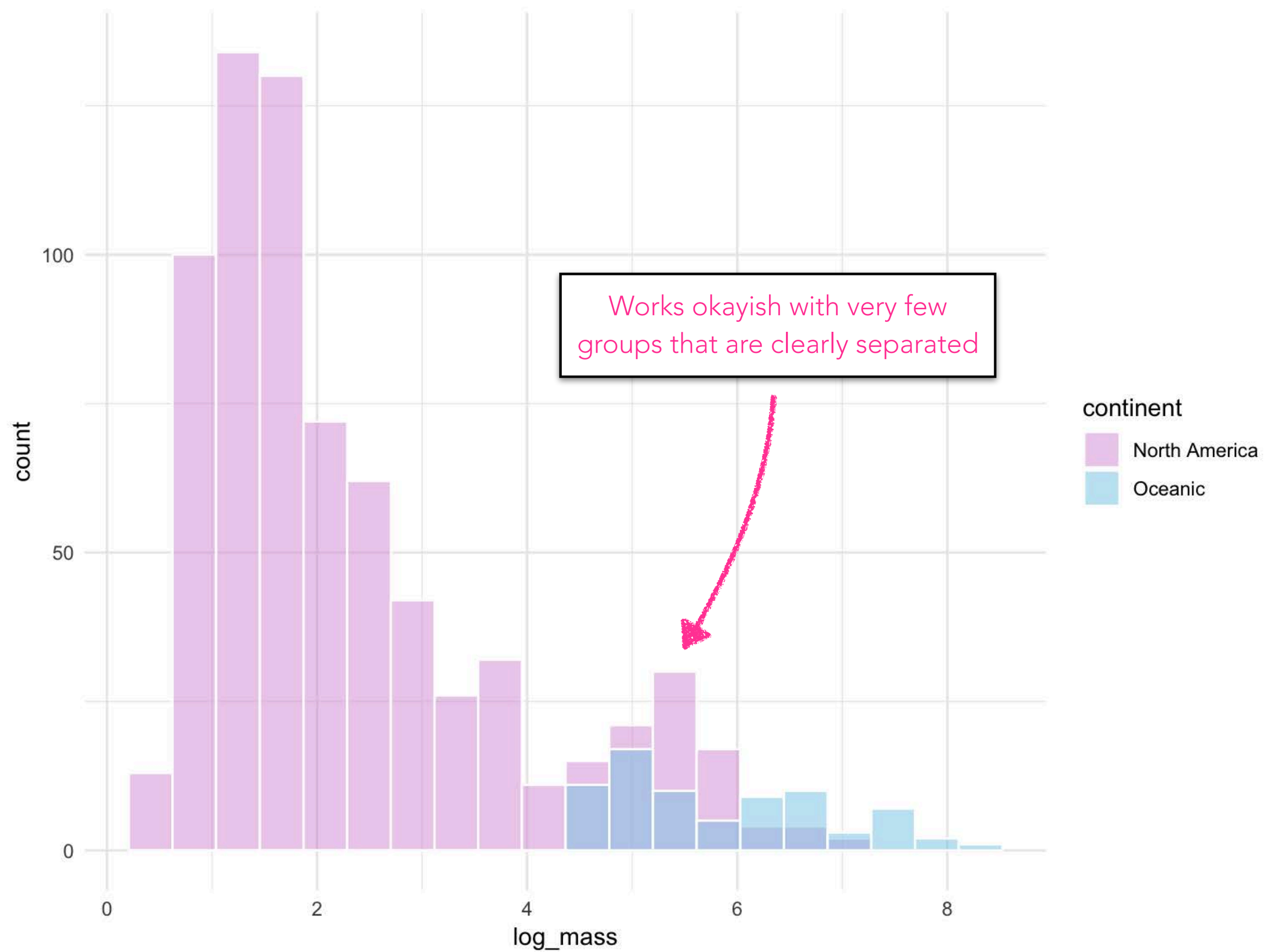
Makes them partially transparent

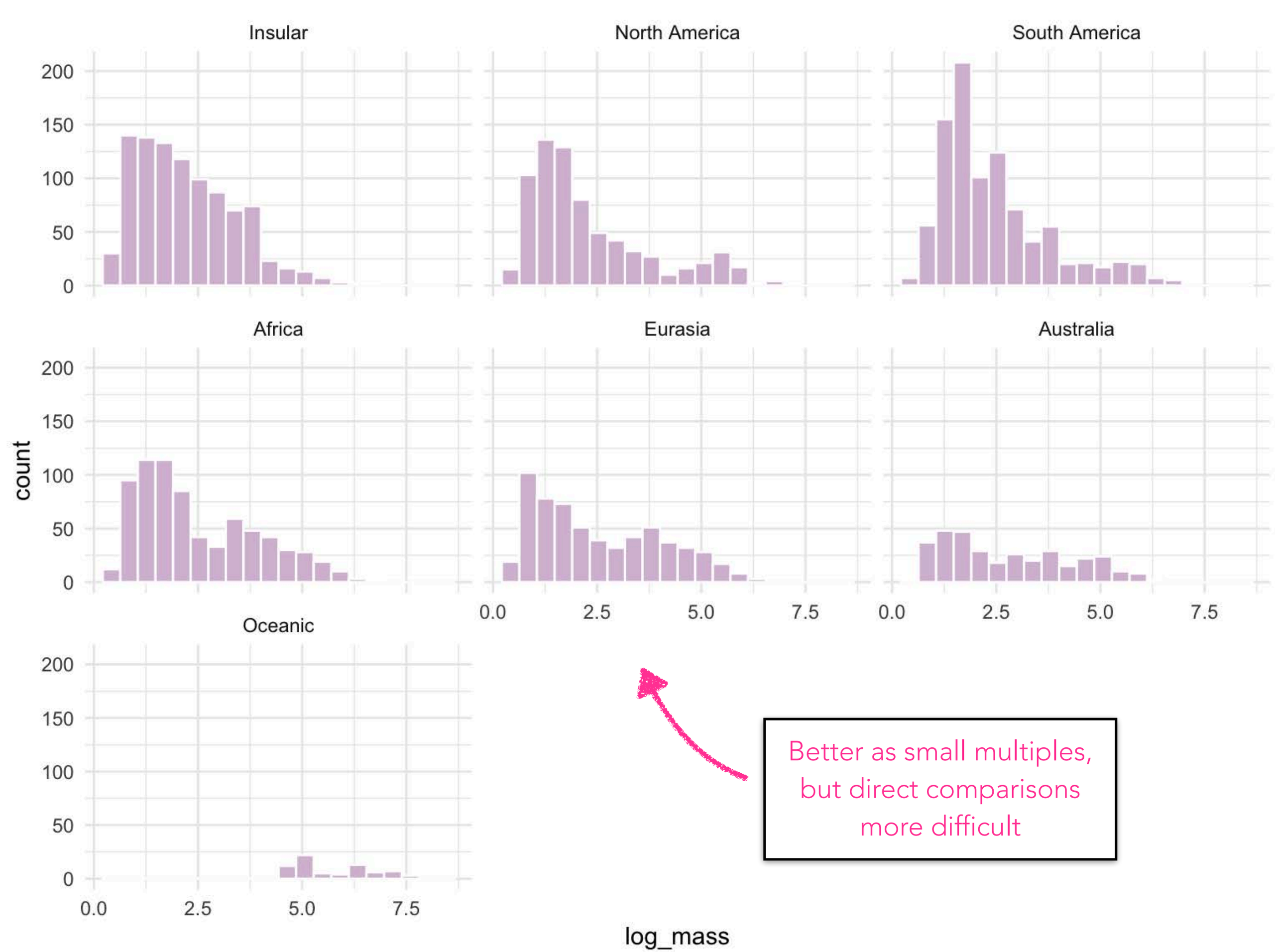
```
ggplot(lqm, aes(x = log_mass, fill = continent)) +
  geom_histogram(color = "white", position = "identity", alpha = 0.5)
```

# YOUR TURN

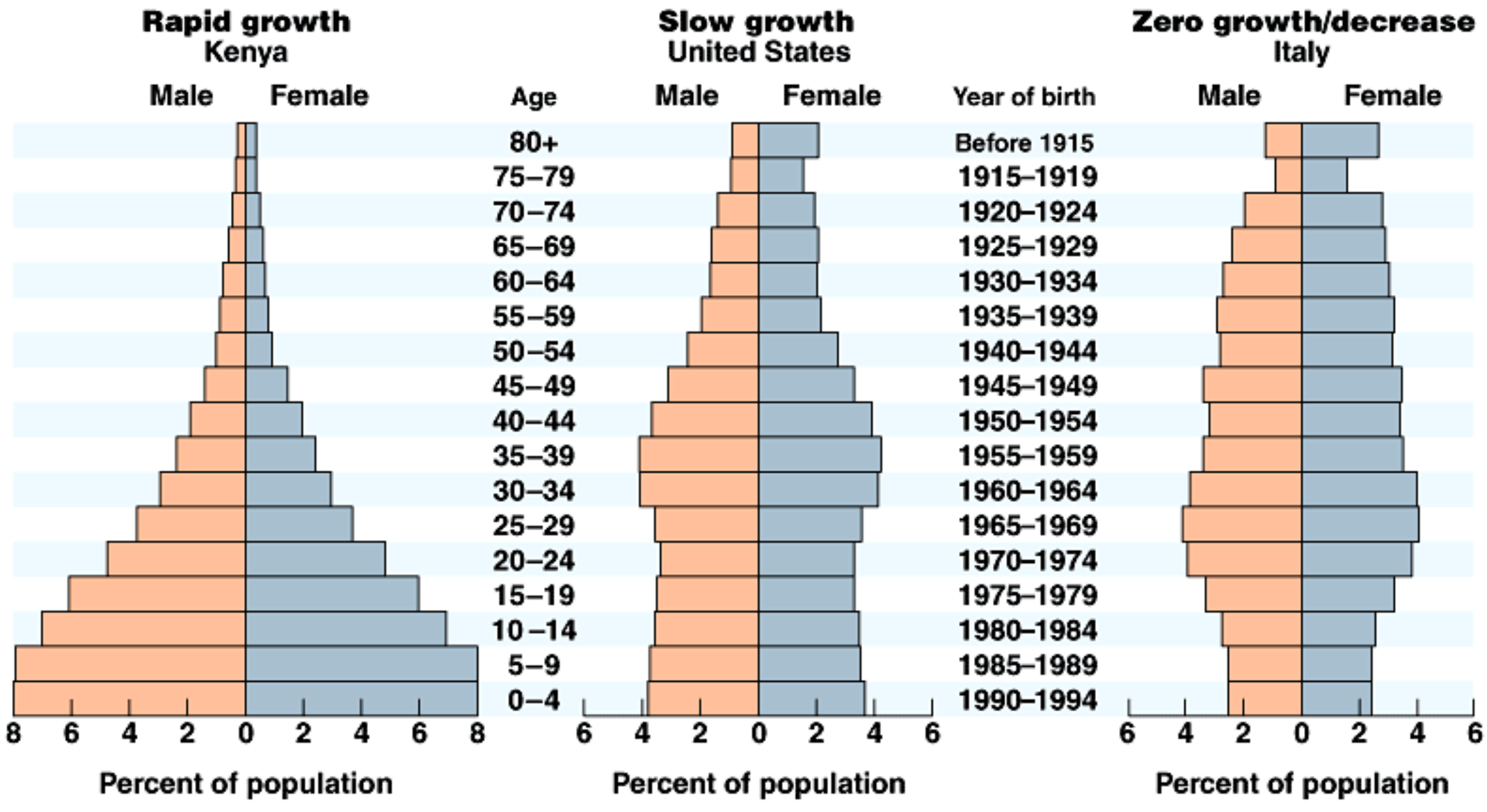



```
ggplot(lqm, aes(x = log_mass, fill = continent)) +
  geom_histogram(color = "white", position = "identity", alpha = 0.5)
```

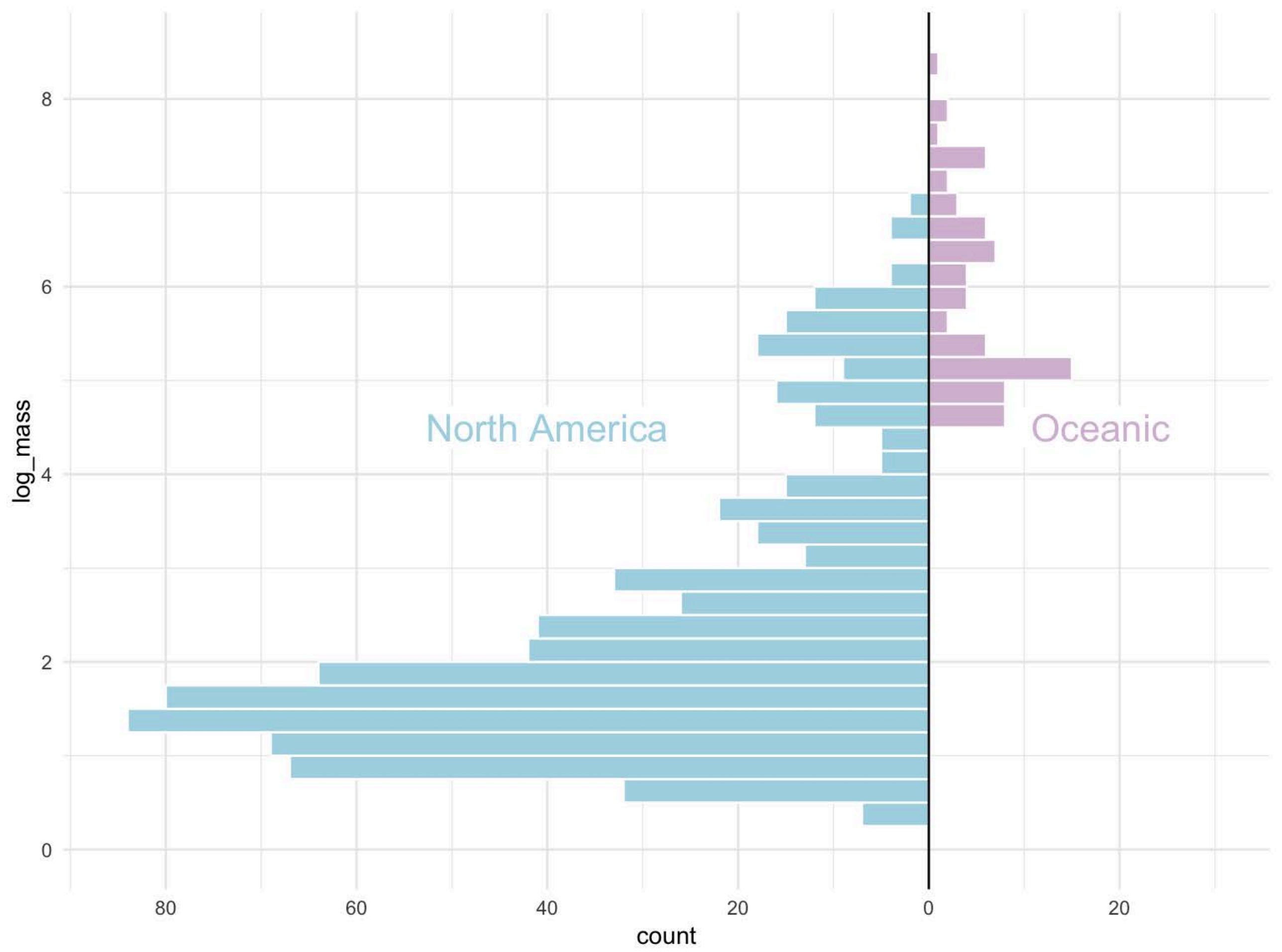

Works okayish with very few groups that are clearly separated
count
100
50
0
0
2
4
6
8
log_mass
continent
North America
Oceanic


Insular
North America
South America
Africa
Eurasia
Australia
Oceanic
count
log_mass
0
50
100
150
200
0.0
2.5
5.0
7.5
Better as small multiples, but direct comparisons more difficult

# "Age pyramid" style plot is ideal for comparing two (and only two) distributions




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log_mass
8
6
4
2
0
North America
Oceanic
80
60
40
20
0
20
count

Works well even if distributions overlap extensively

North America

Africa

log_mass

6

4

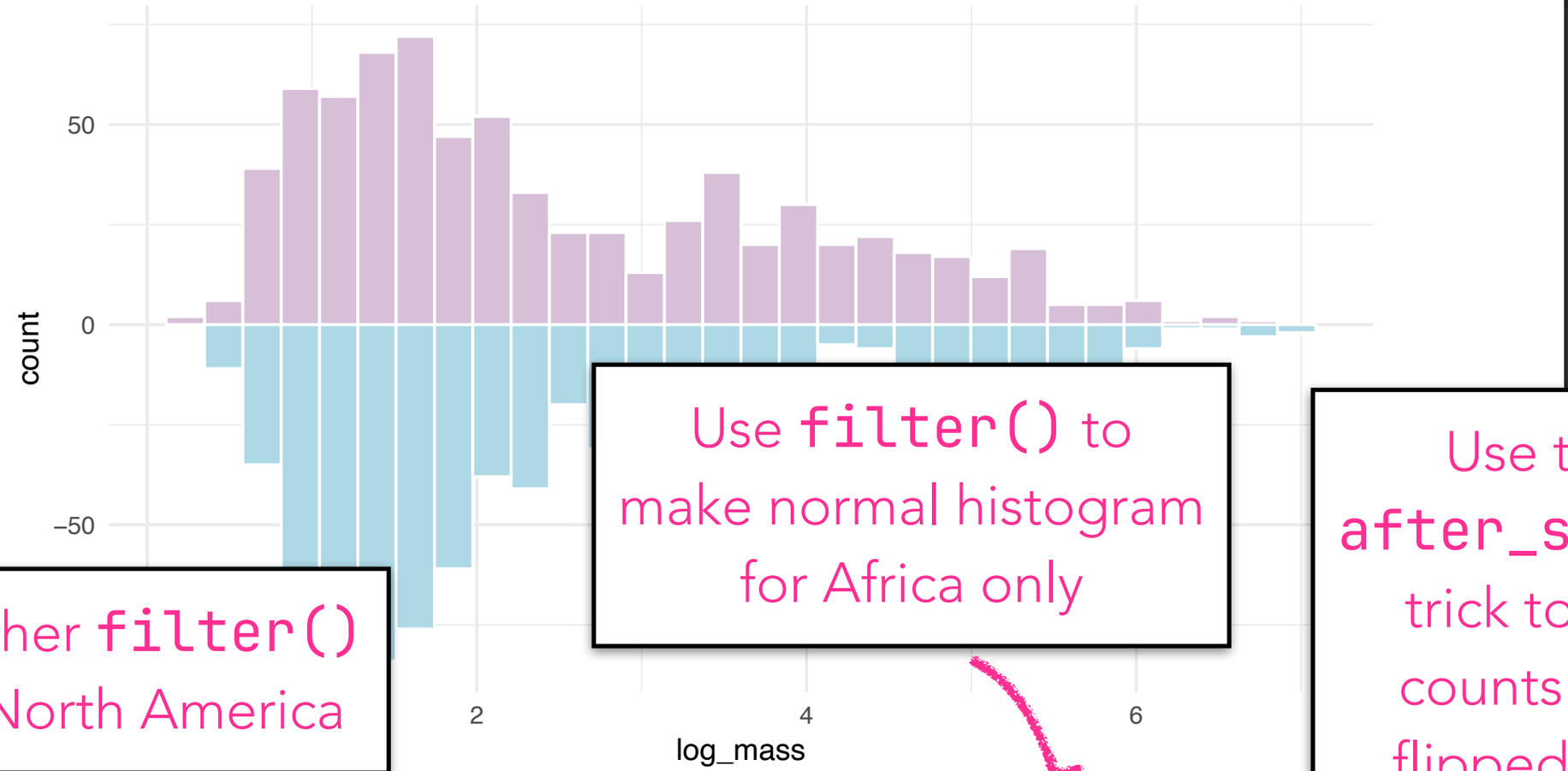
2

0

80 60 40 20 0 20 40 60 80

count

# A STARTER...



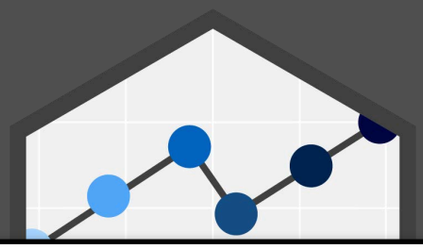

Use `filter()` to make normal histogram for Africa only

Use the `after_stat()` trick to get counts with flipped sign

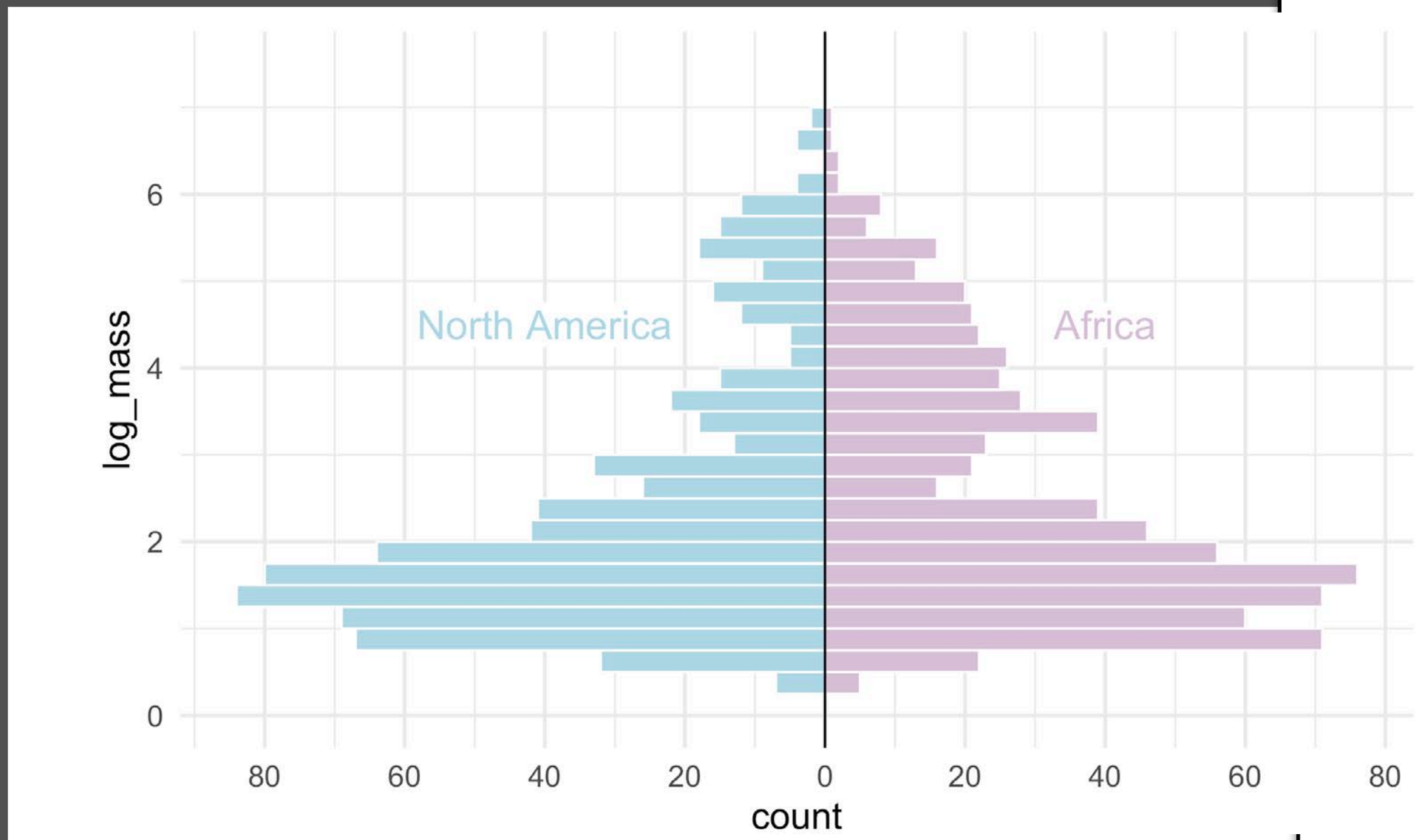
Another `filter()` for North America

```
ggplot() +
  geom_histogram(data = filter(lqm, continent == "Africa"),
                 aes(x = log_mass),
                 fill = "thistle", color = "white") +
  geom_histogram(data = filter(lqm, continent == "North America"),
                 aes(x = log_mass, y = -after_stat(count)),
                 fill = "lightblue", color = "white")
```

# A STARTER...



Add text here using `rotate()`

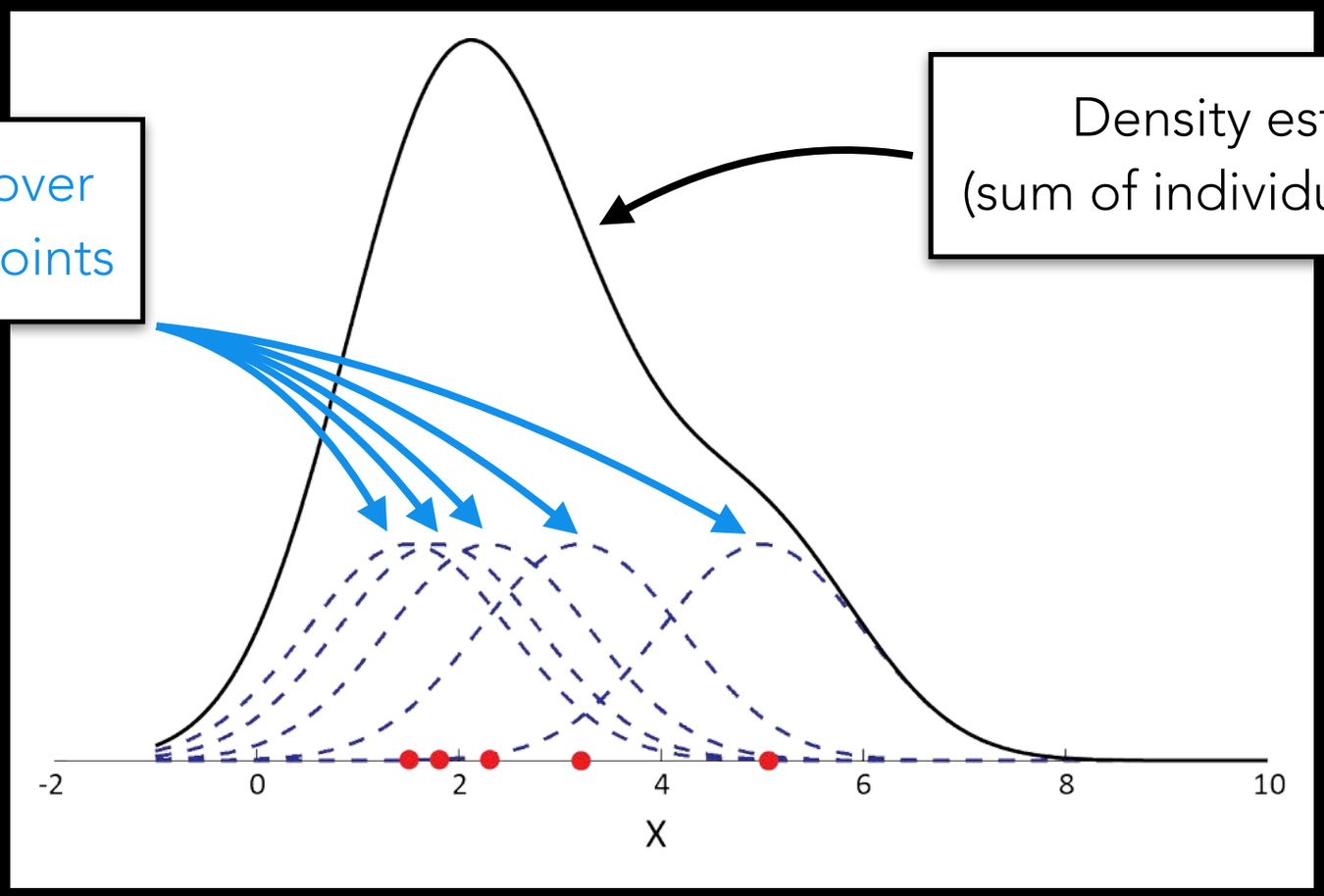



to make it vertical

```
...   +
annotate(geom = "text", y = 40, x = 4.5, label = "Africa",
         color = "thistle", size = 6) +
annotate(geom = "text", y = -40, x = 4.5, label = "North America",
          color = "lightblue", size = 6) +
coord_flip()
```

# DENSITY PLOTS

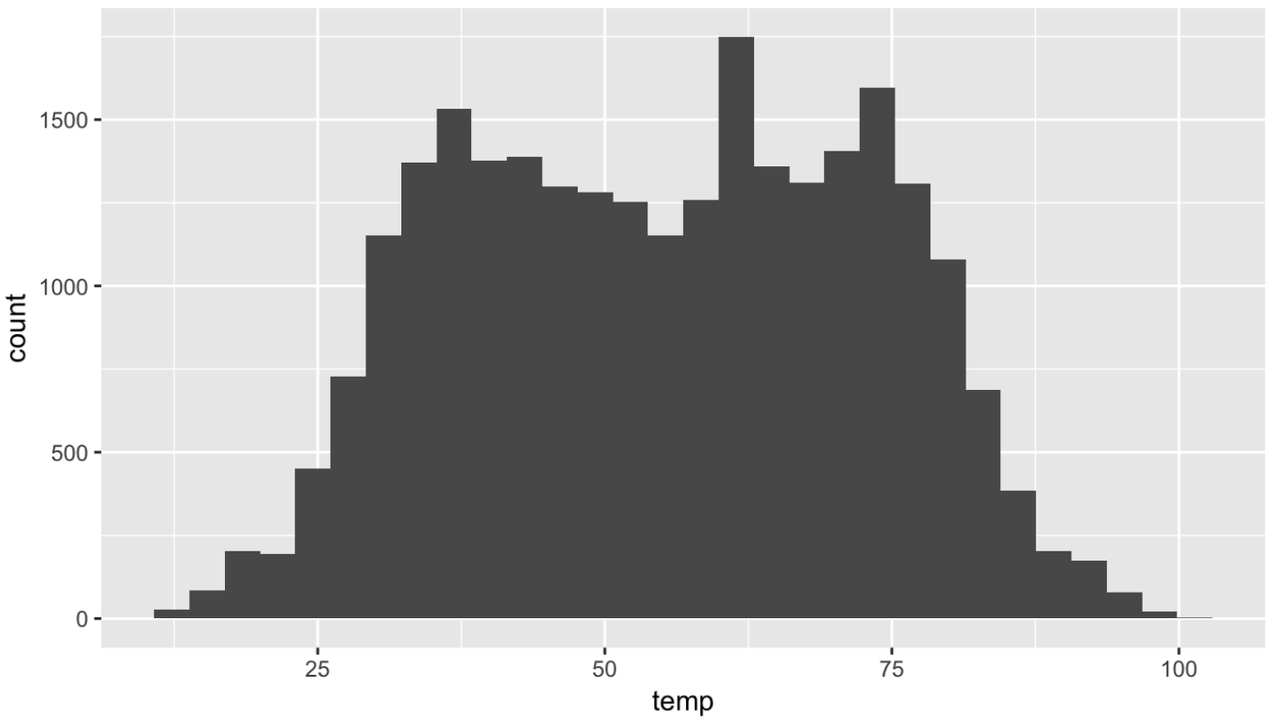
- Like histograms, density plots show the distribution of a single variable
- Uses smoothing to estimate the *density* of points over a *continuous* range of data values.



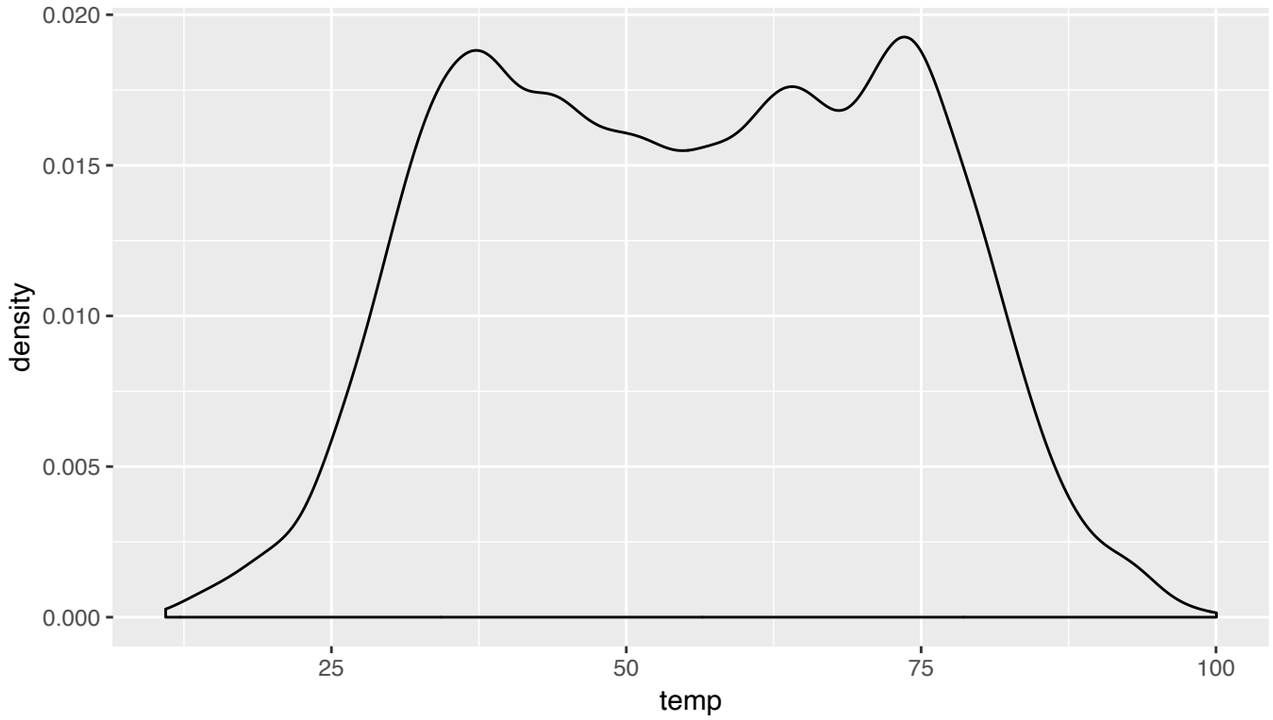
# DENSITY PLOTS

- Geometric object is `geom_density()`
- Aesthetic mappings are same as for histogram

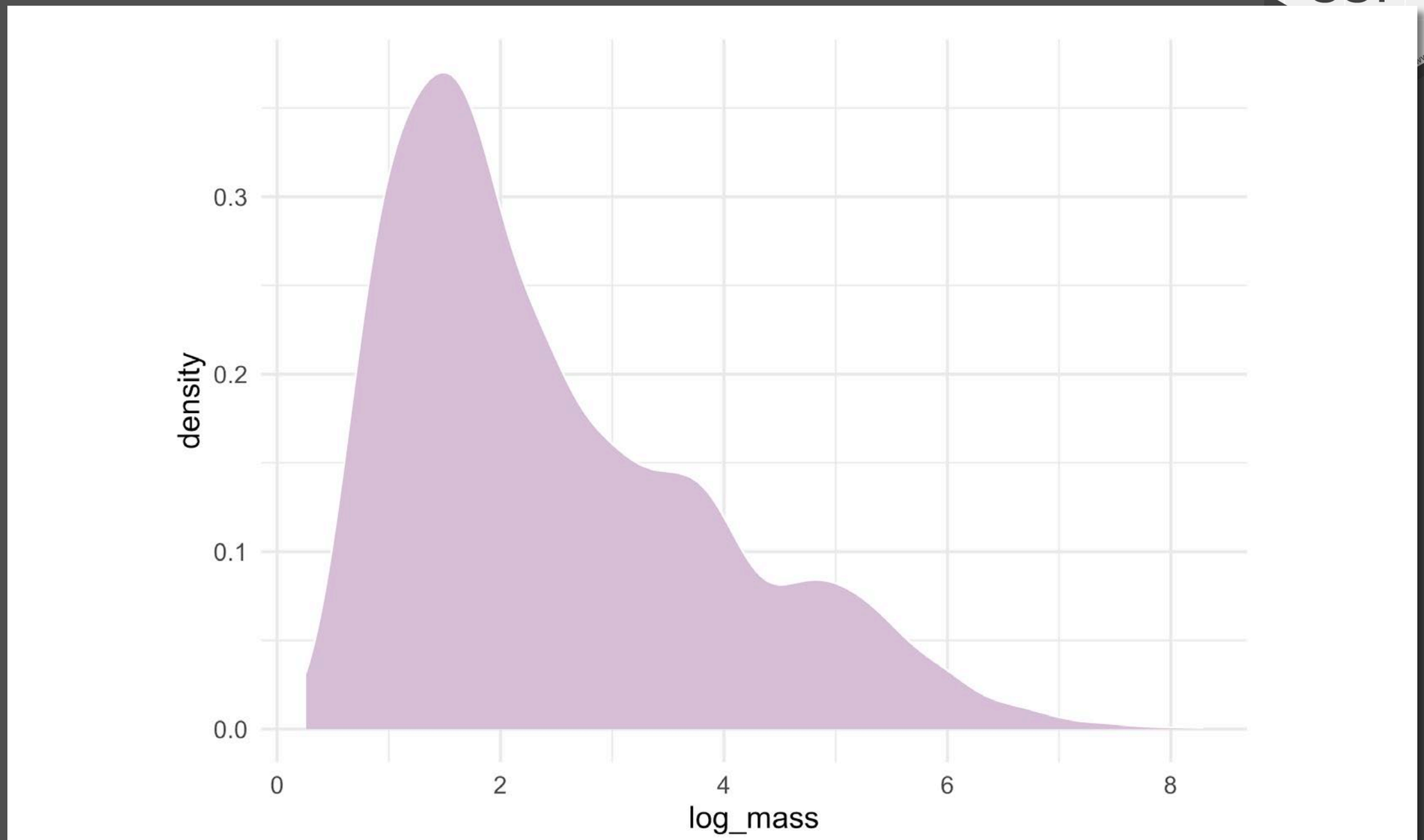
Histogram




Density plot

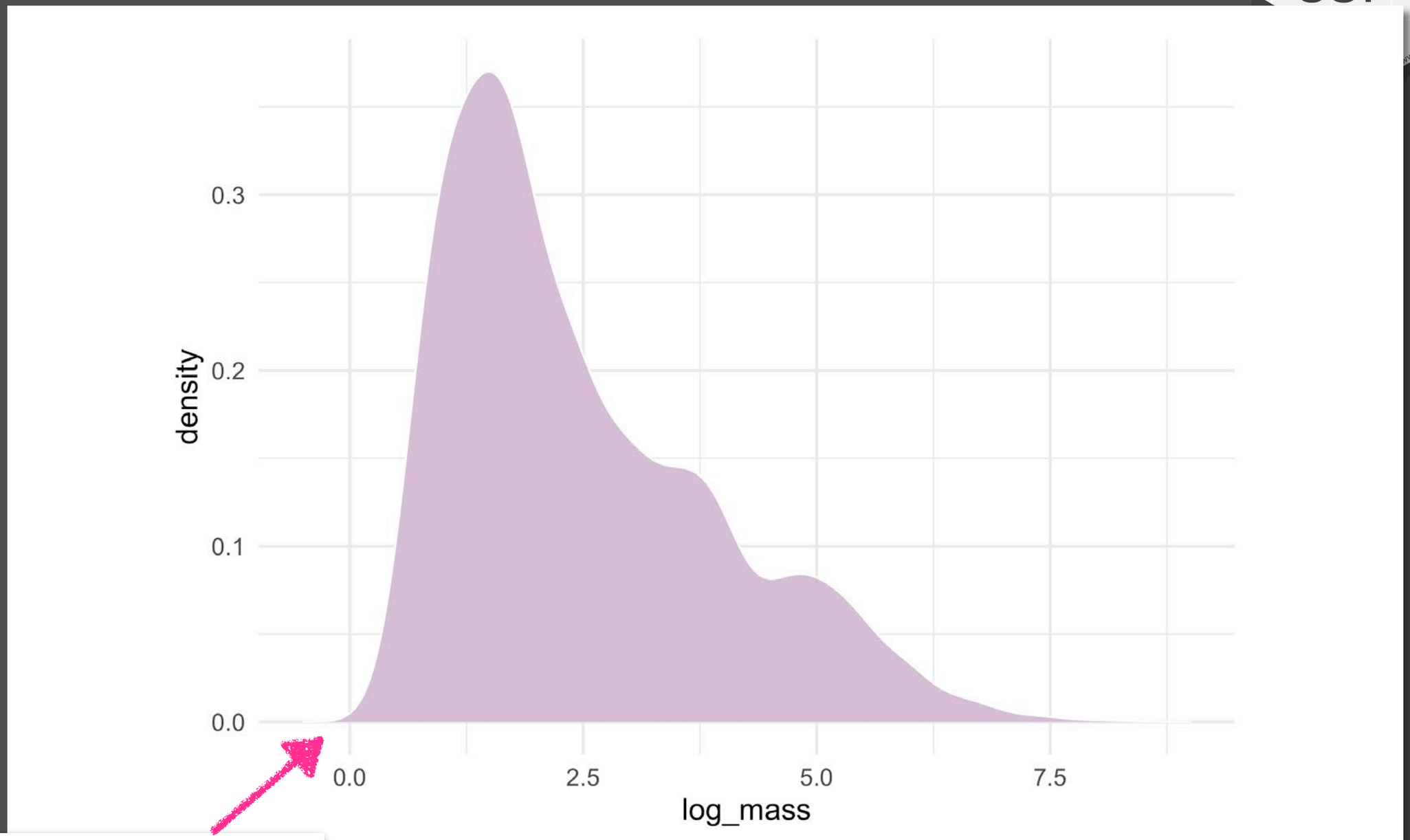


# YOUR TURN



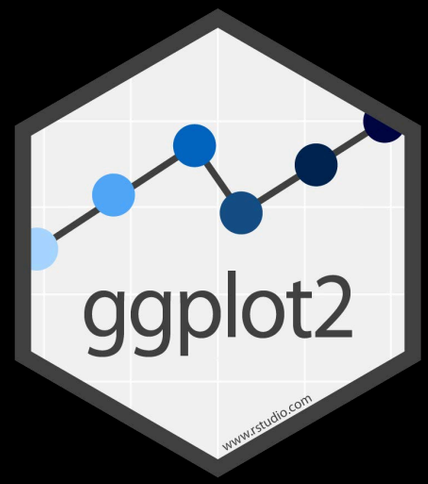

```
ggplot(lqm, aes(x = log_mass)) +
  geom_density(fill = "thistle", color = "white")
```

# YOUR TURN




Be careful with density plots—they can show nonsensical data.

# DENSITY PLOTS

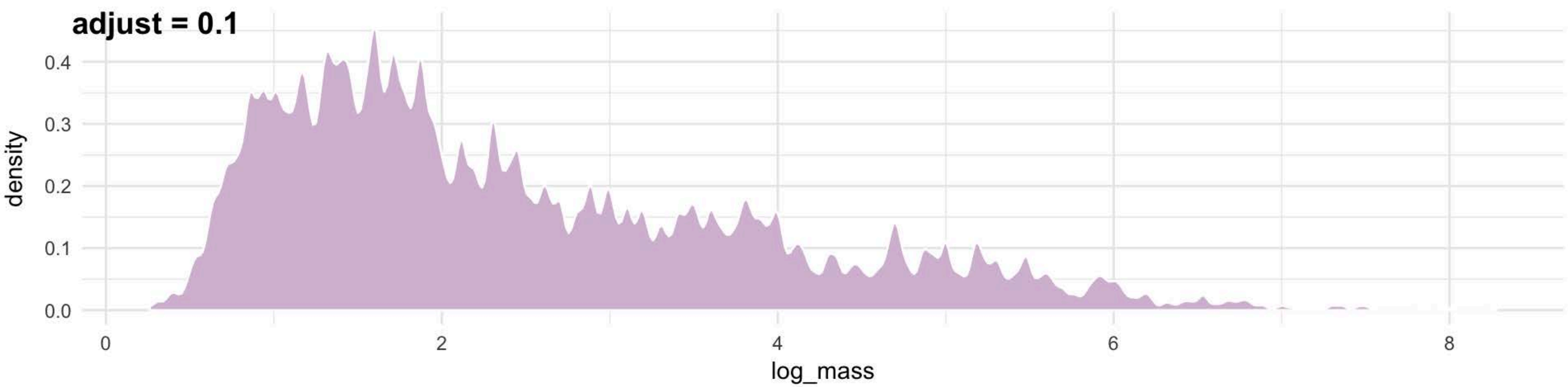


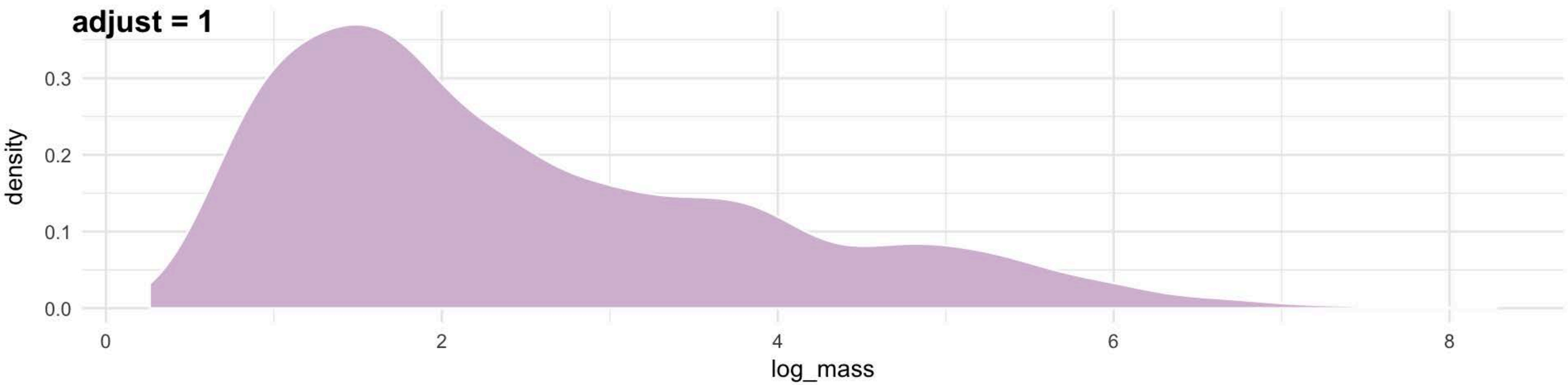

- Like histograms, the visual appearance of density plots depends on the amount of smoothing.

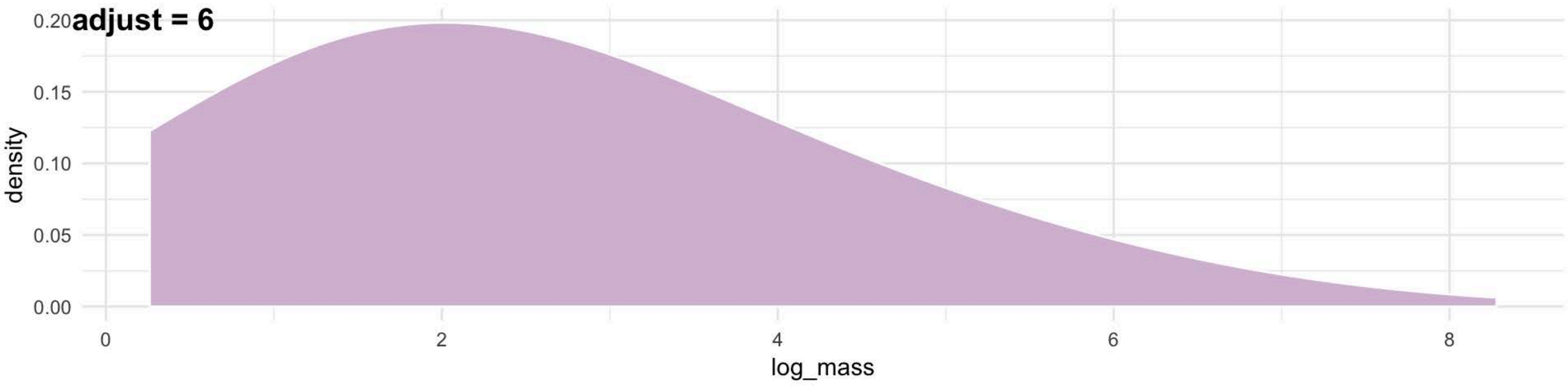
- Smoothing behavior can be adjusted in various complex ways, but the only one you're likely to use is:

  - adjust: `geom_density(adjust = 1)`

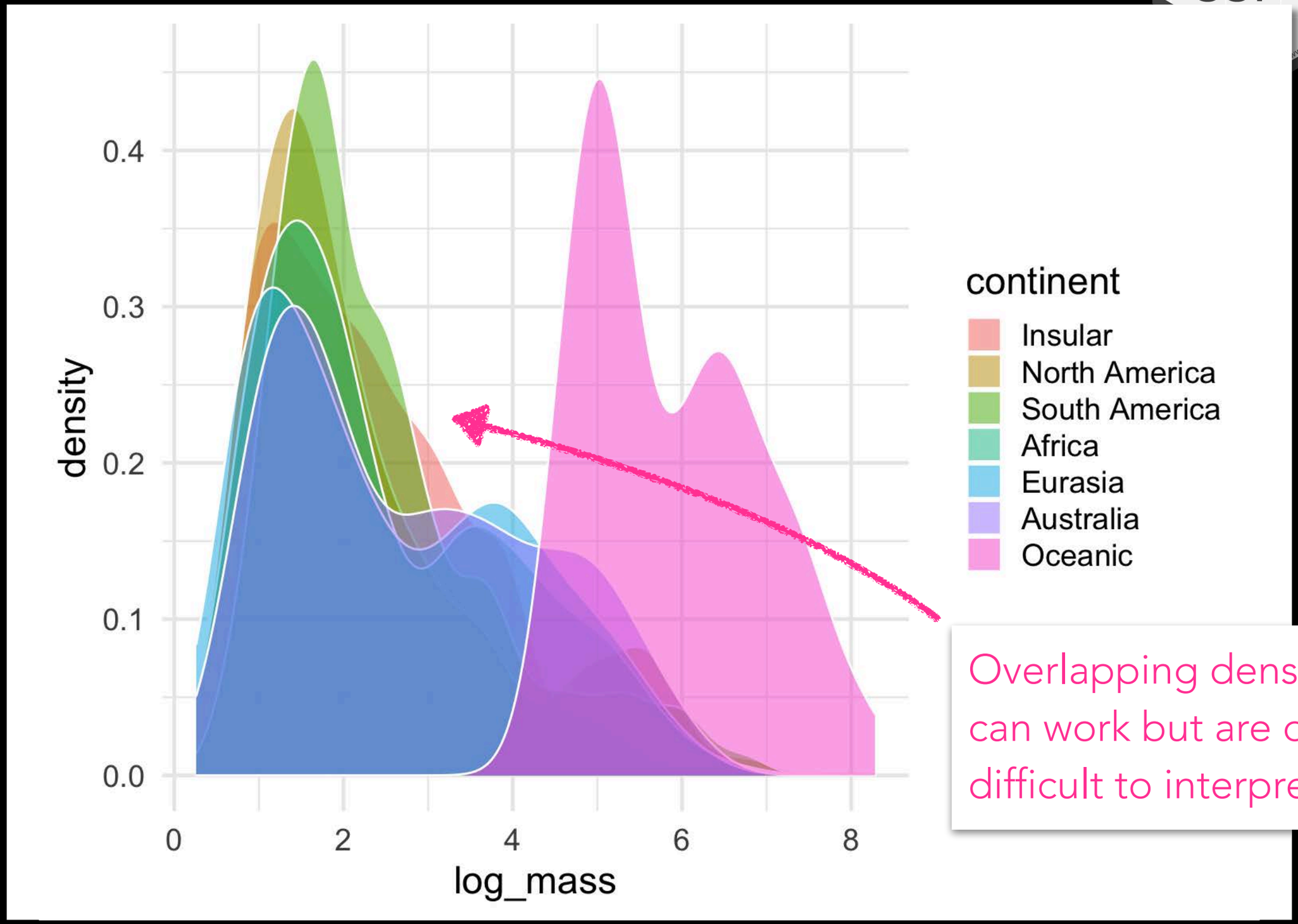
Multiplier for the amount of smoothing (1 = default)


adjust = 0.1
density
0.4
0.3
0.2
0.1
0.0
0
2
4
6
8
log_mass


adjust = 1
density
0.3
0.2
0.1
0.0
0
2
4
6
8
log_mass

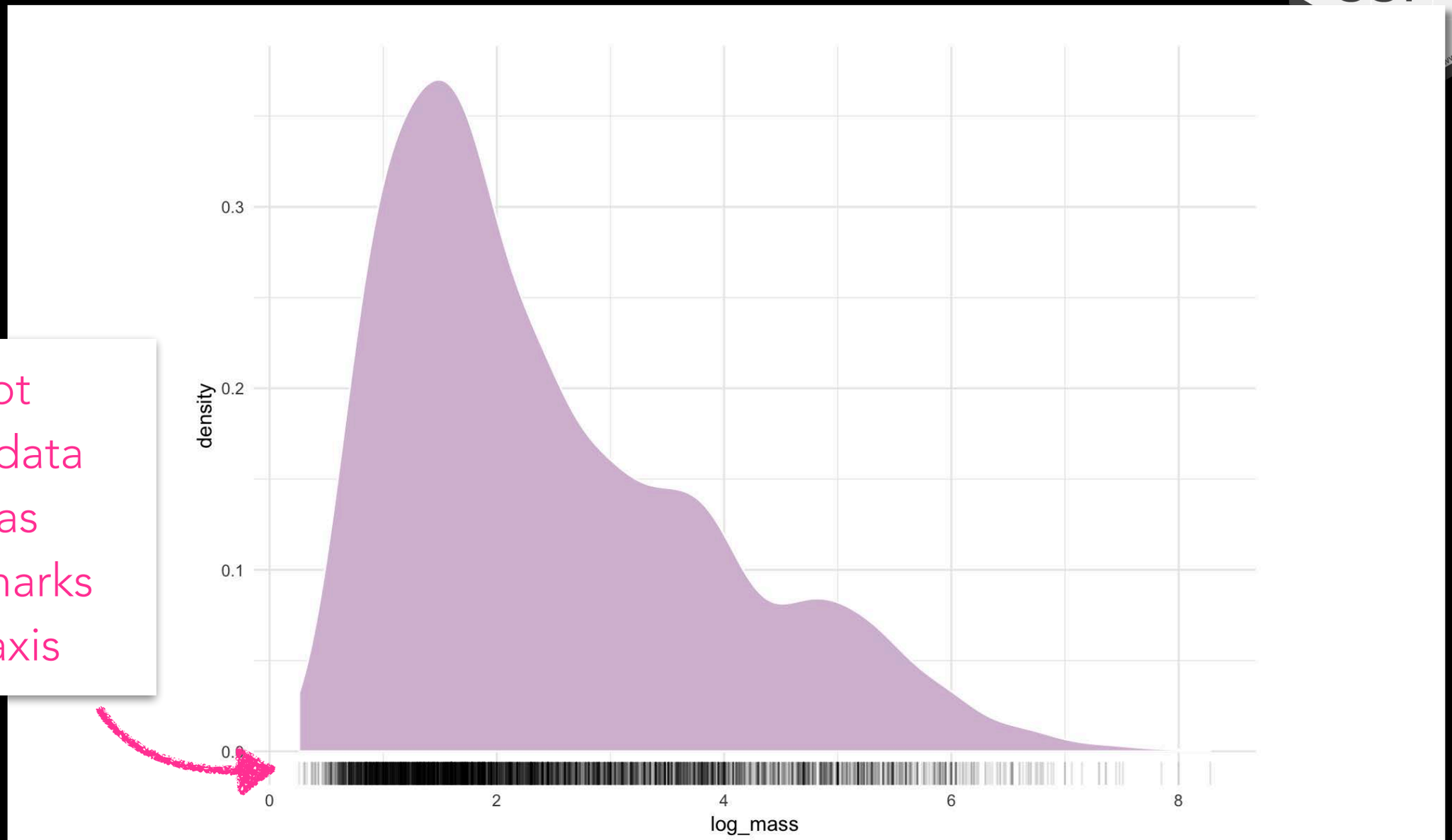

adjust = 6
density
0.20
0.15
0.10
0.05
0.00
0
2
4
6
8
log_mass

# DENSITY PLOTS




Overlapping densities can work but are often difficult to interpret

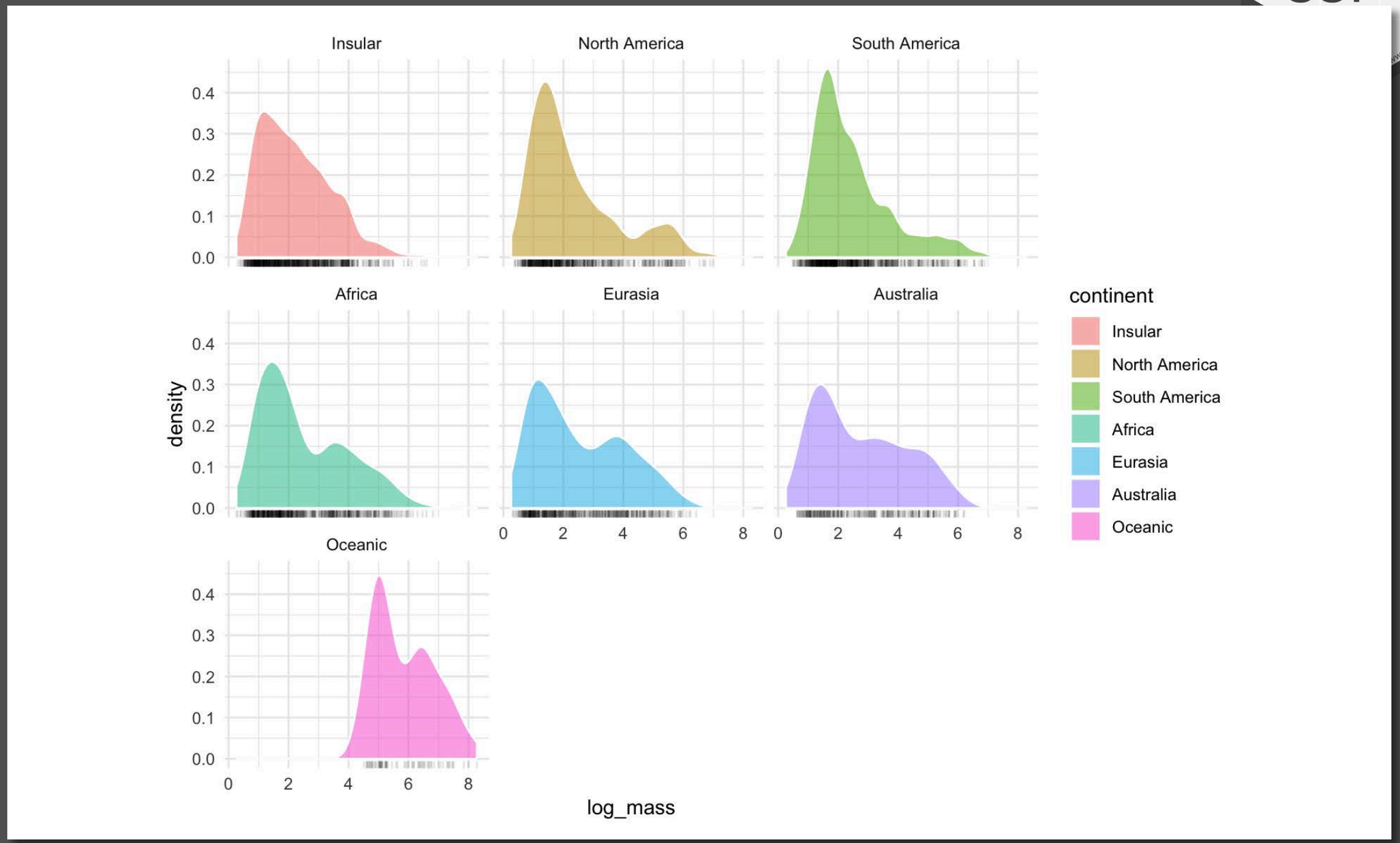
# DENSITY PLOTS




Rug plot shows data points as small marks along axis

```
ggplot(lqm, aes(x = log_mass)) +
  geom_density(fill = "thistle", color = "white") +
  geom_rug(alpha = 0.1)
```

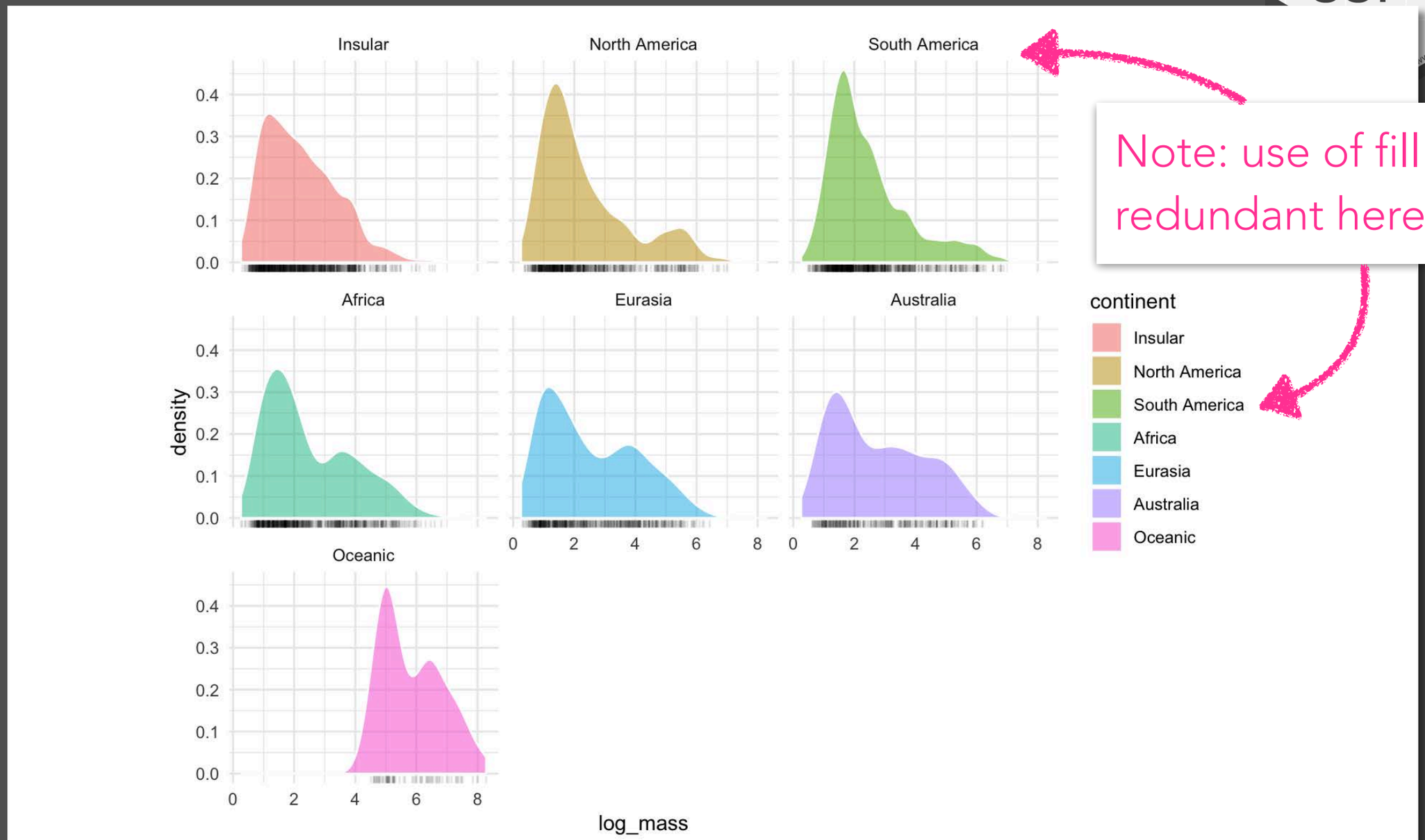
# YOUR TURN




Try to reverse-engineer this plot.

5:00

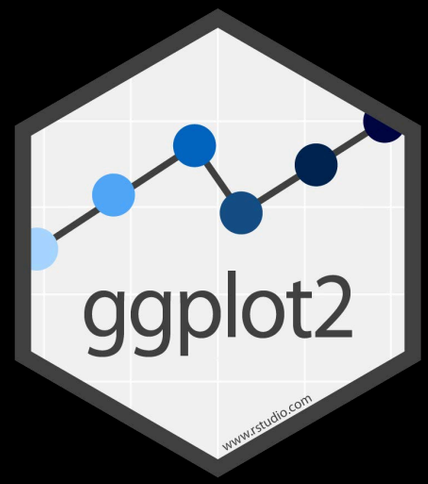
# YOUR TURN




Note: use of fill colors redundant here

```
ggplot(lqm, aes(x = log_mass, fill = continent)) +
  geom_density(color = NA, alpha = 0.5) +
  geom_rug(alpha = 0.1) +
  facet_wrap(vars(continent))
```

# HISTOGRAMS VS. DENSITY PLOTS

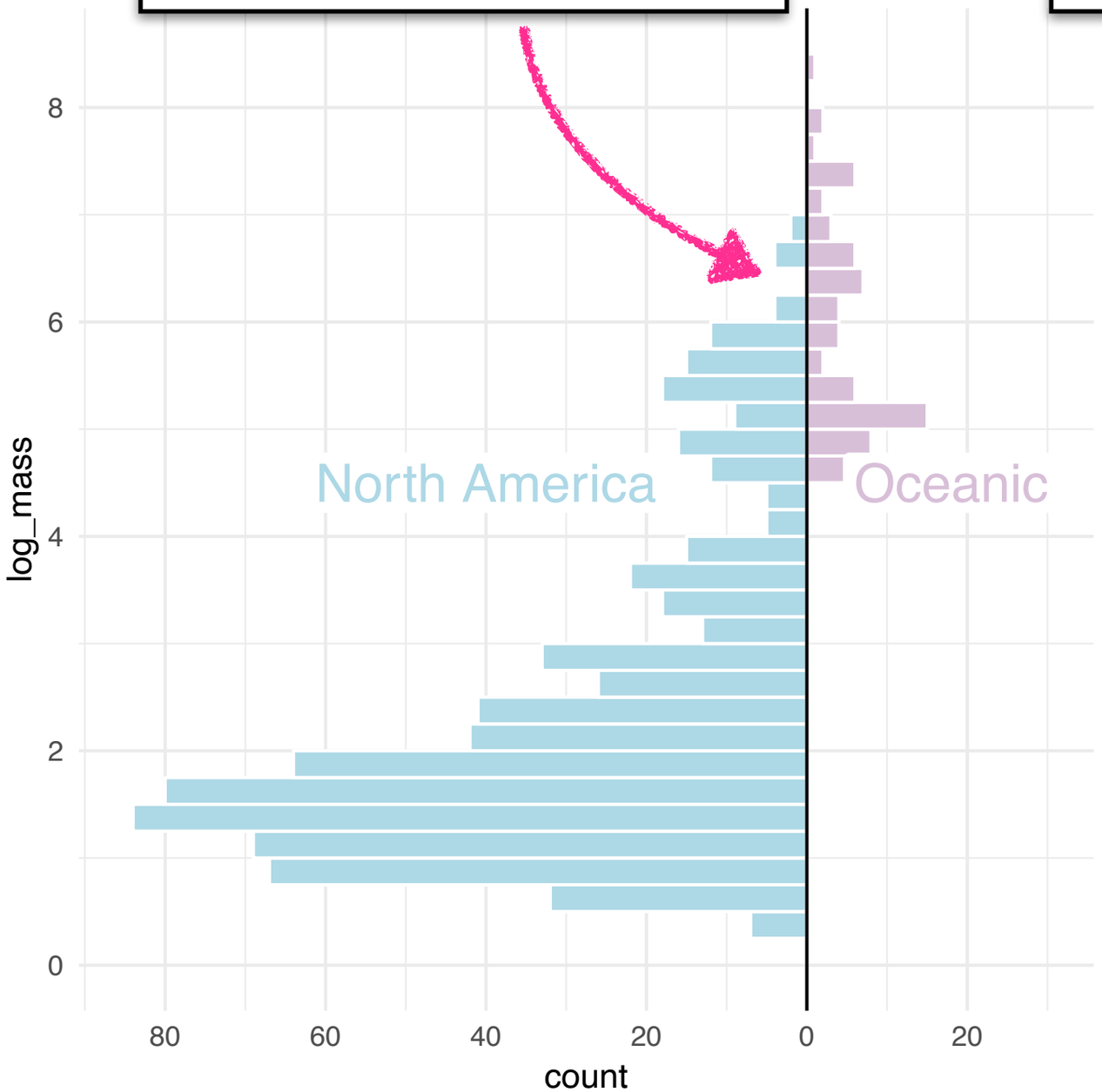



- Histograms are much more common, and are therefore more familiar to a wide audience

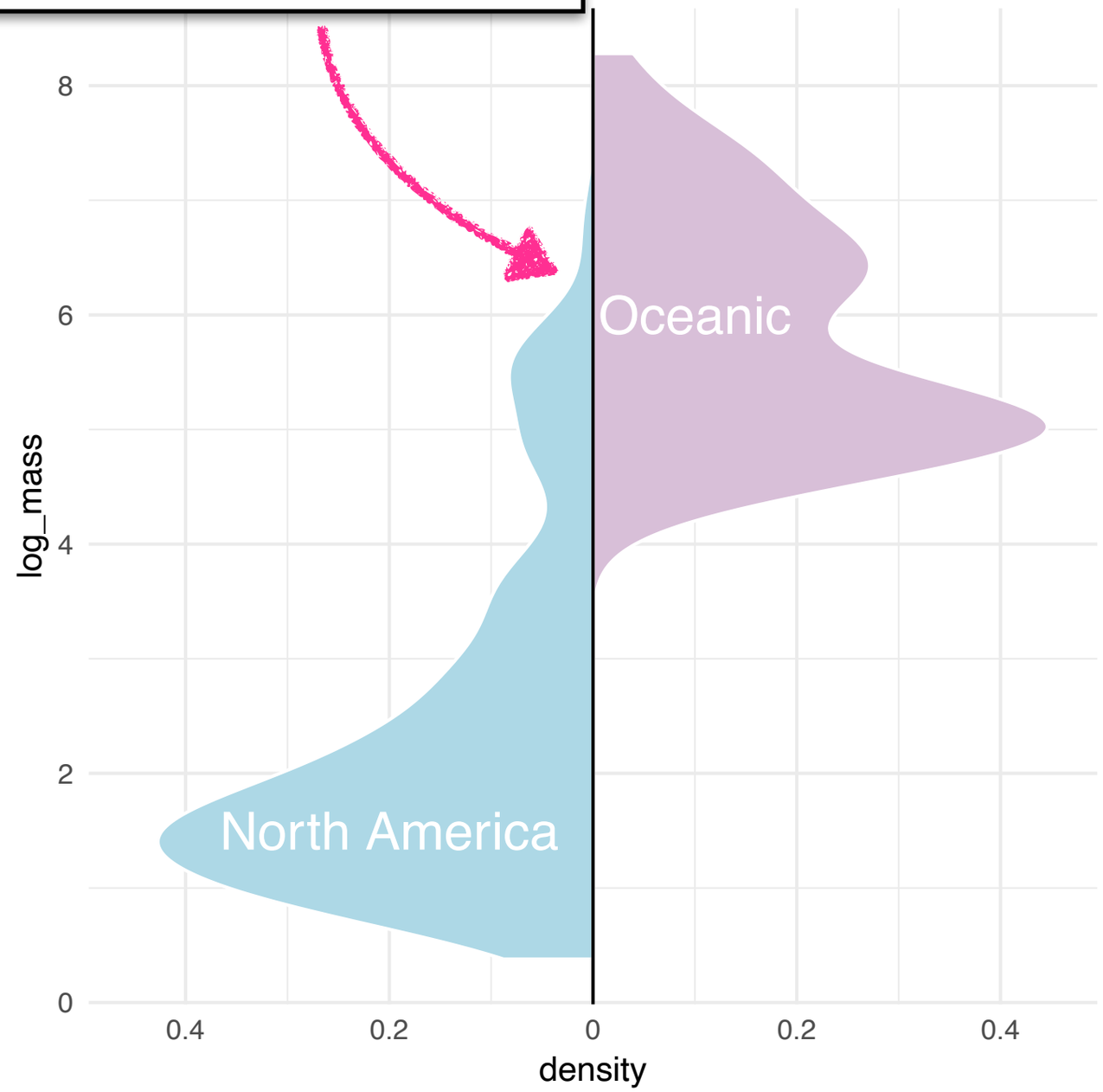
- Histograms use a more intuitive stat (count vs. density)

- Whereas the appearance of a histograms depends on sample size, densities are *scaled*

Smaller sample size for "Oceanic" means bars are small

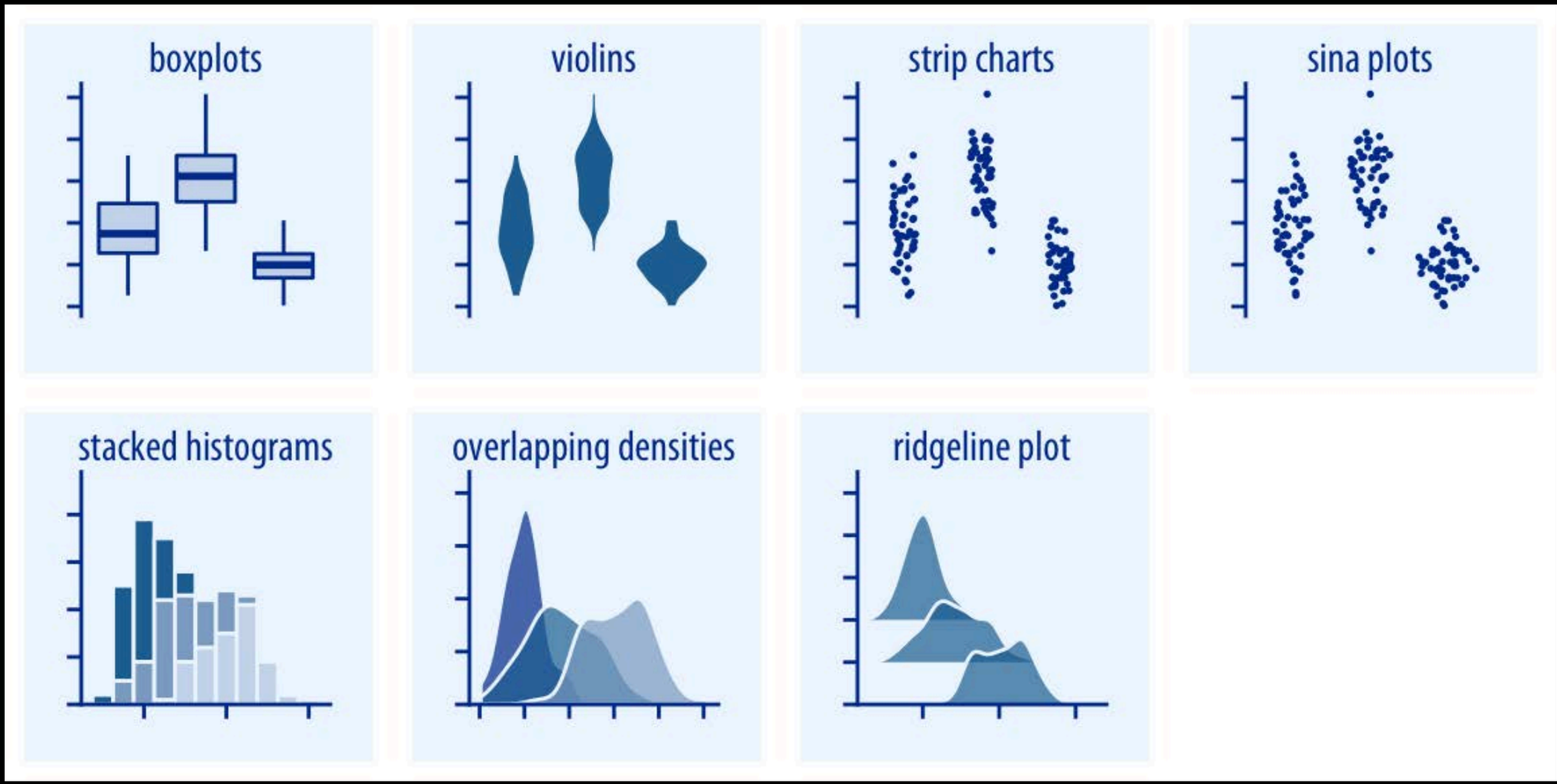



Densities have equal areas (area under the curve sums to 1)

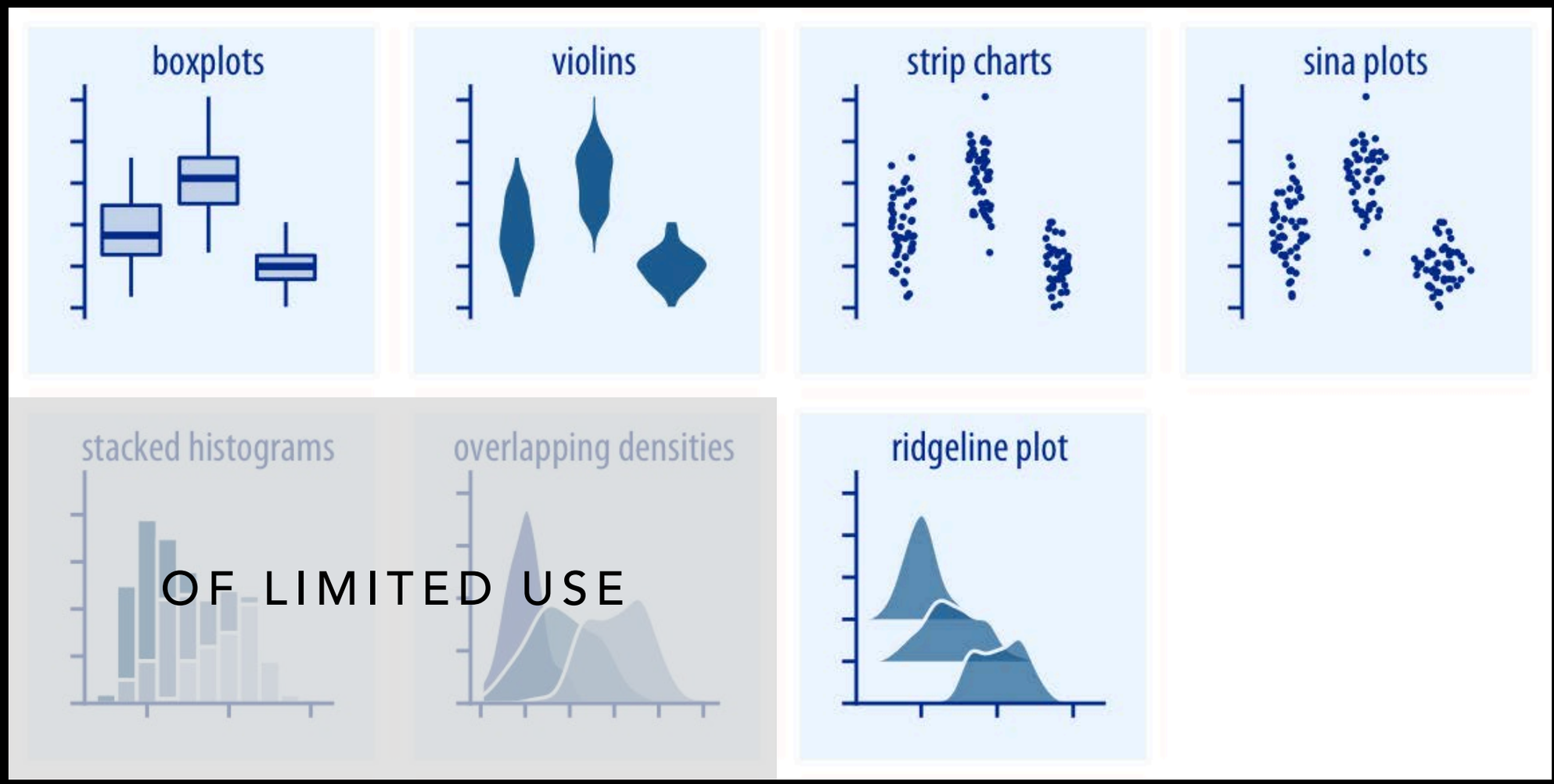


# WHAT IF YOU NEED TO SHOW MULTIPLE DISTRIBUTIONS TOGETHER?

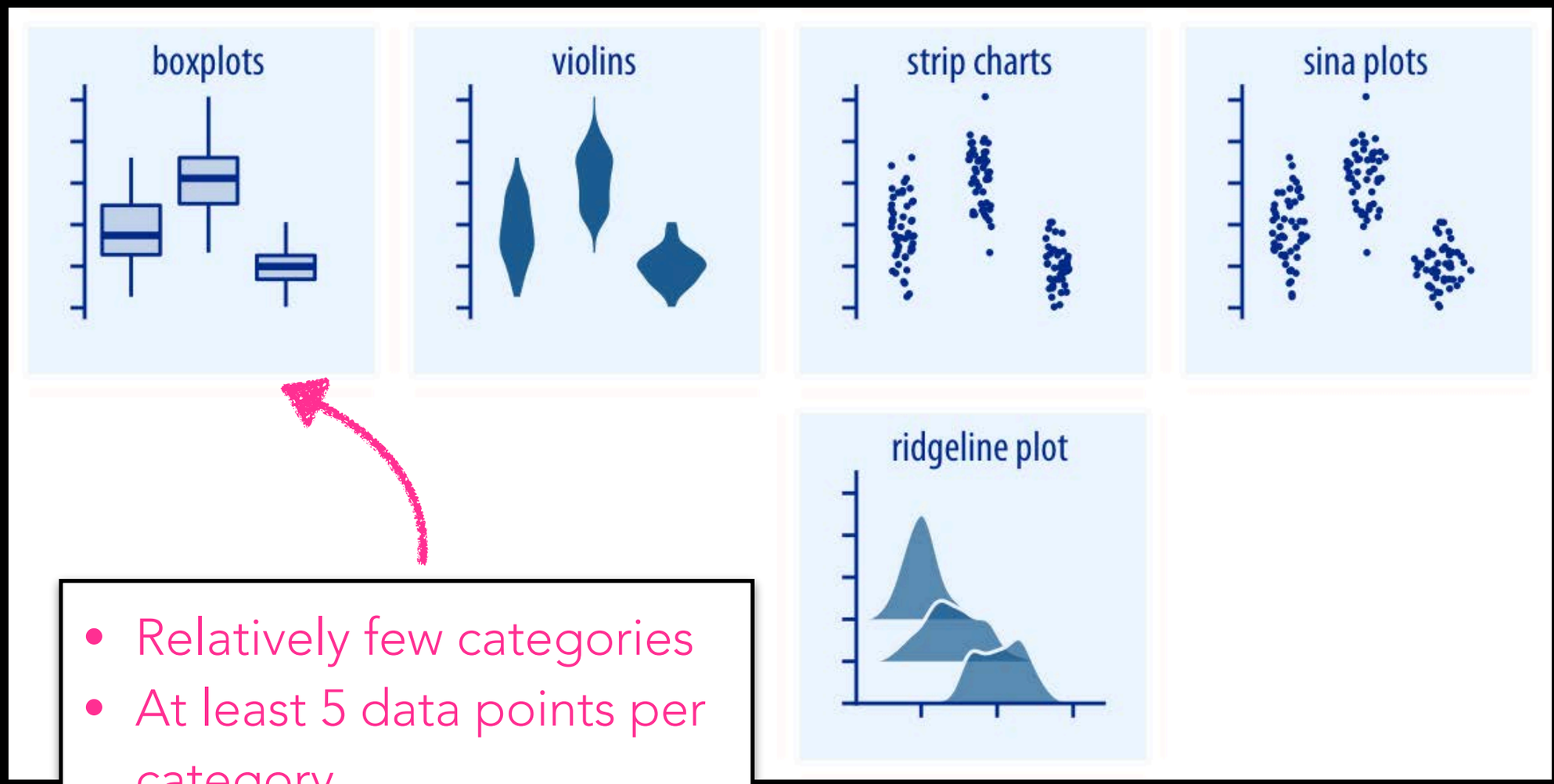
# WHAT IF YOU NEED TO SHOW MULTIPLE DISTRIBUTIONS TOGETHER?



# WHAT IF YOU NEED TO SHOW MULTIPLE DISTRIBUTIONS TOGETHER?

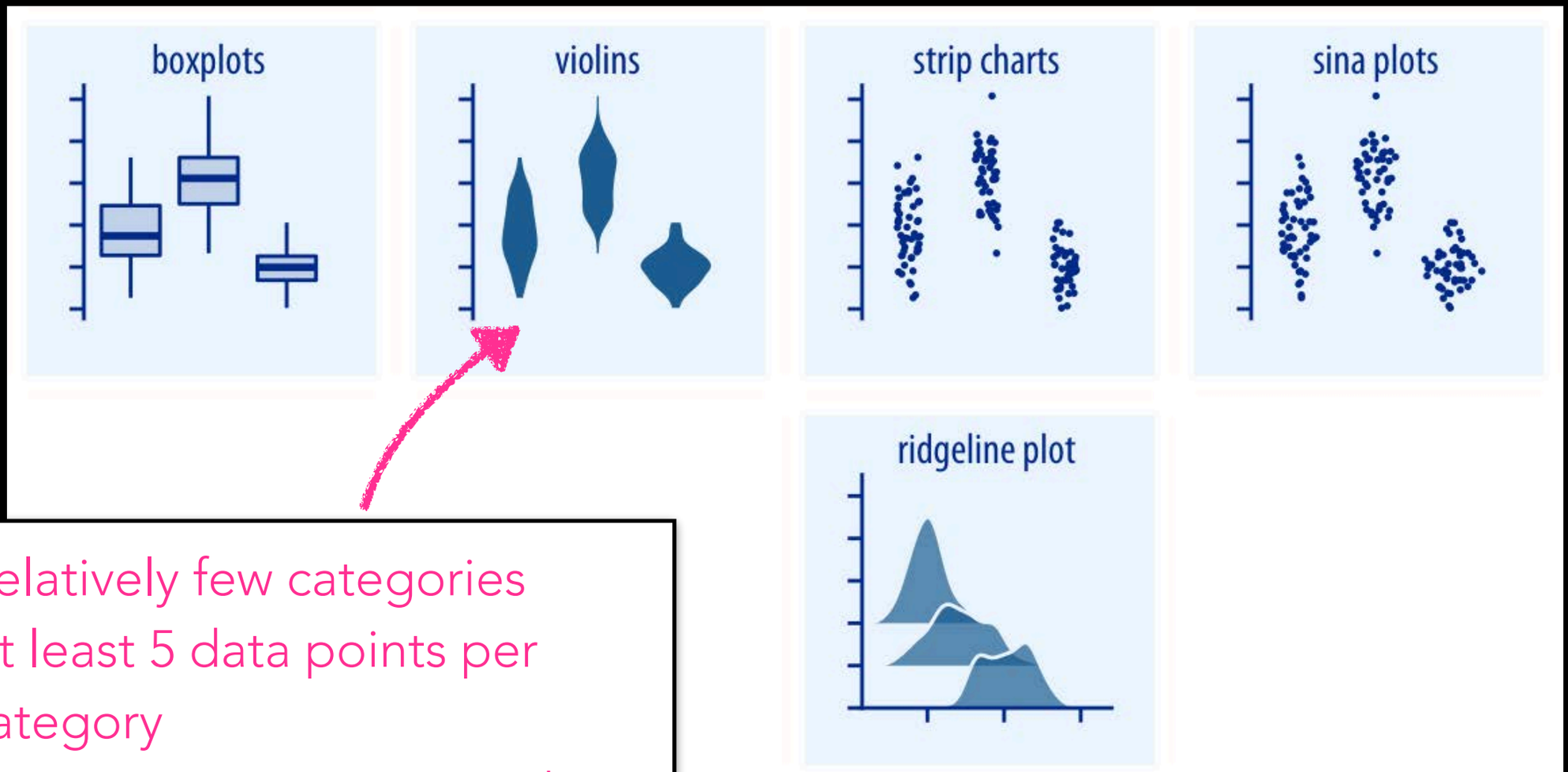


# USAGE GUIDELINES



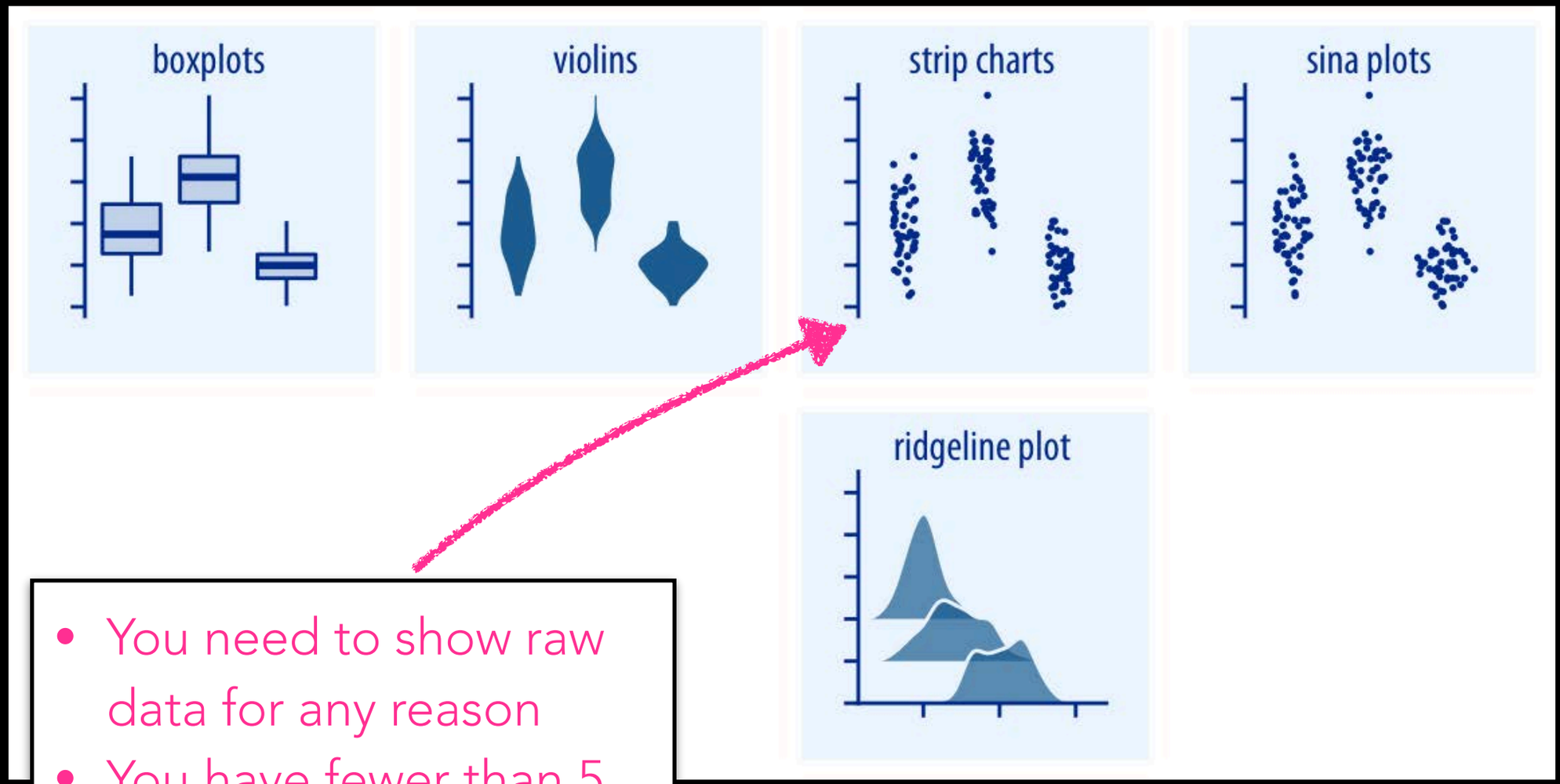

- Relatively few categories
- At least 5 data points per category
- Statistical comparisons are important

# USAGE GUIDELINES



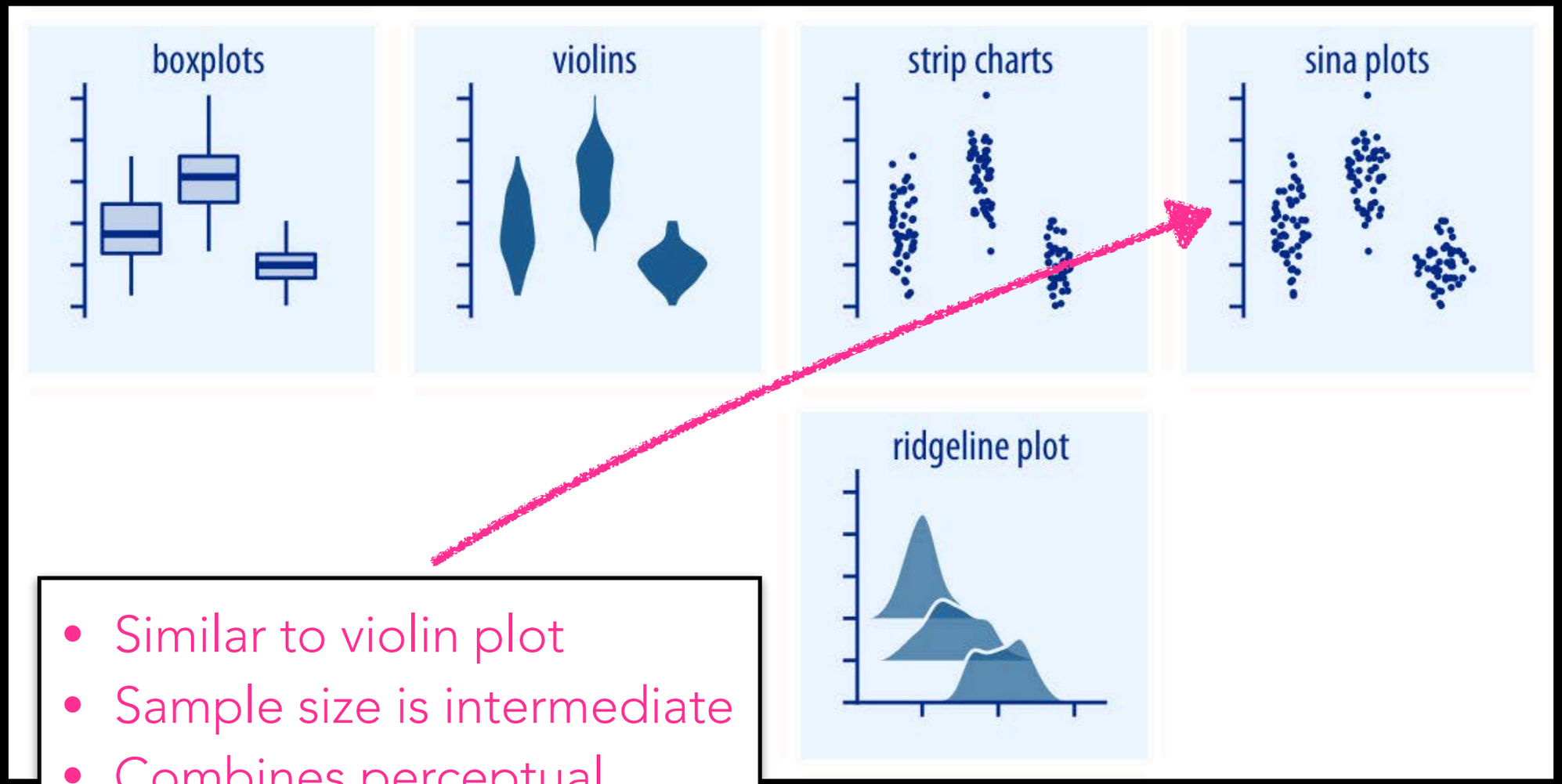

- Relatively few categories
- At least 5 data points per category
- You want a more nuanced comparison of the *shapes* of the distributions rather than statistical values

# USAGE GUIDELINES



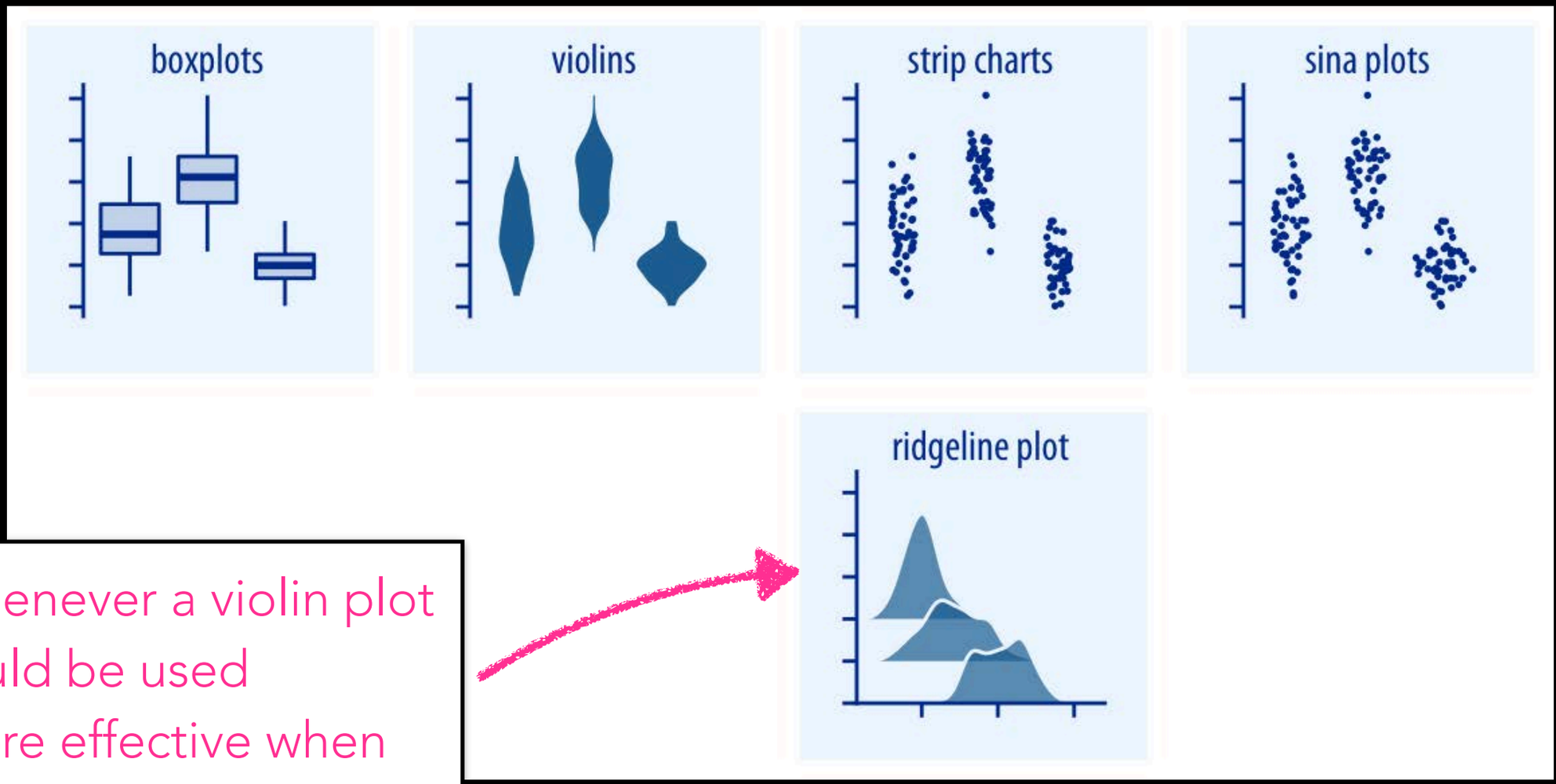

- You need to show raw data for any reason
- You have fewer than 5 data points in some categories

# USAGE GUIDELINES




- Similar to violin plot
- Sample size is intermediate
- Combines perceptual properties of violins and strip charts

# USAGE GUIDELINES

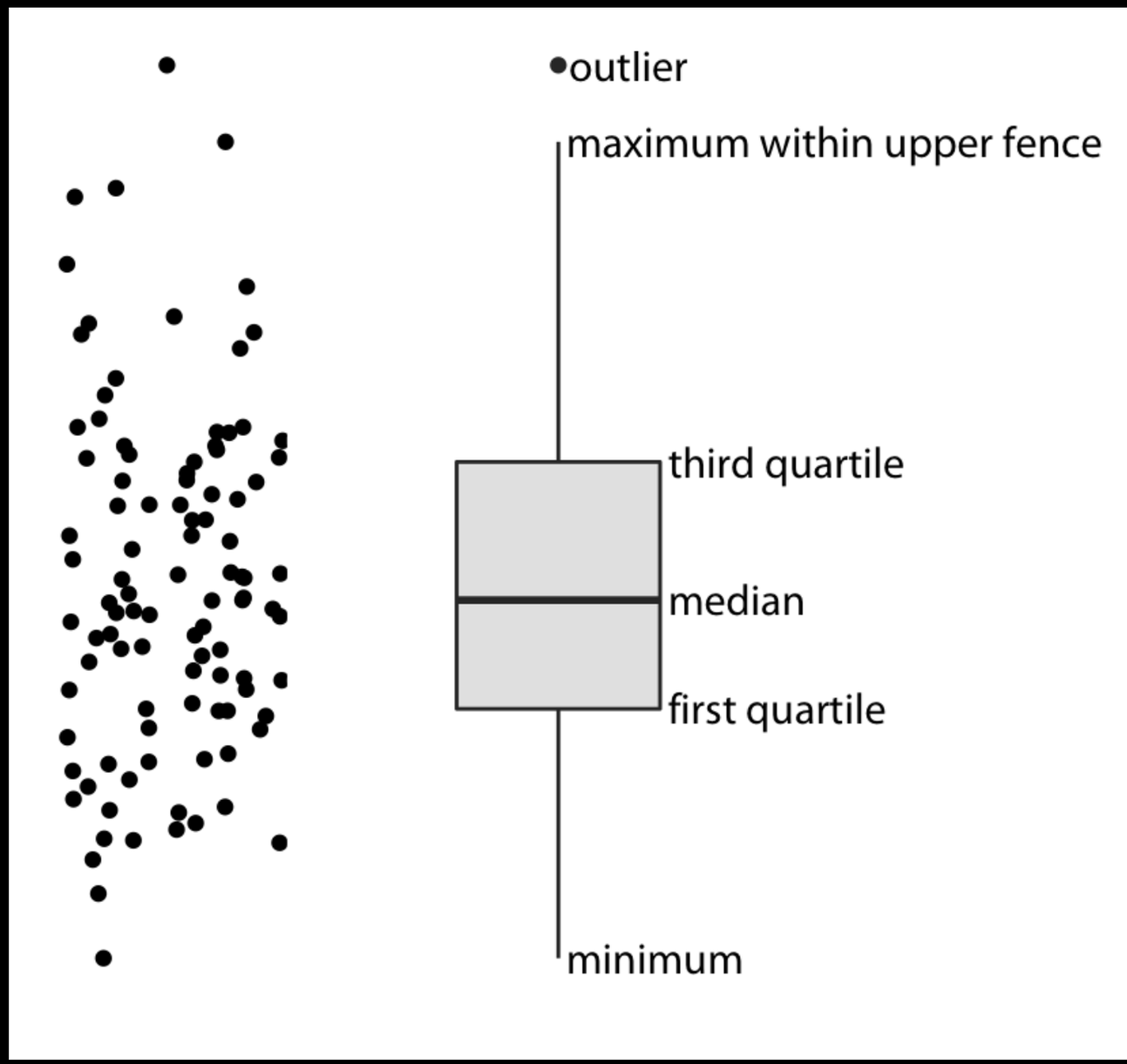



- Whenever a violin plot could be used
- More effective when there are *many* categories that you want to compare

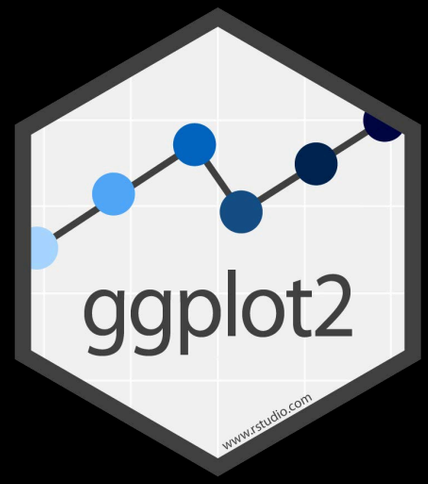
# BOXPLOTS

# BOXPLOTS

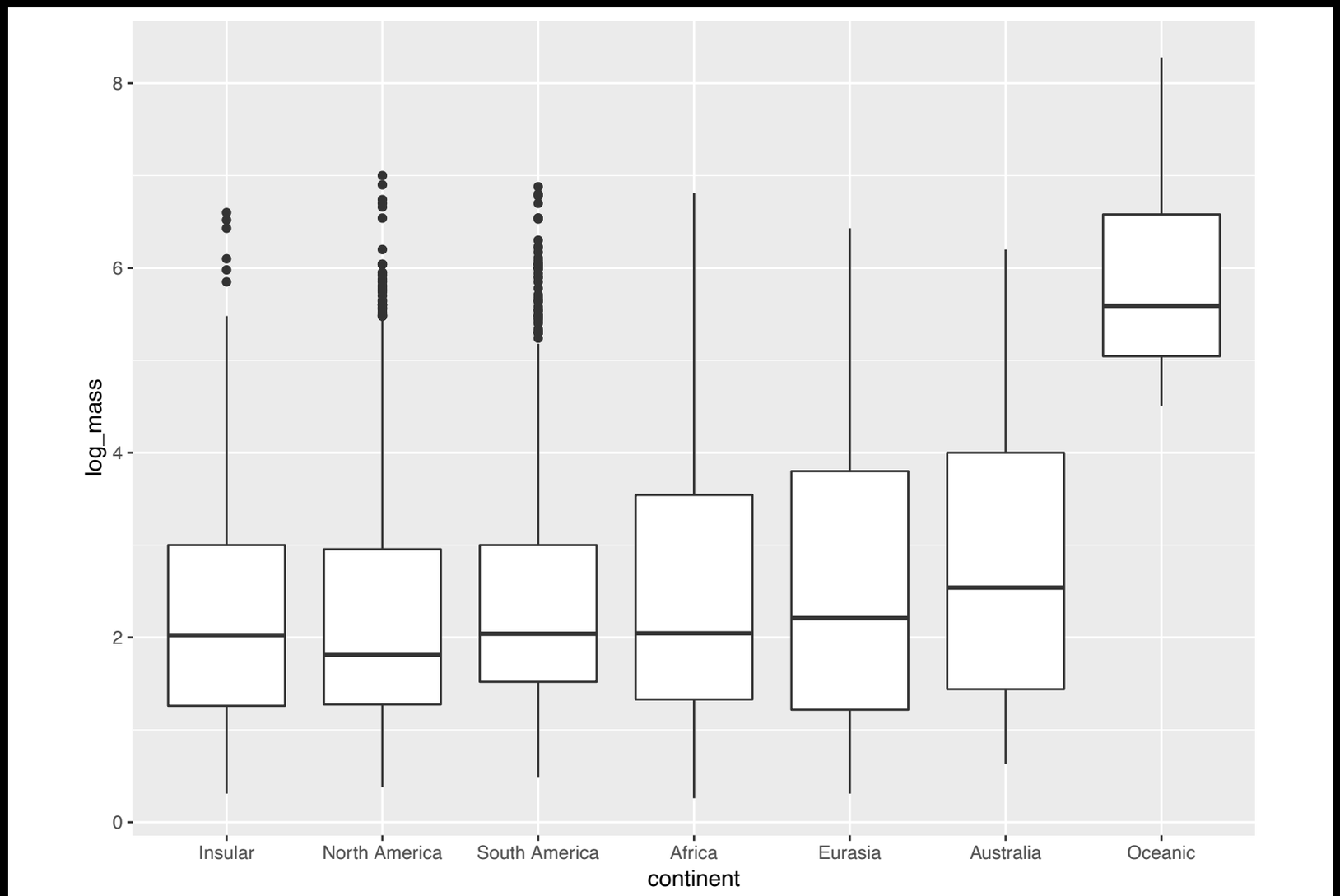
- Each box plot shows a standardized set of 5 statistical summaries of the distribution, as well as individual outliers
- Very common in scientific lit
- But beware of using with lay audience (e.g., “what does the line sticking out of the top show exactly?”)



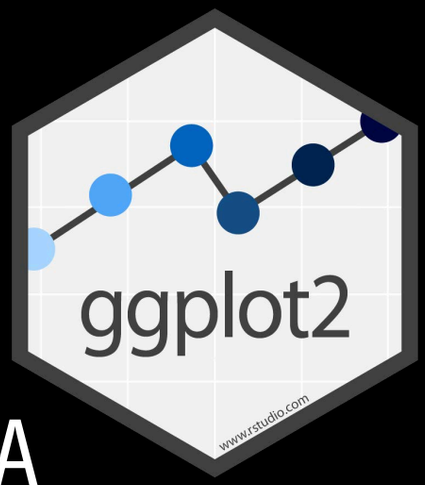
# BOXPLOTS



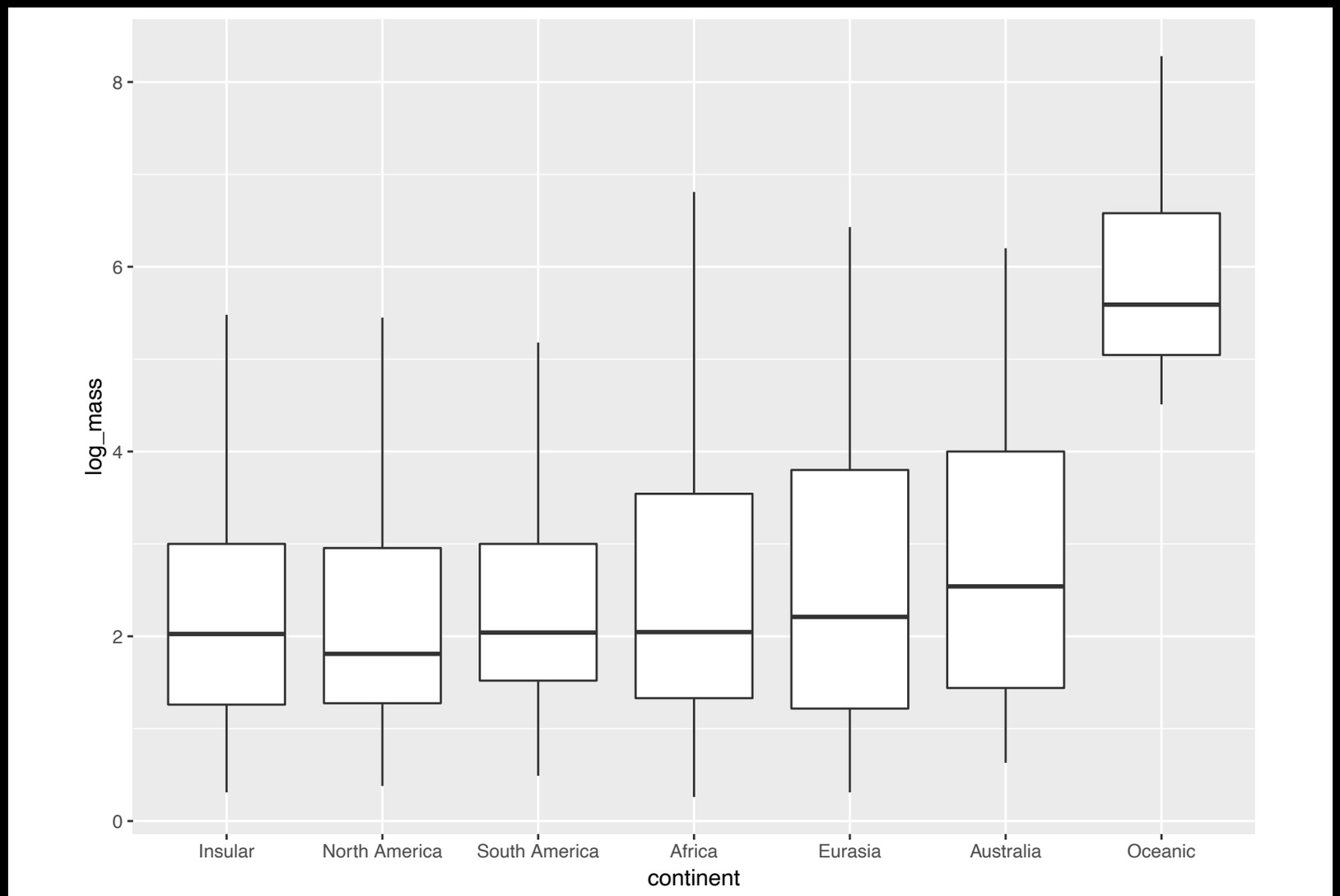

- Geometric object is `geom_boxplot()`
- Categorical variable to mapped to x, and numeric variable mapped to y



# BOXPLOTS




- Suppress outliers with `outlier.shape = NA`

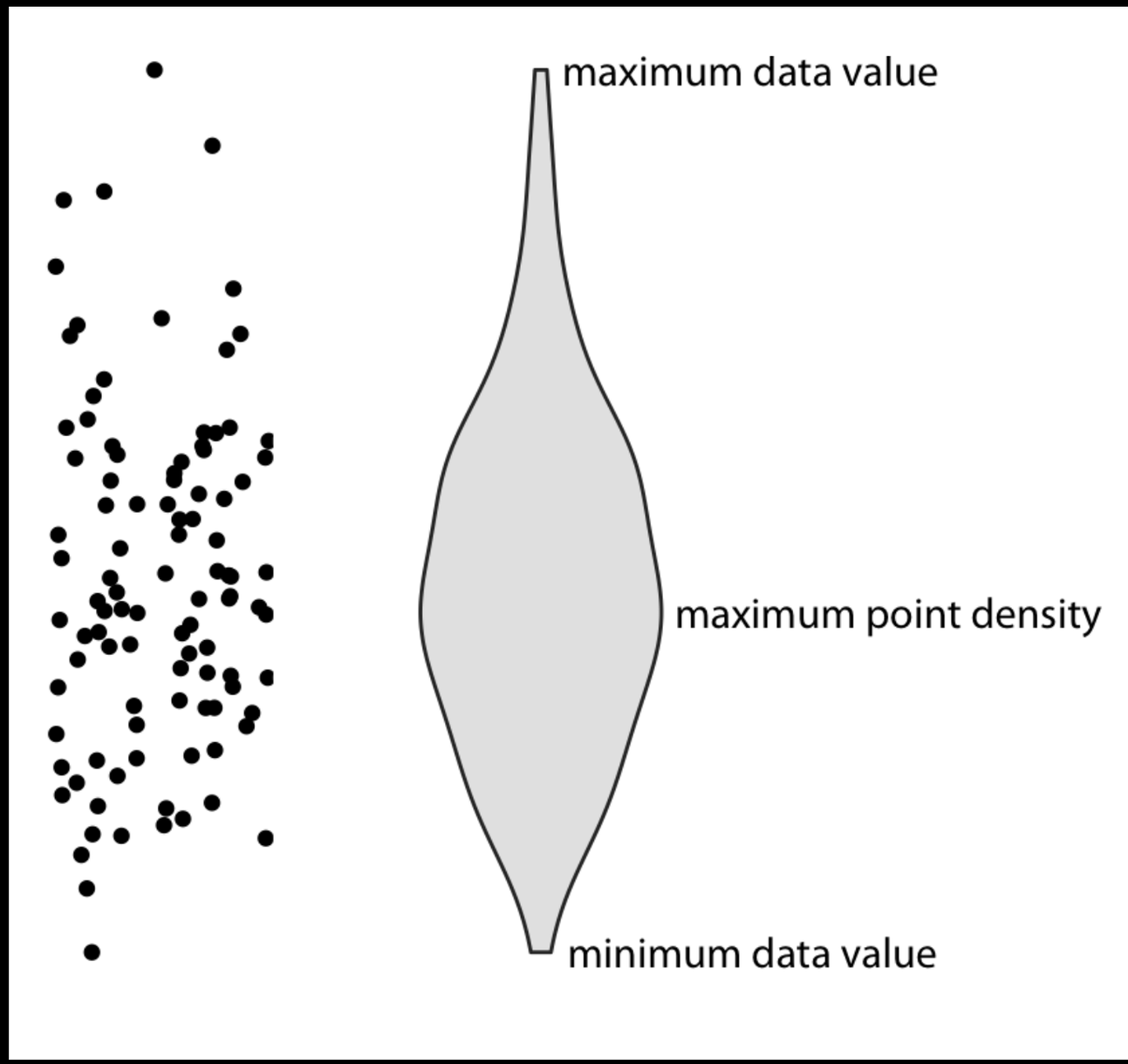


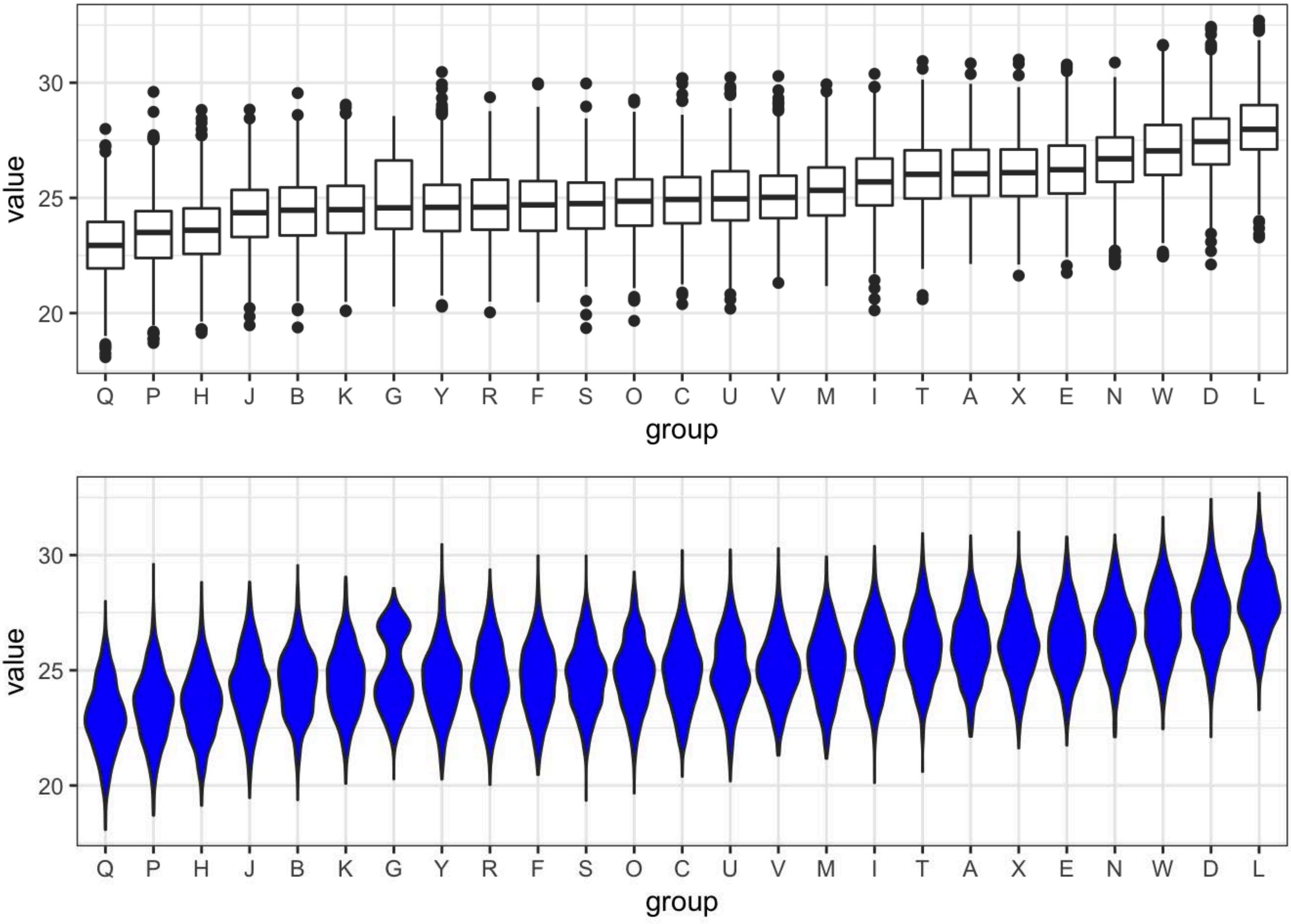

```
ggplot(lqm, aes(x = continent, y = log_mass)) +
  geom_boxplot(outlier.shape = NA)
```

# VIOLIN PLOTS

# VIOLIN PLOTS

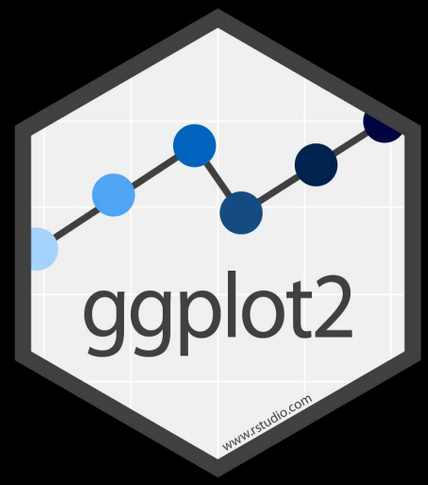
- Each violin is essentially a density plot rotated vertically and mirrored across the vertical
- Less standardized and precise, but also easier to intuit
- Better at showing multi-modal distributions



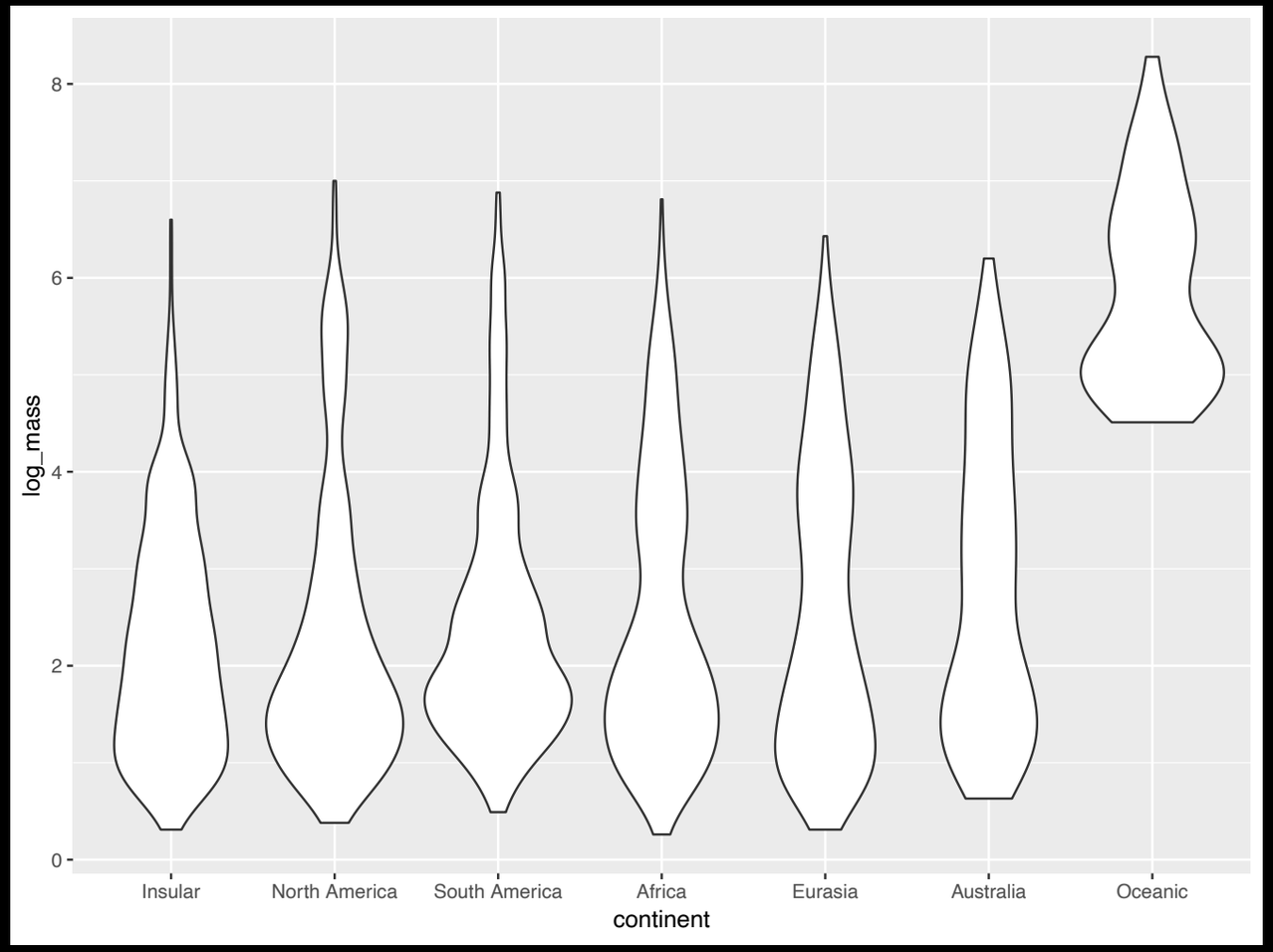

value
30
25
20
Q P H J B K G Y R F S O C U V M I T A X E N W D L
group


value
30
25
20
Q P H J B K G Y R F S O C U V M I T A X E N W D L
group

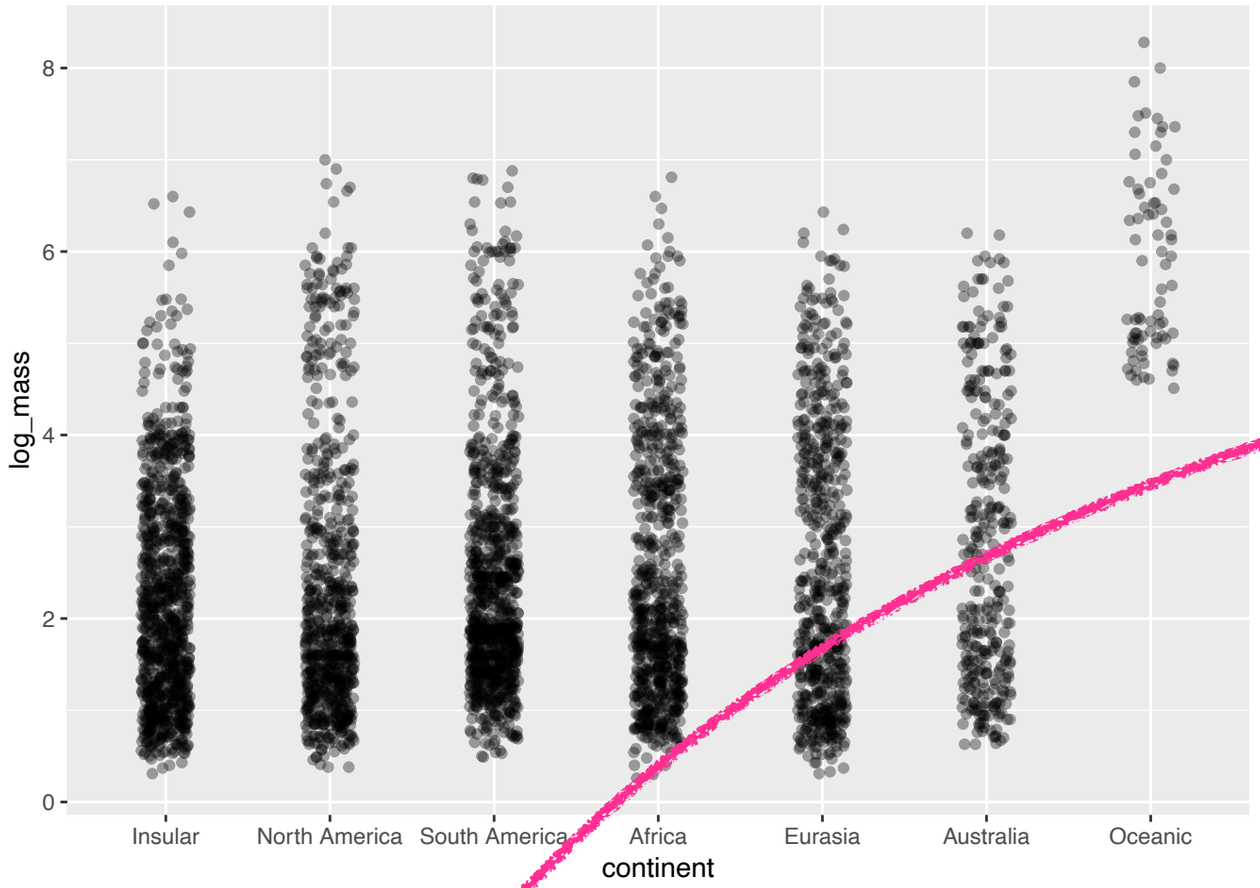
# VIOLIN PLOTS




- Geometric object is `geom_violin()`
- Mappings same as boxplot



# STRIP PLOTS




`geom_jitter()` makes randomly displaced points

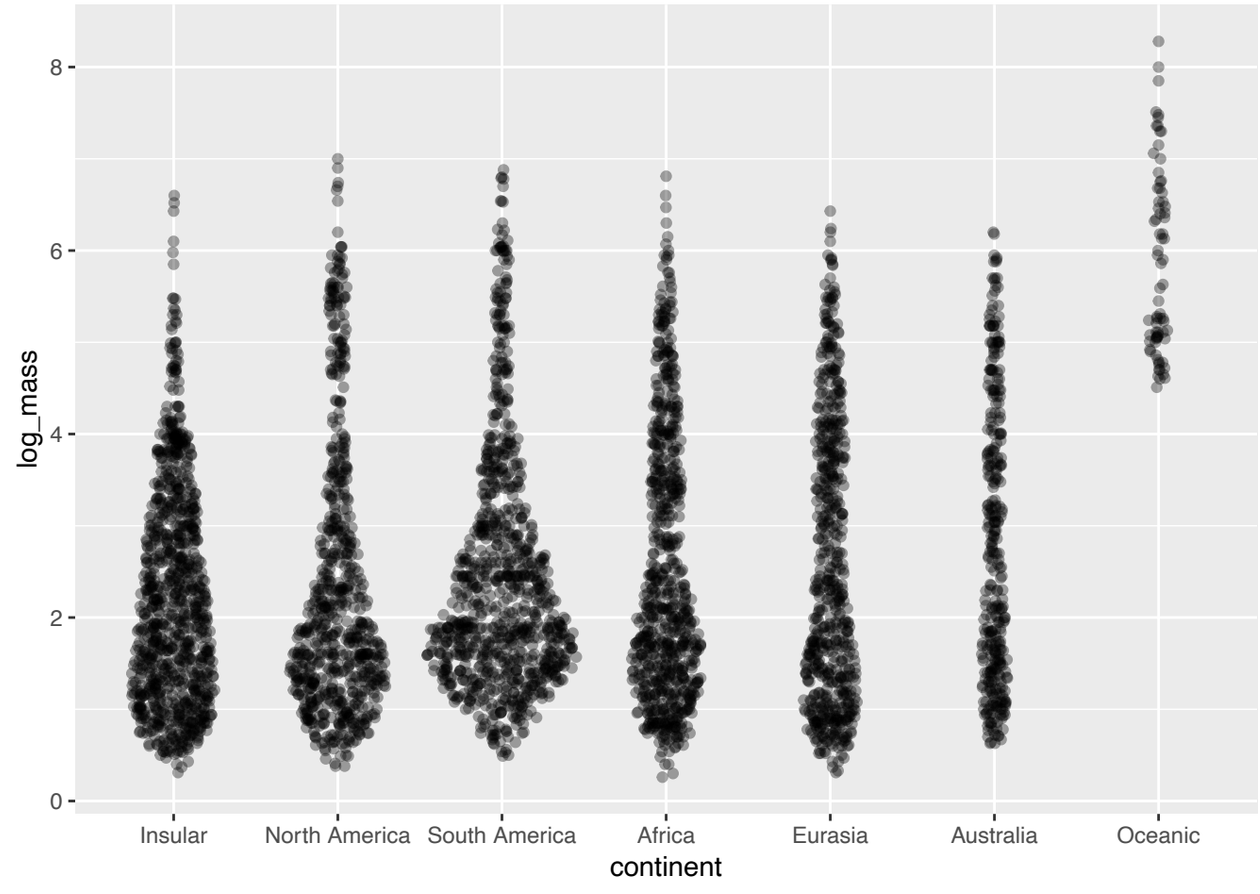
`height` and `width` parameters control amount of jitter

```
ggplot(lqm, aes(x = continent, y = log_mass)) +
  geom_jitter(width = 0.15, height = 0, alpha = 0.35)
```

# SINA PLOTS

Not a default geom!

Provided by the "ggforce" package.

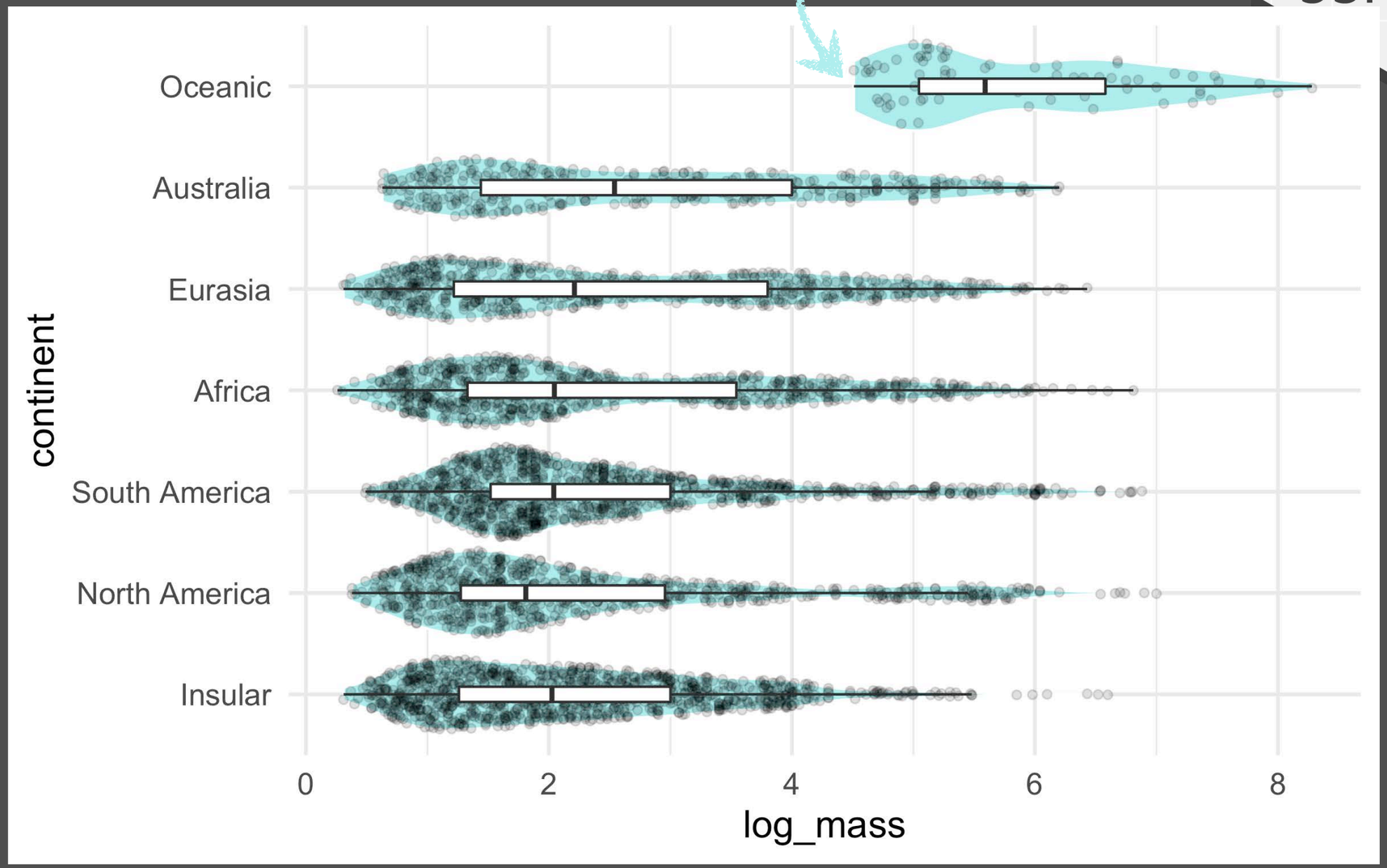



```
library("ggforce")

ggplot(lqm, aes(x = continent, y = log_mass)) +
  geom_sina(alpha = 0.15)
```

# YOUR TURN

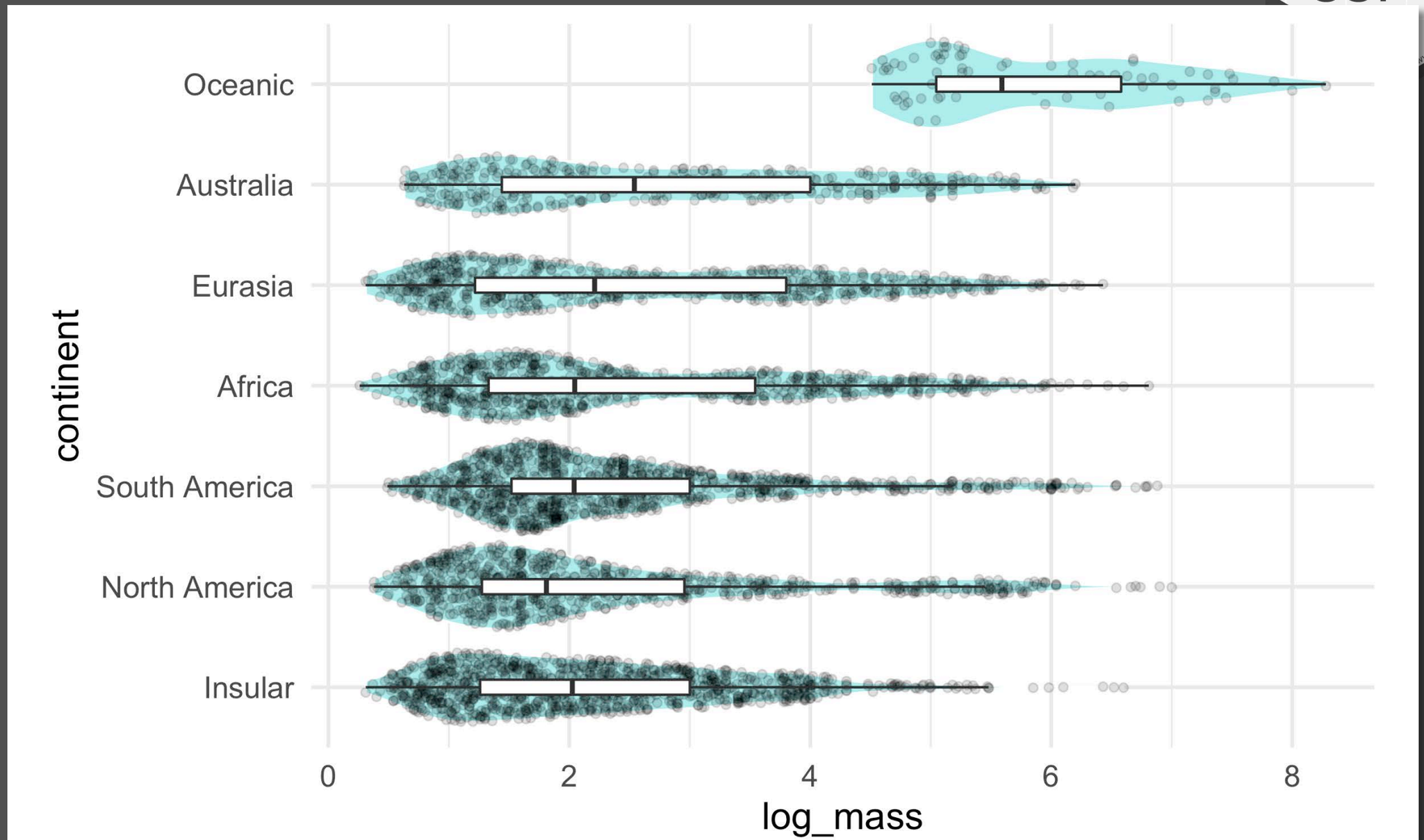
fill is "paleturquoise"




Try to reverse-engineer this (rather busy) plot.

5:00

# YOUR TURN

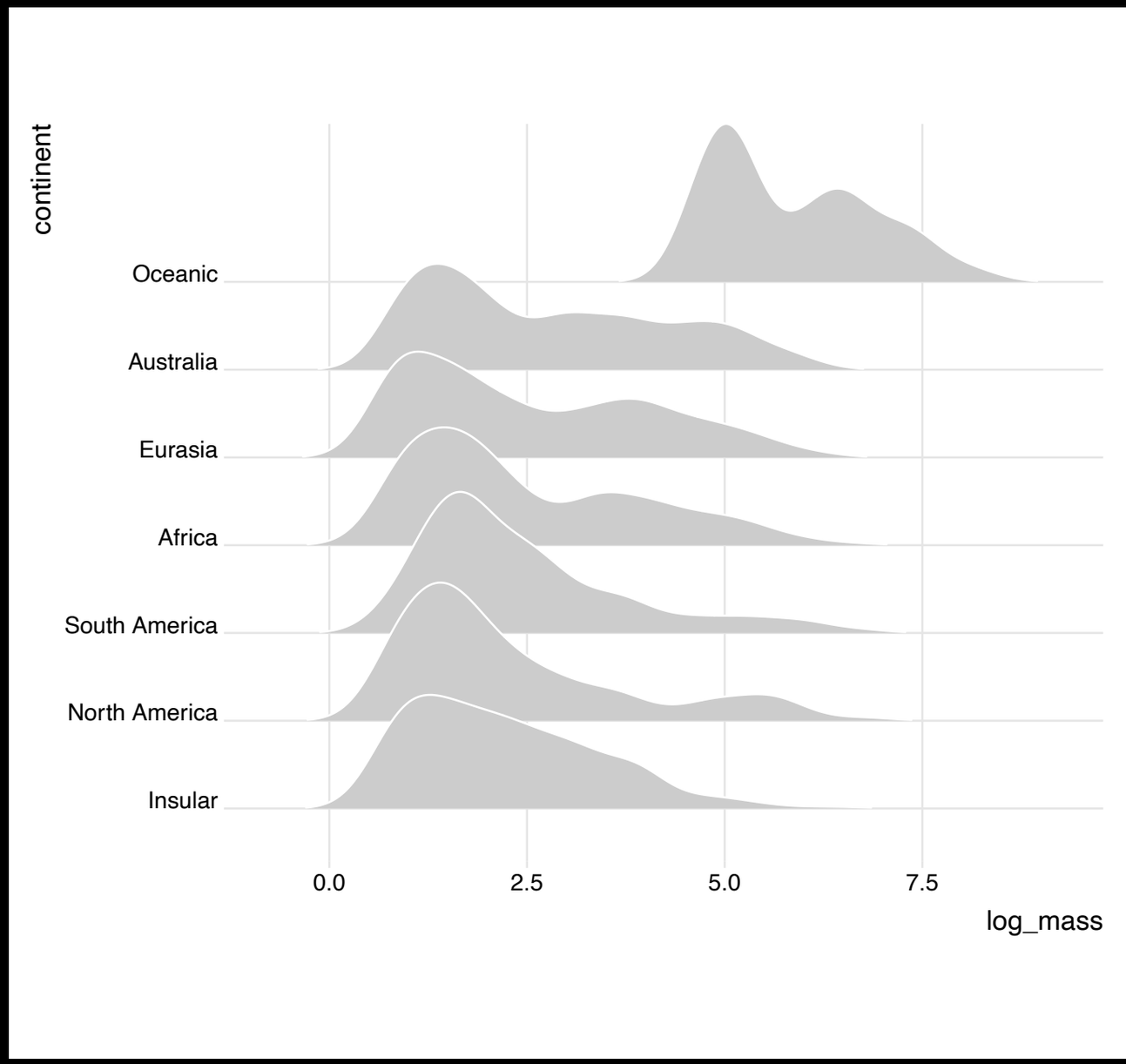


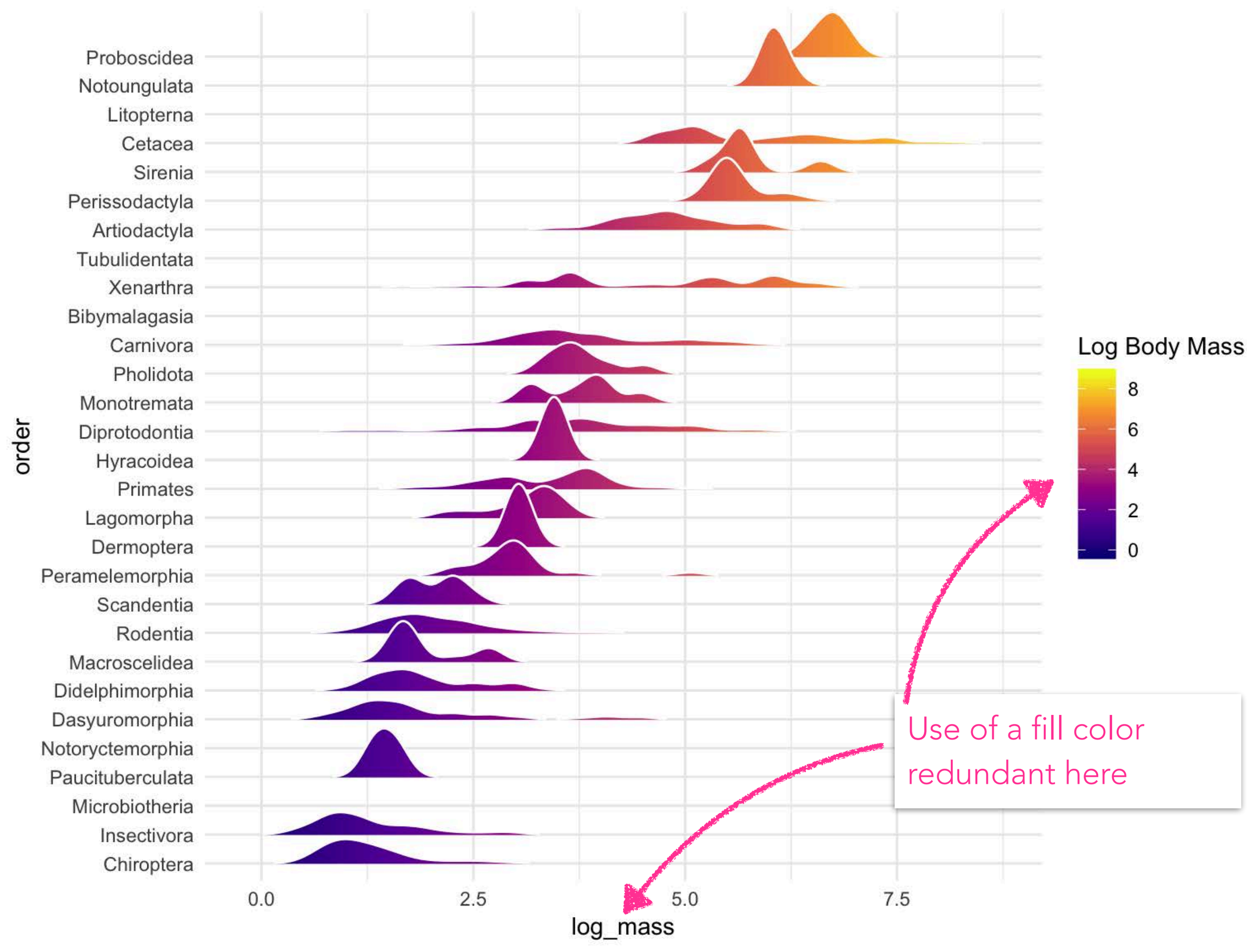

```
ggplot(lqm, aes(x = continent, y = log_mass)) +
  geom_violin(fill = "paleturquoise", color = NA) +
  geom_sina(alpha = 0.15) +
  geom_boxplot(fill = "white", width = 0.15, outlier.shape = NA) +
  coord_flip()
```

# RIDGELINE PLOTS

# RIDGELINE PLOTS

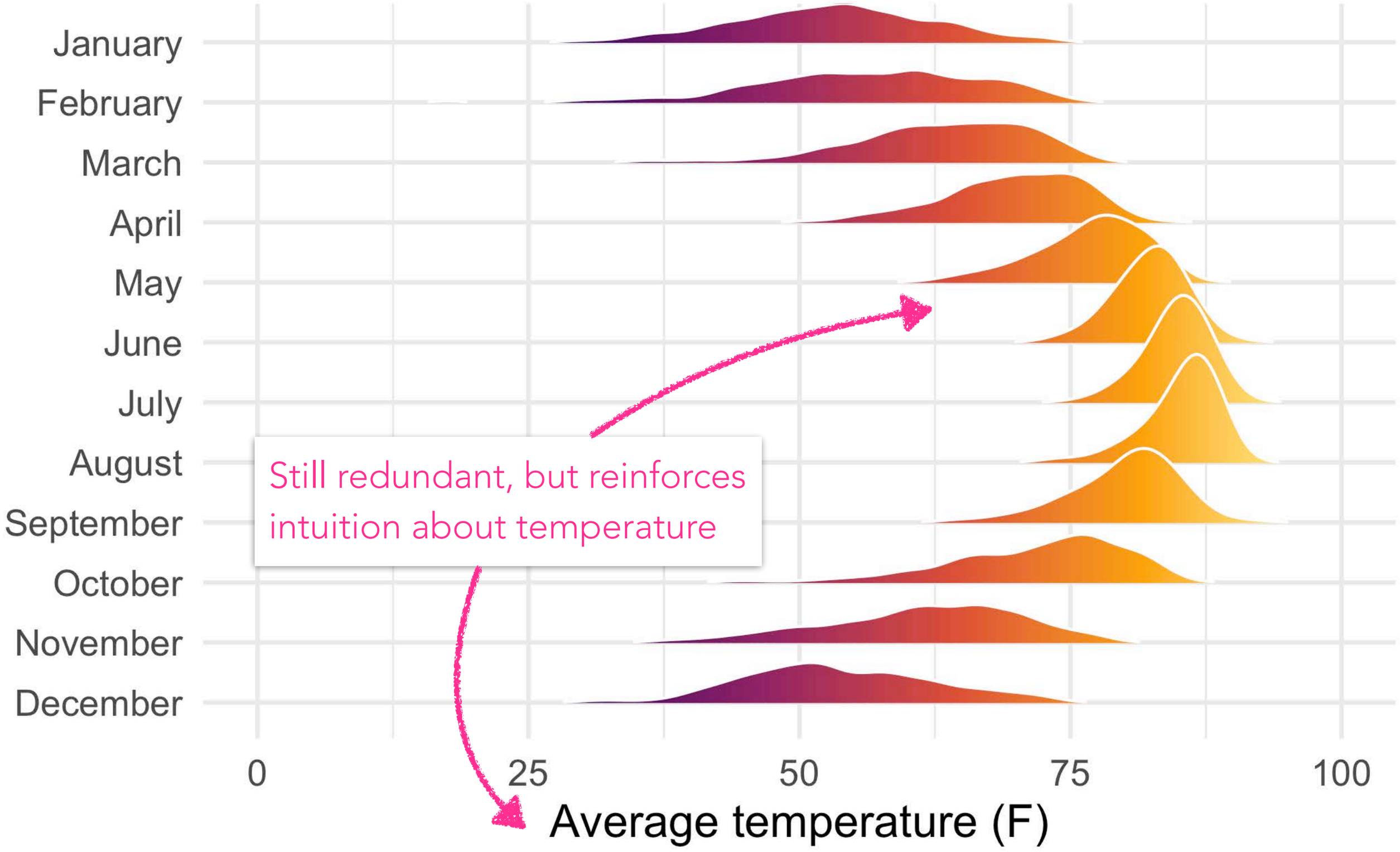
- Each ridge is a vertically offset density plot
- Rarely used, but easy to compare shapes and relative heights
- Not good if specific statistical values are of interest.




order
Proboscidea
Notoungulata
Litopterna
Cetacea
Sirenia
Perissodactyla
Artiodactyla
Tubulidentata
Xenarthra
Bibymalagasia
Carnivora
Pholidota
Monotremata
Diprotodontia
Hyracoidea
Primates
Lagomorpha
Dermoptera
Peramelemorphia
Scandentia
Rodentia
Macroscelidea
Didelphimorphia
Dasyuromorphia
Notoryctemorphia
Paucituberculata
Microbiotheria
Insectivora
Chiroptera
0.0
2.5
5.0
7.5
log_mass
Log Body Mass
8
6
4
2
0
Use of a fill color redundant here

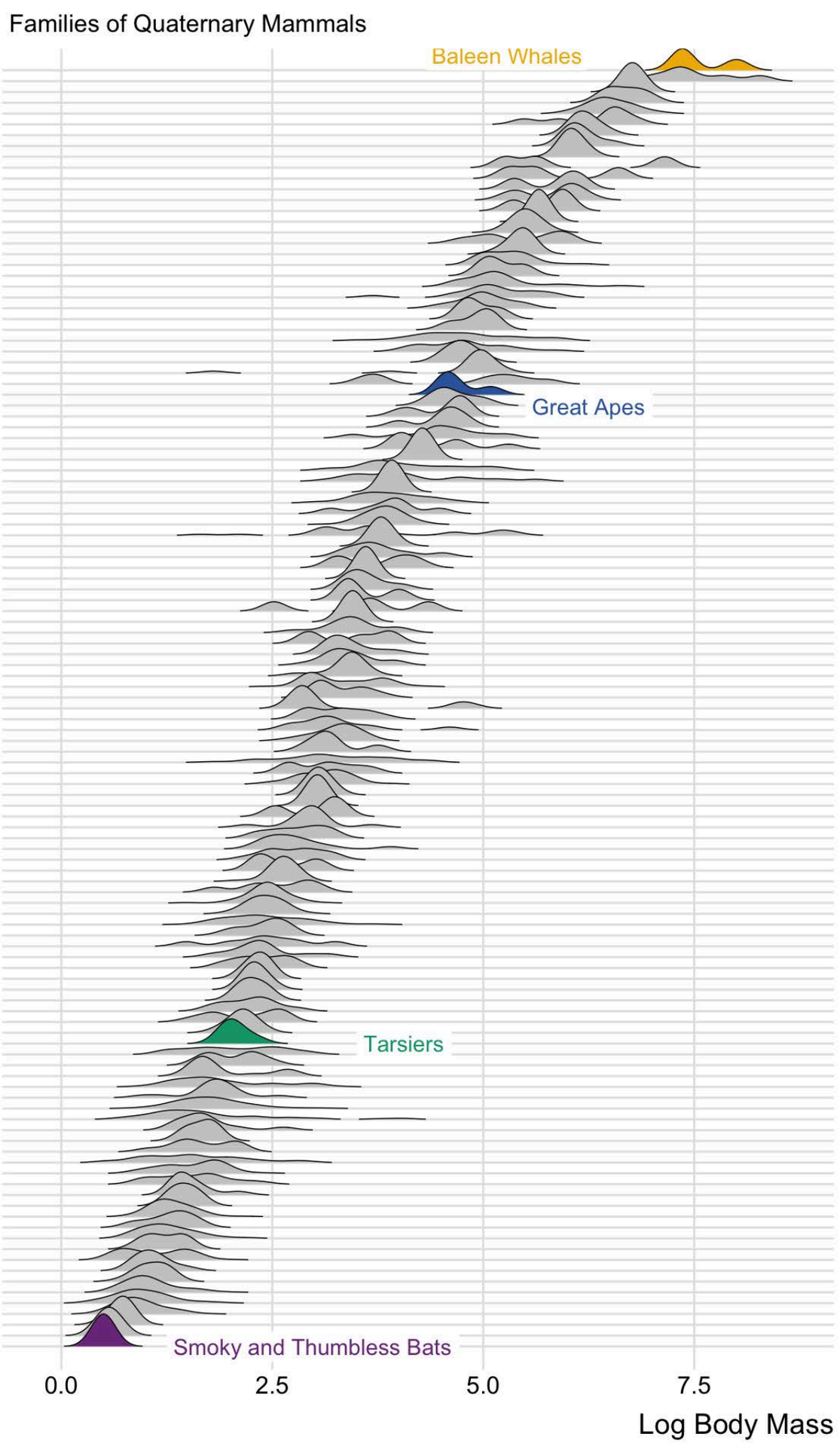
# SATX Climate

## 1946 - 2021

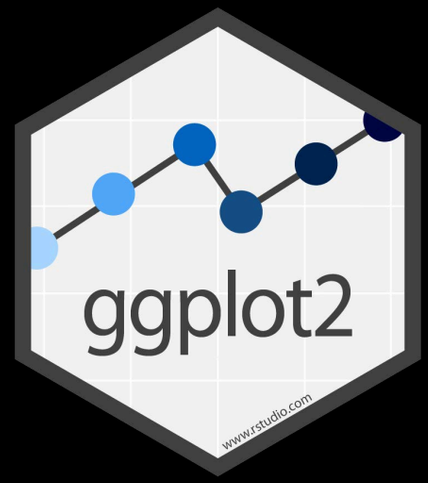


# RIDGELINE PLOT

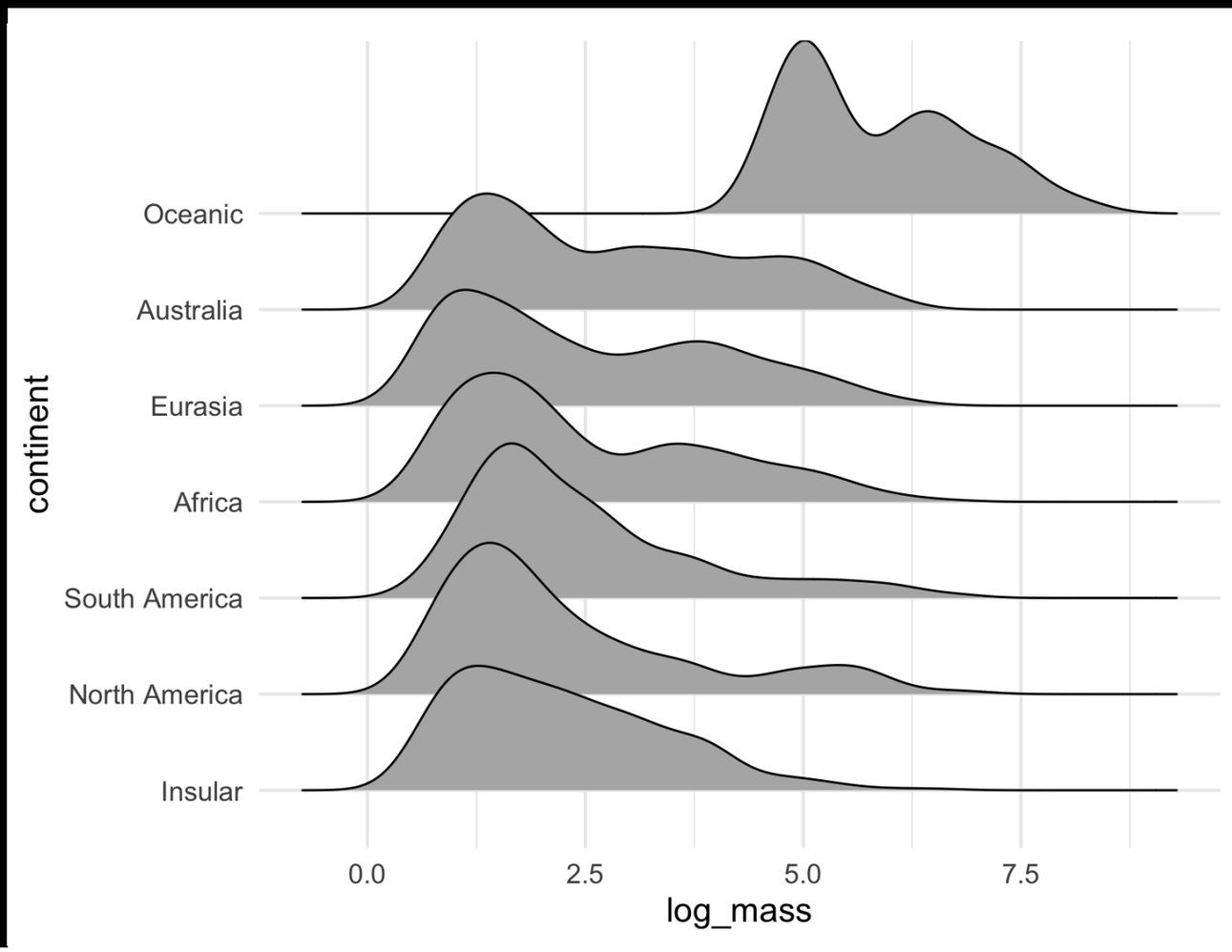
- Only method we have discussed that remains readable with large number of distributions



# RIDGELINE PLOTS



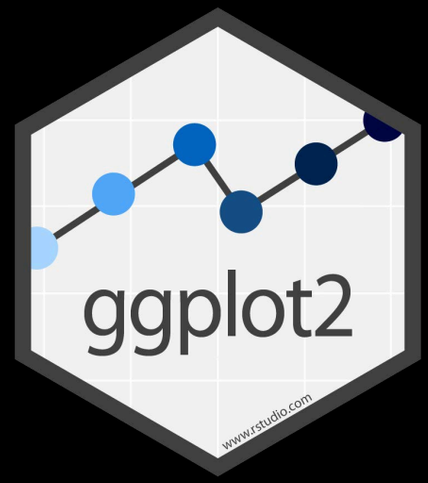

- Not a default geom—provided by "ggridges" package
- Basic geometric object is `geom_density_ridges()`
- Like density plot, numeric variable mapped to x but categorical variable to mapped to y



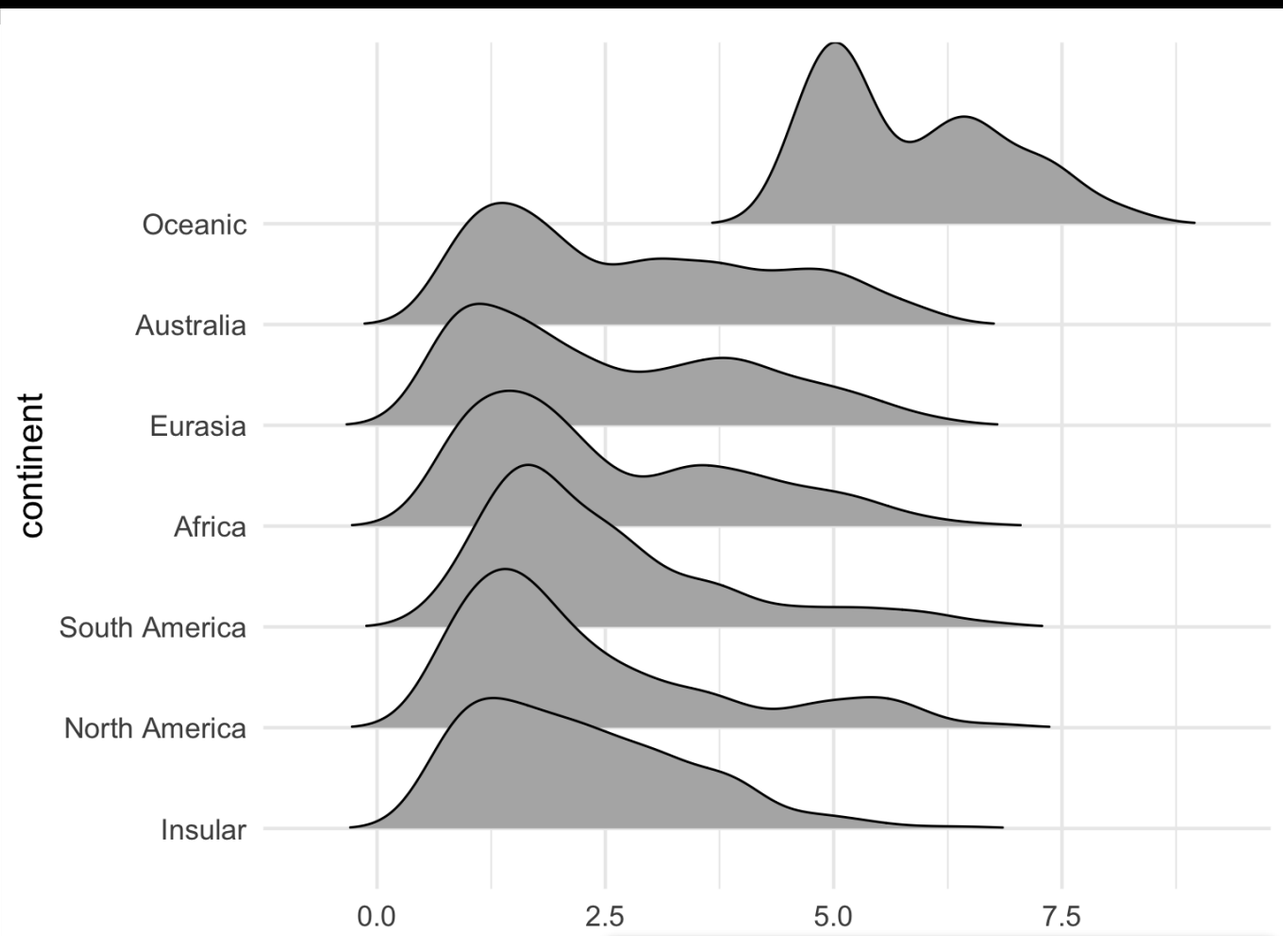

```
library("ggridges")

ggplot(lqm, aes(x = log_mass, y = continent)) +
  geom_density_ridges()
```

# RIDGELINE PLOTS




- Not a default geom—provided by "ggridges" package
- Basic geometric object is `geom_density_ridges()`
- Like density plot, numeric variable mapped to x but categorical variable to mapped to y

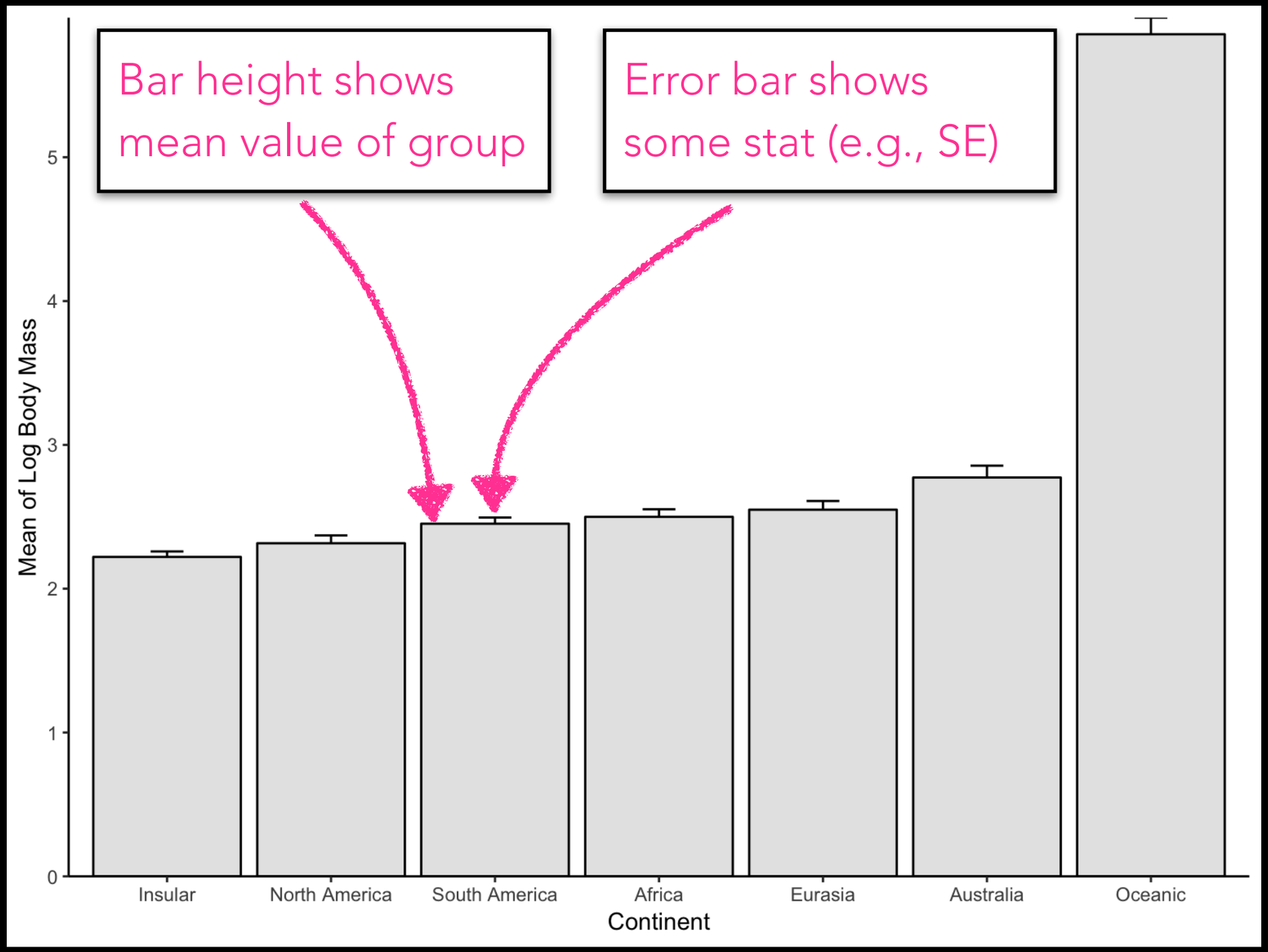



```
library("ggridges")

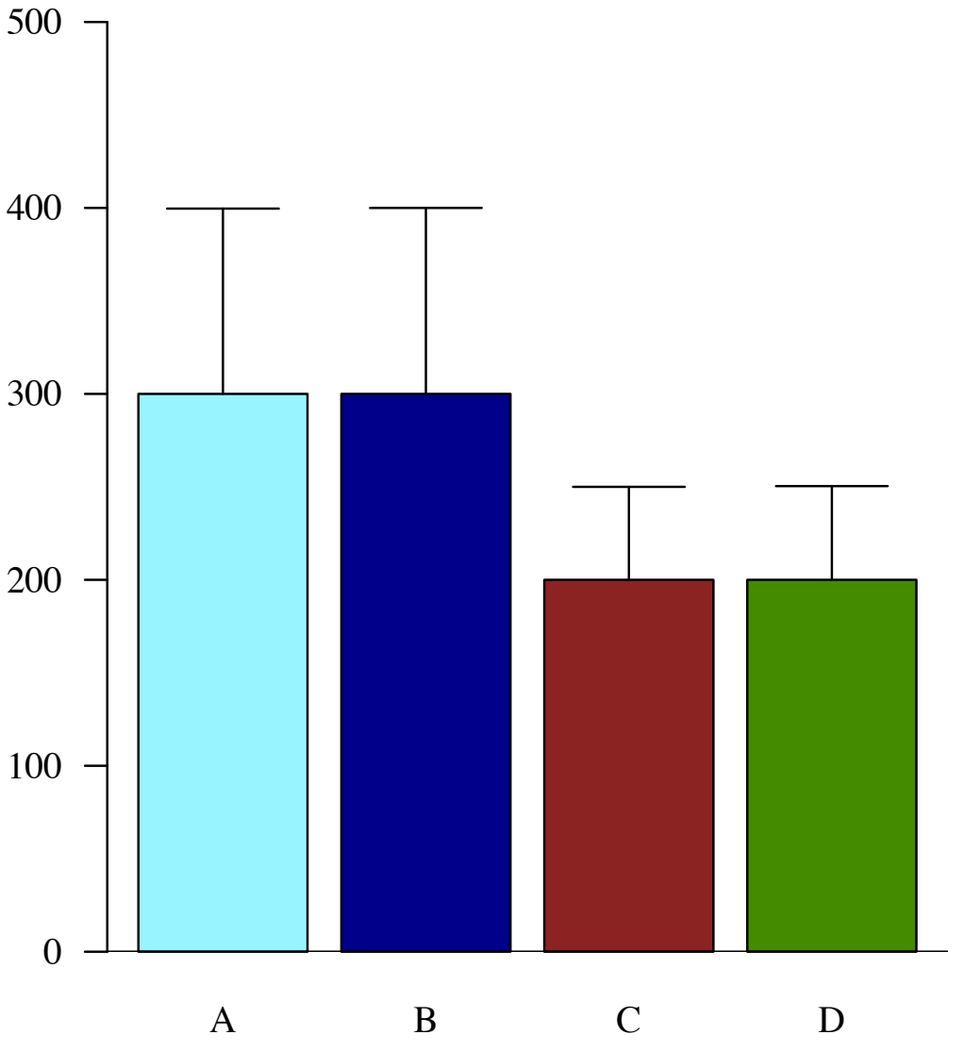
ggplot(lqm, aes(x = log_mass, y = continent)) +
  geom_density_ridges(rel_min_height = 0.005)
```

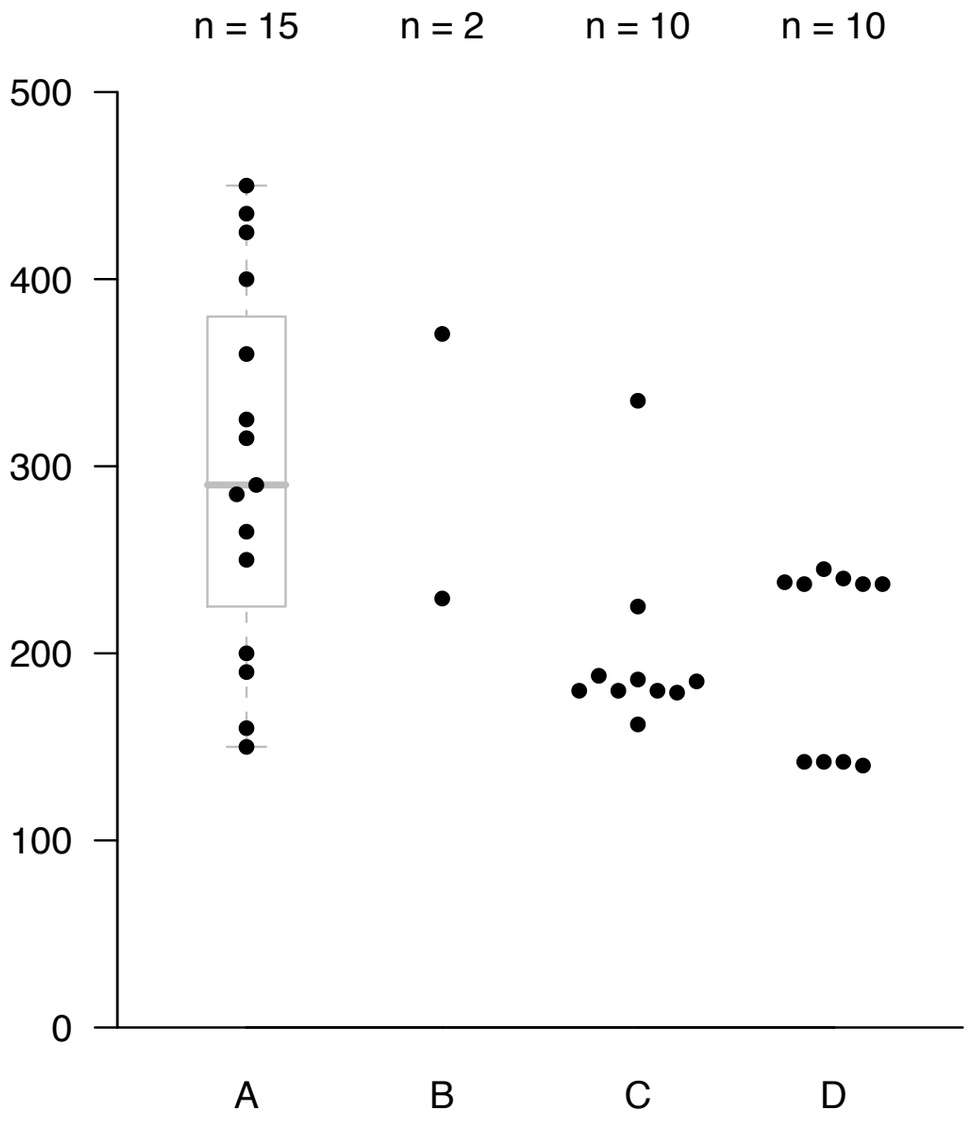
Cleaner look: cuts off long tails.

# WHAT ABOUT A "DYNAMITE PLOT"?



# DON'T USE THEM FOR DISTRIBUTIONS



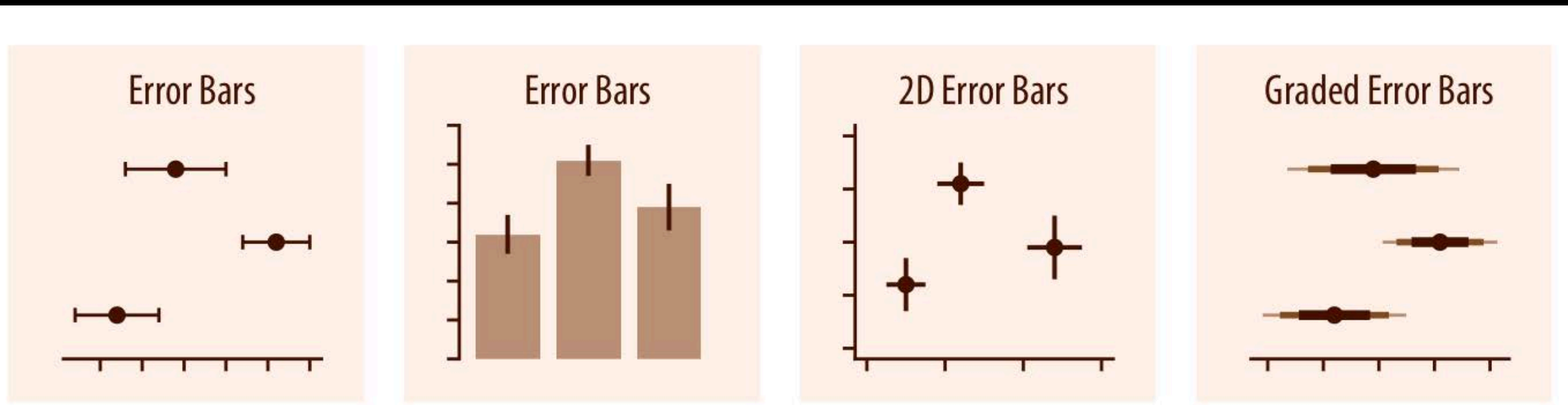




http://biostat.mc.vanderbilt.edu/wiki/pub/Main/TatsukiRcode/Poster3.pdf

# VISUALIZING UNCERTAINTY

# VISUALIZING UNCERTAINTY

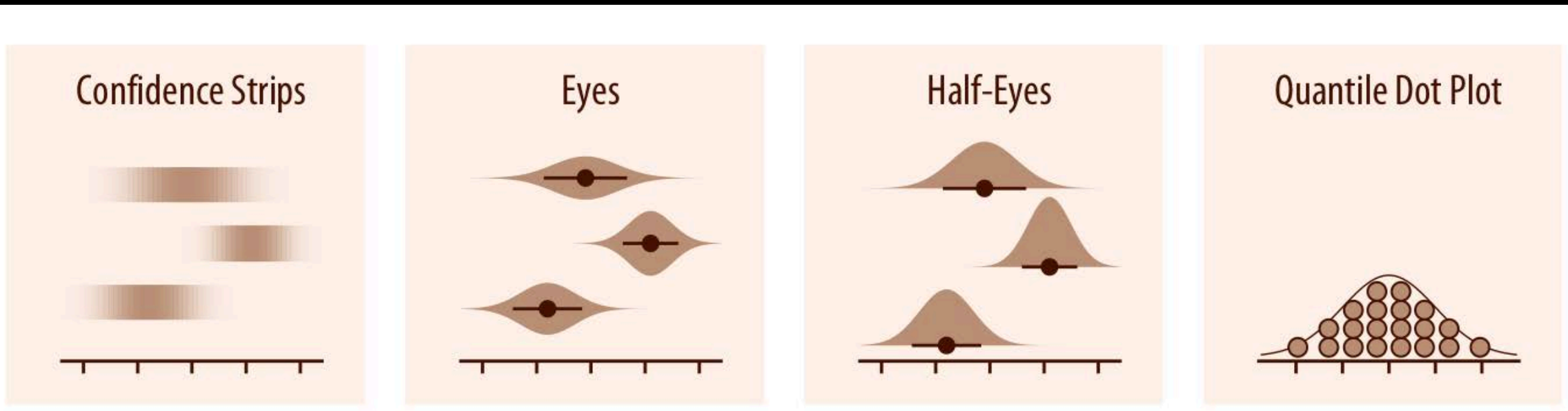
Showing uncertainty intervals around a measurement/estimate




Very common approaches

# VISUALIZING UNCERTAINTY

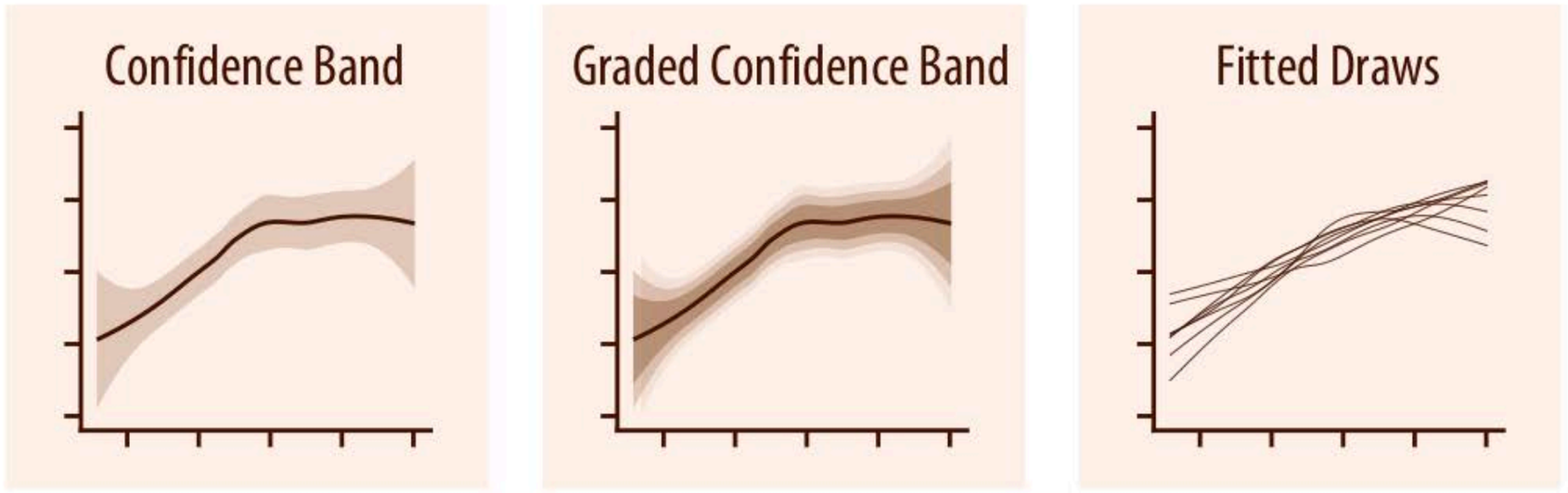
Showing continuous uncertainty around an estimate

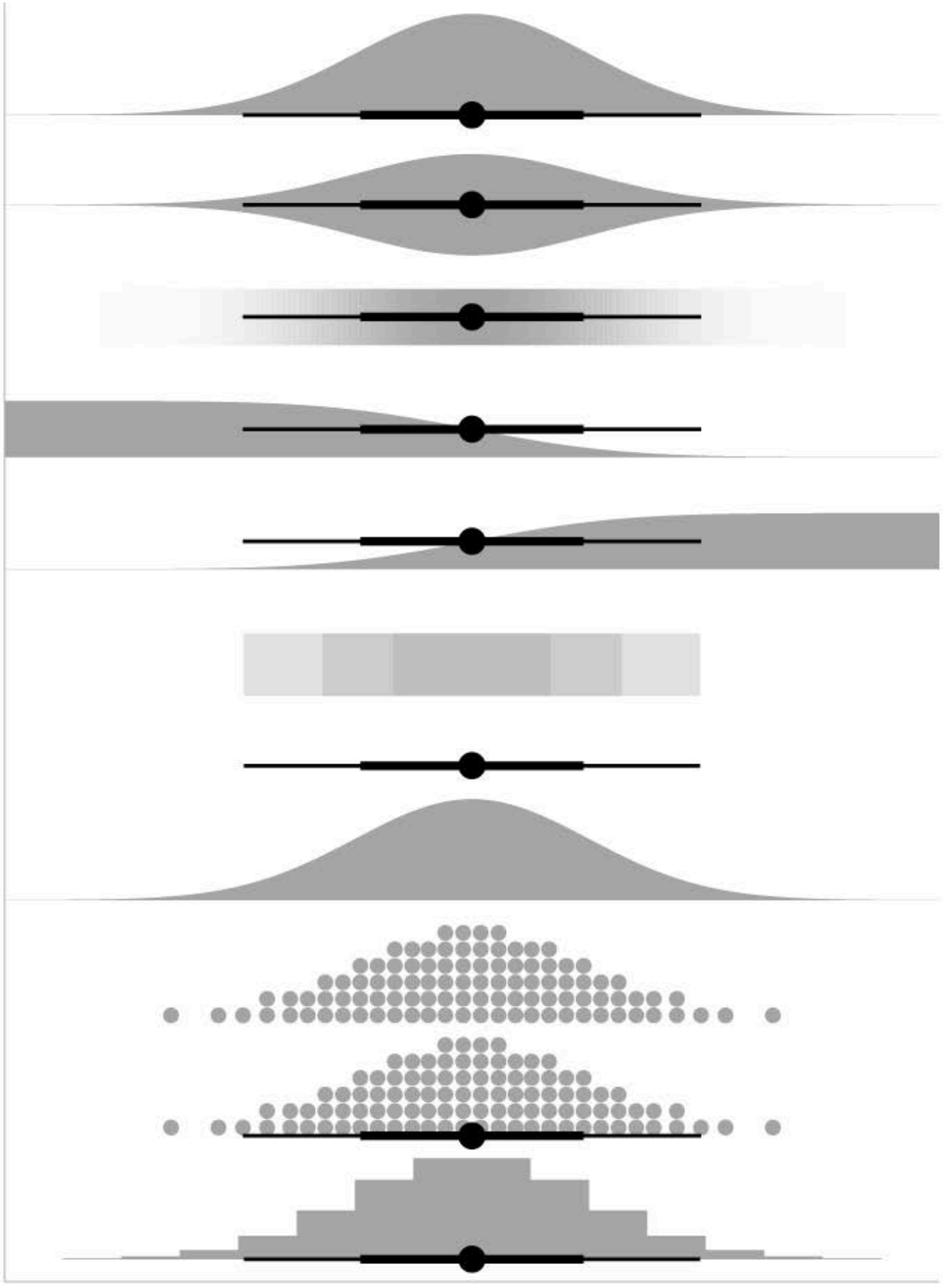



Less-common approaches

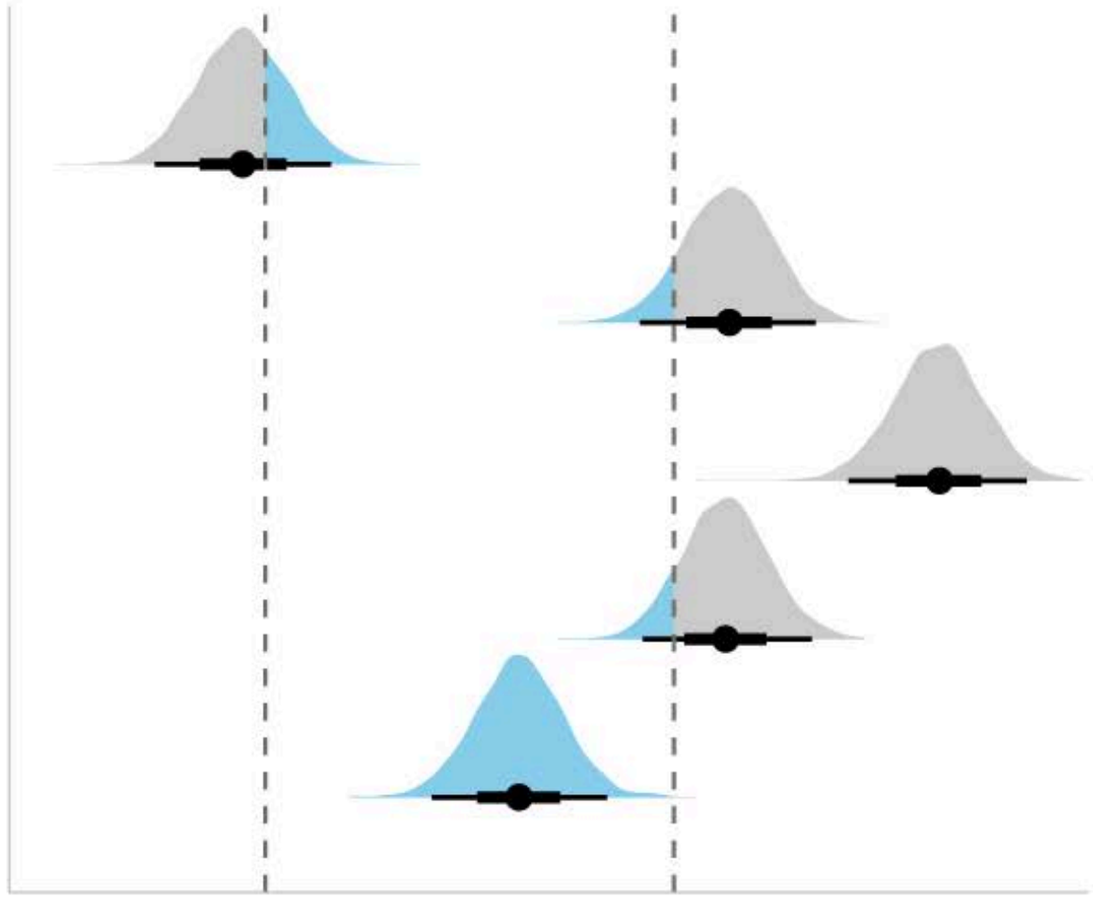
# VISUALIZING UNCERTAINTY

Showing uncertainty around smooth lines



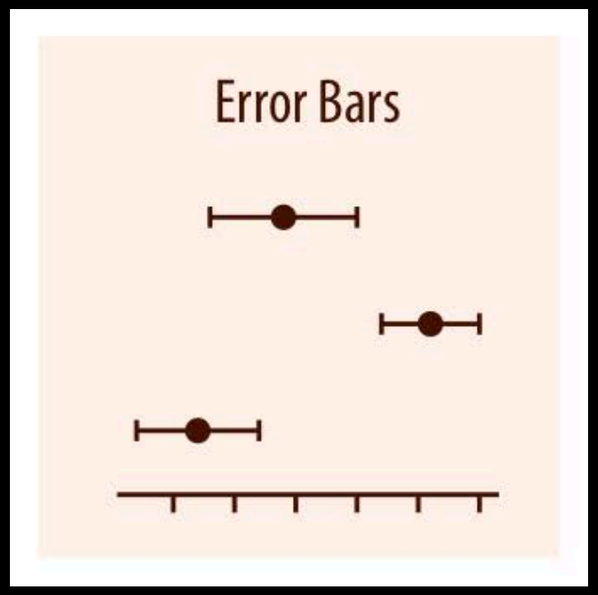



GGDIST



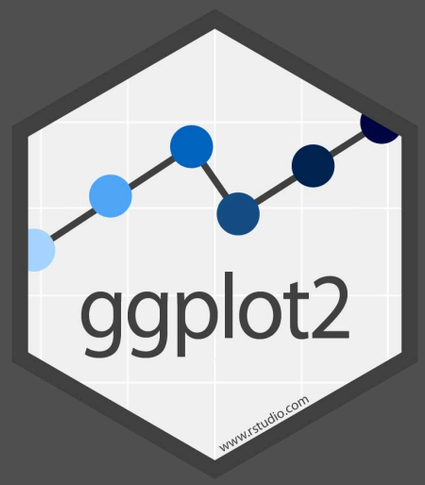
# VISUALIZING UNCERTAINTY

Showing uncertainty intervals around a measurement/estimate




The workhorse: error bars

# YOUR TURN



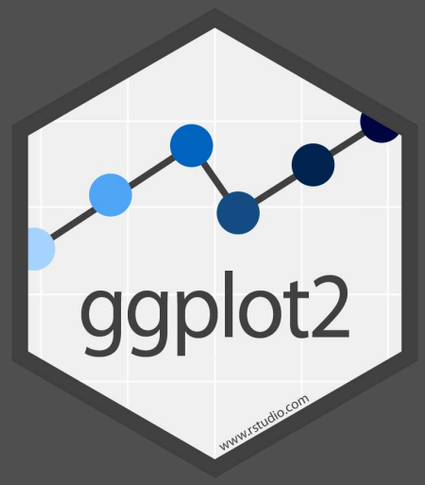

- Download the model summary data set from the course website ("model_summary.csv").

- Read the file into R using:

```
model_summary ← read_csv("your-path/model_summary.csv")
```

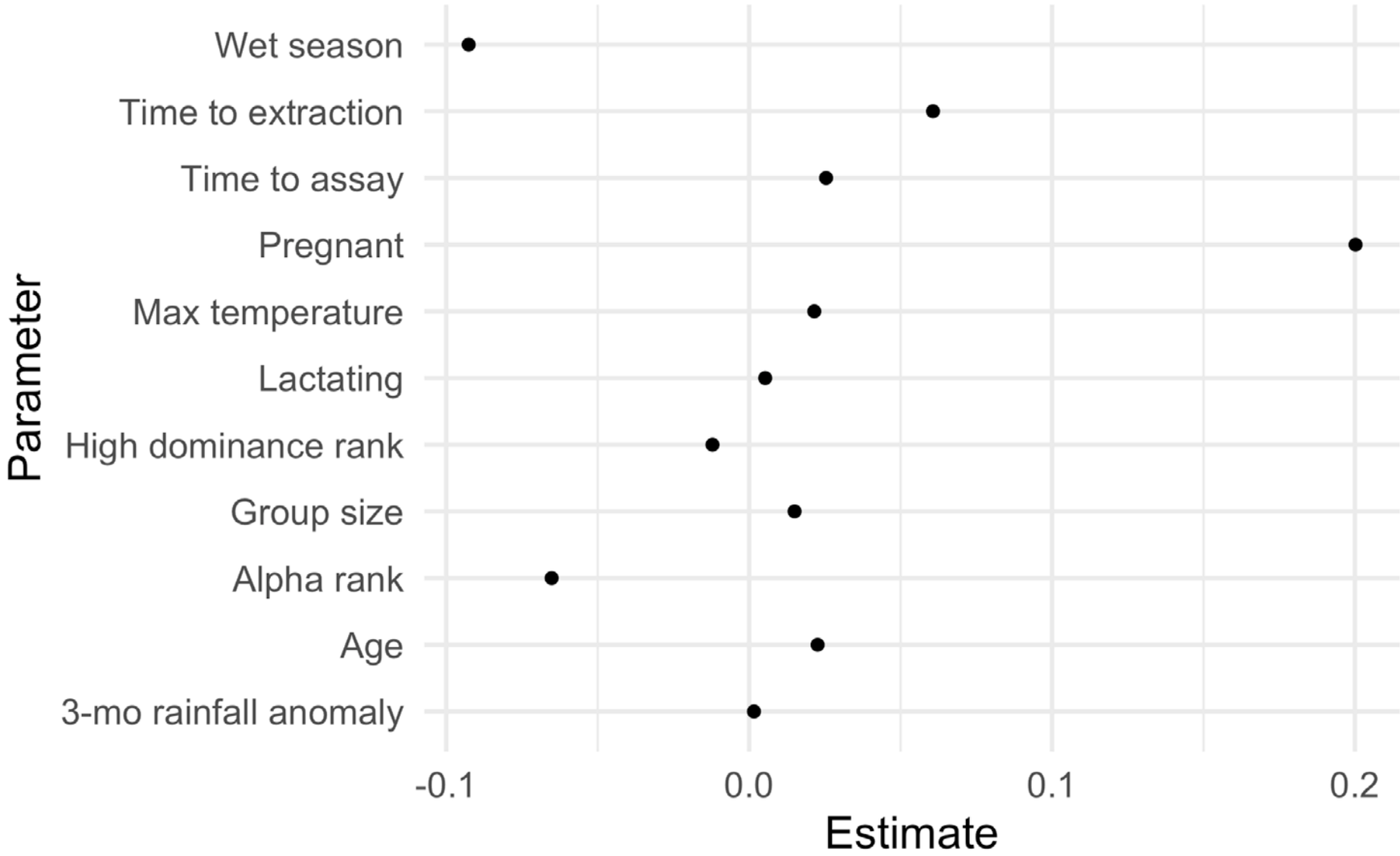


# YOUR TURN




| parameter | mean | std_error | lower | upper |
|---|---|---|---|---|
| Lactating | 0.0053 | 0.0089 | -0.0094 | 0.0198 |
| Pregnant | 0.2002 | 0.0097 | 0.1842 | 0.2163 |
| Wet season | -0.0926 | 0.0084 | -0.1063 | -0.0786 |
| Group size | 0.0150 | 0.0071 | 0.0033 | 0.0267 |
| Alpha rank | -0.0652 | 0.0316 | -0.1171 | -0.0133 |
| High dominance rank | -0.0121 | 0.0104 | -0.0293 | 0.0050 |
| Max temperature | 0.0215 | 0.0043 | 0.0144 | 0.0286 |
| 3-mo rainfall anomaly | 0.0016 | 0.0037 | -0.0044 | 0.0077 |
| Time to extraction | 0.0607 | 0.0044 | 0.0534 | 0.0680 |
| Time to assay | 0.0254 | 0.0040 | 0.0189 | 0.0321 |

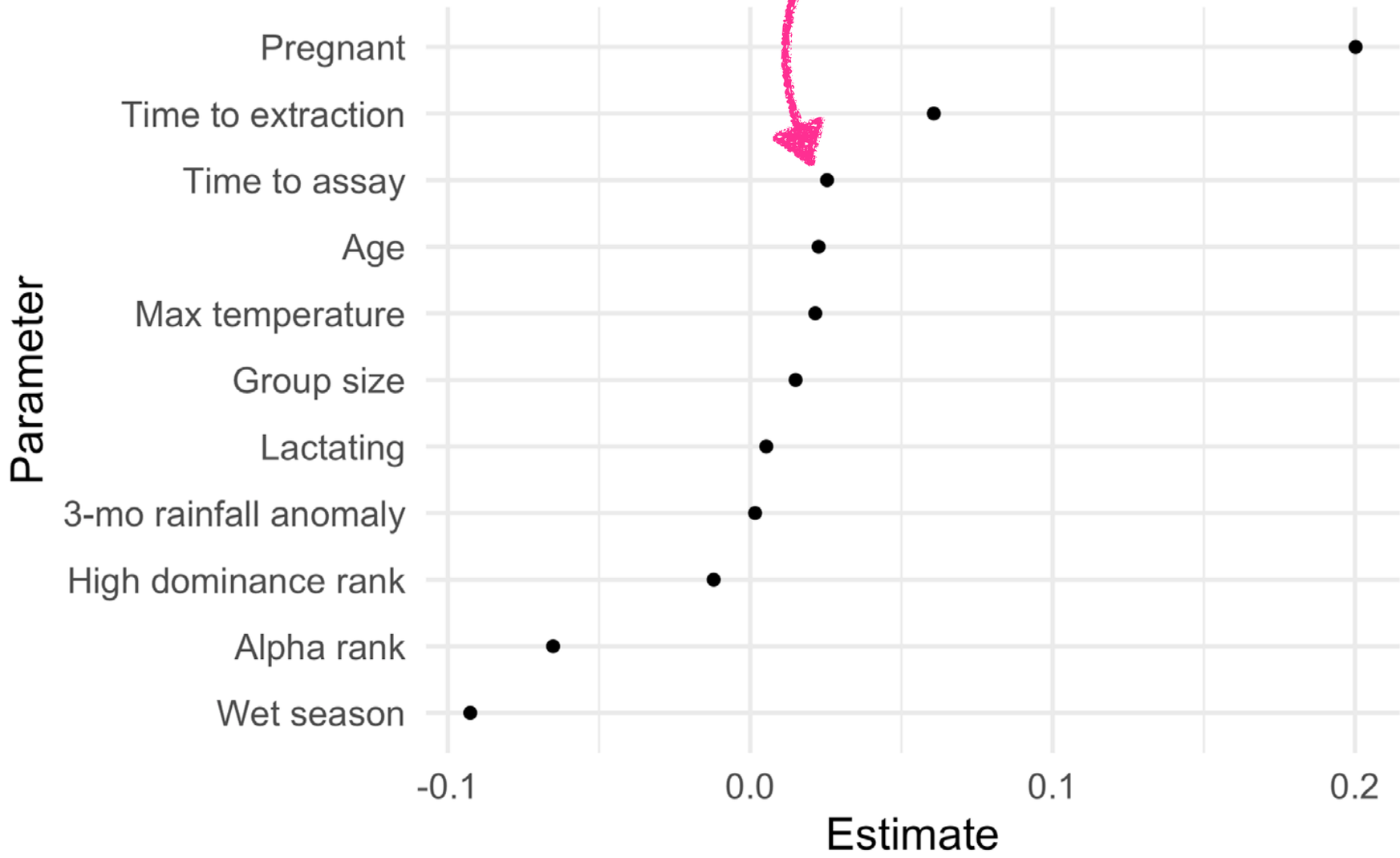
# YOUR TURN




```
ggplot(model_summary, aes(x = mean, y = parameter)) +
  geom_point(size = 2) +
  labs(x = "Estimate", y = "Parameter")
```

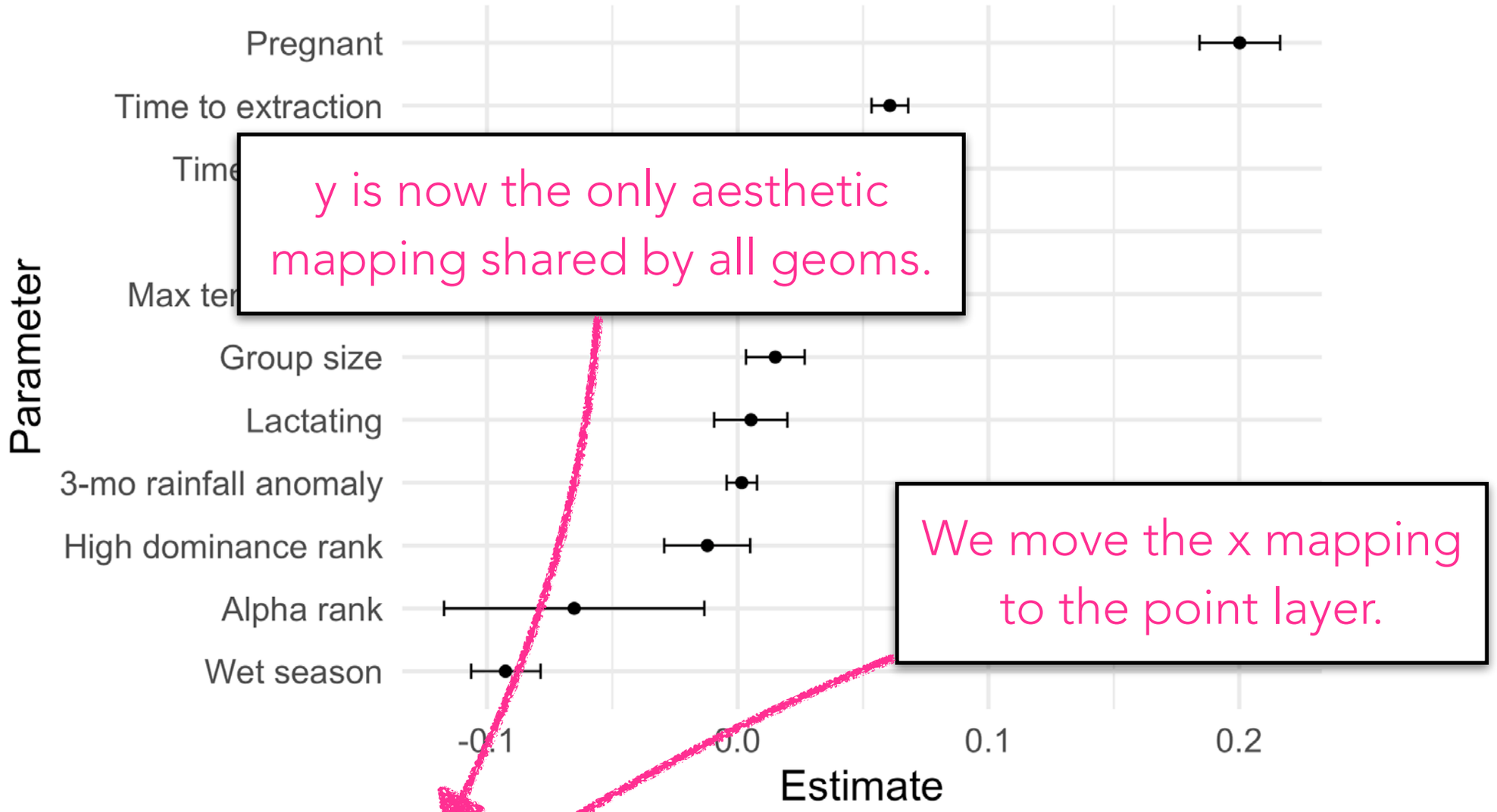
# YOUR TURN

Reordering by mean looks cleaner and improves interpretation.



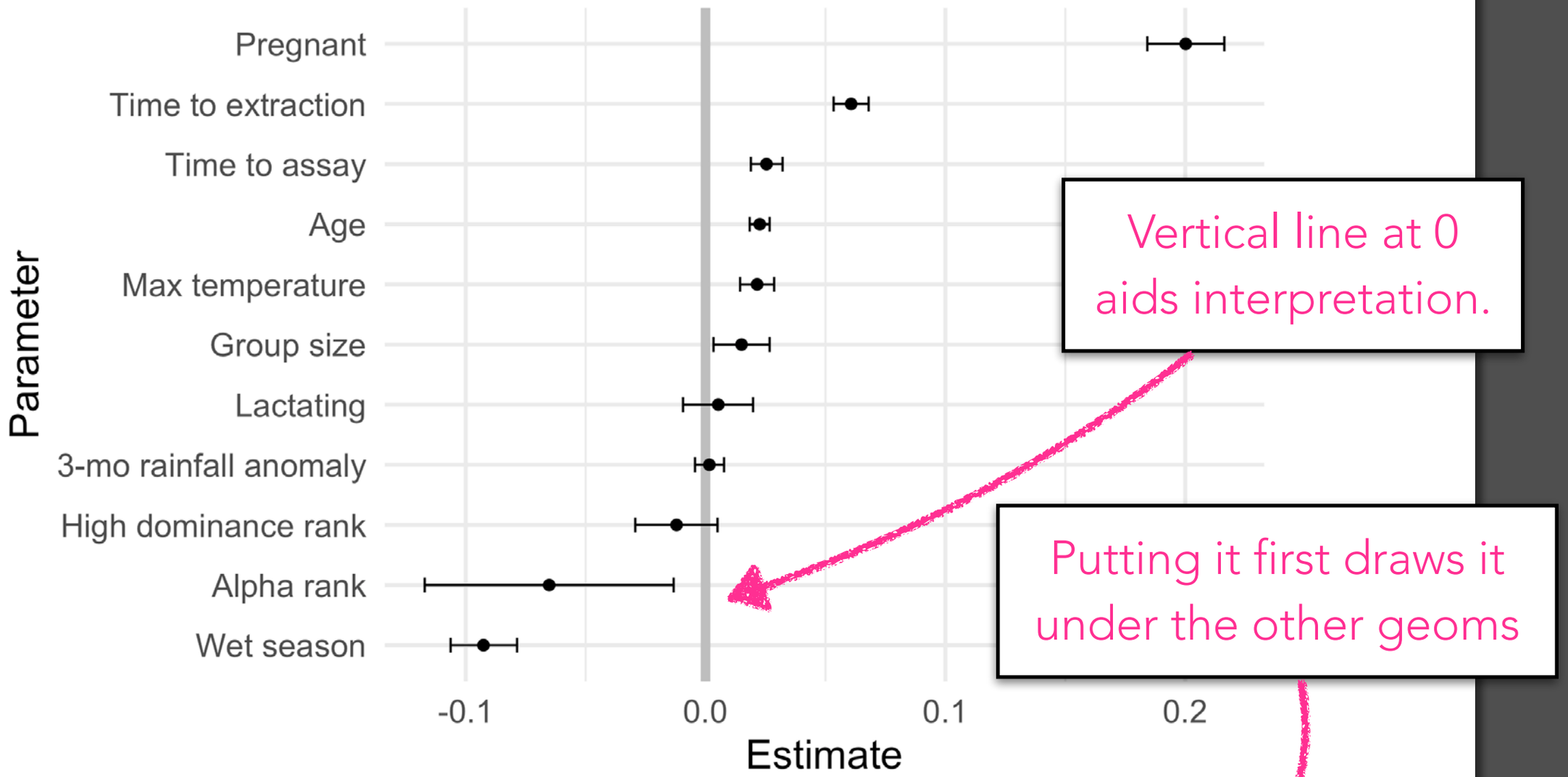

```
ggplot(model_summary, aes(x = mean, y = fct_reorder(parameter, mean))) +
  geom_point(size = 2) +
  labs(x = "Estimate", y = "Parameter")
```

# YOUR TURN




```
ggplot(model_summary, aes(y = fct_reorder(parameter, mean))) +
  geom_errorbarh(aes(xmin = lower, xmax = upper), height = 0.25) +
  geom_point(aes(x = mean), size = 2) +
  labs(x = "Estimate", y = "Parameter")
```

# YOUR TURN

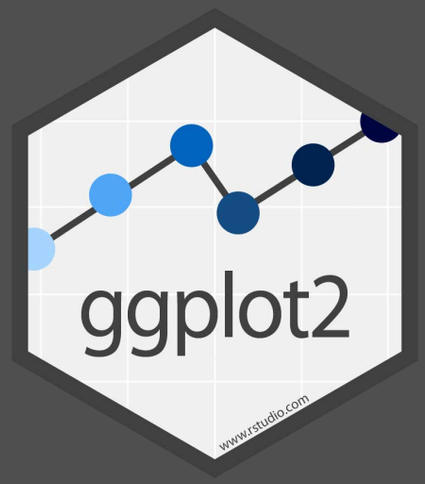



Vertical line at 0 aids interpretation.

Putting it first draws it under the other geoms

```
ggplot(model_summary, aes(y = fct_reorder(parameter, mean))) +
  geom_vline(xintercept = 0, color = "gray", size = 2) +
  geom_errorbarh(aes(xmin = lower, xmax = upper), height = 0.25) +
  geom_point(aes(x = mean), size = 2) +
  labs(x = "Estimate", y = "Parameter")
```

# YOUR TURN




- Go to this week's assignments on the course website.

- Download the **quaternary_mammals.qmd** quarto file.

- Follow the instructions to visualize distributions.

